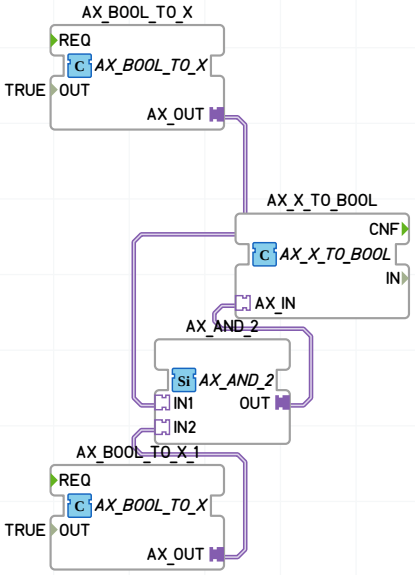


Uebung_000_AX

Einfache Addition



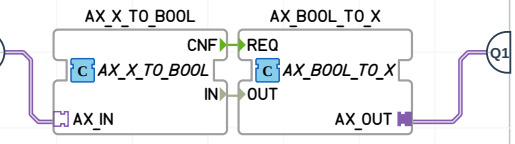
Uebung_000b_AX

AX_AND



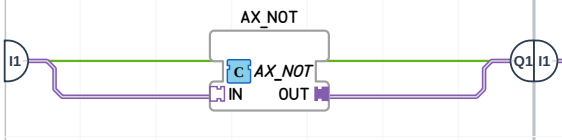
Uebung_001_AX

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



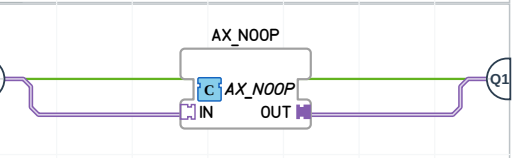
Uebung_001_AX_b

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



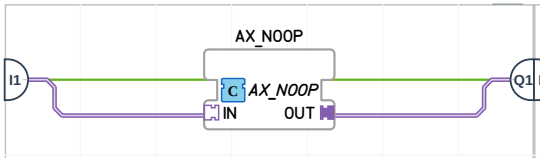
Uebung_001c2_AX

DigitalInput_I1 negiert auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



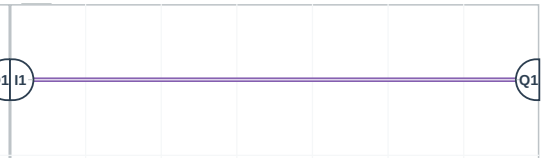
Uebung_001c3_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



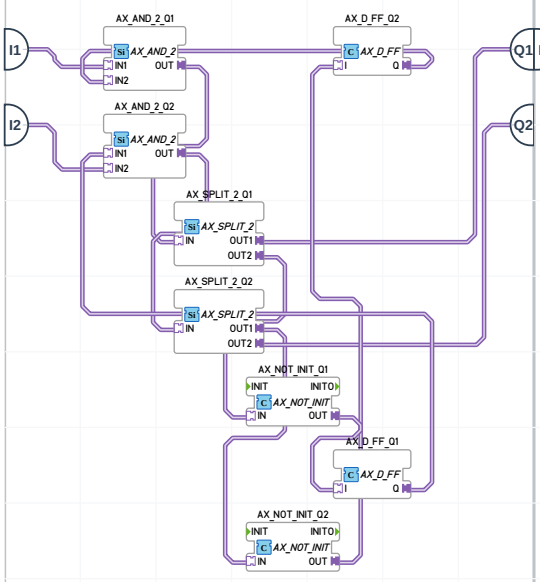
Uebung_001c4_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



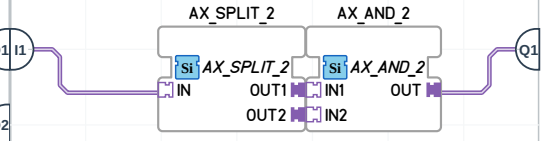
Uebung_001c_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



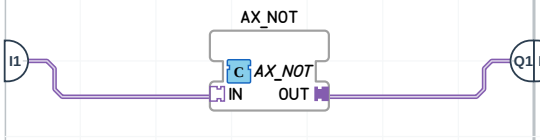
Uebung_001d2_AX

DigitalInput_I1/2 auf DigitalOutput_Q1/2, als Alternative(Verriegelt) mit Plug and Socket, ohne ECC



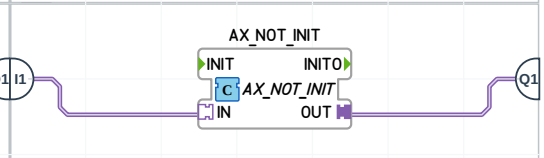
Uebung_001d_AX

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



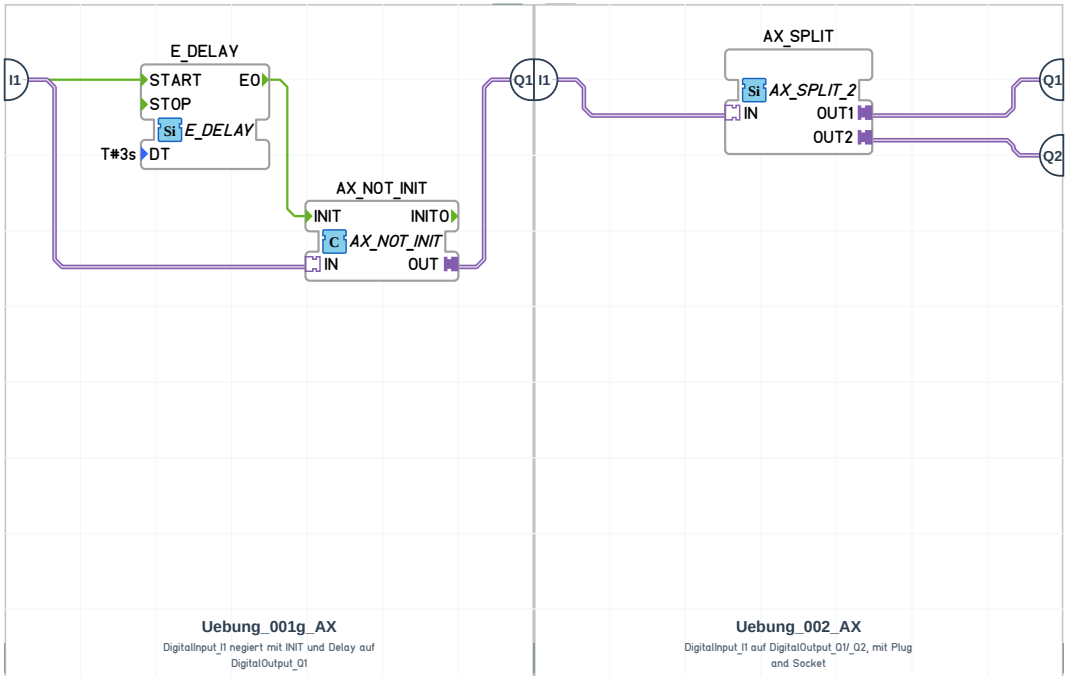
Uebung_001e_AX

DigitalInput_I1 negiert auf DigitalOutput_Q1



Uebung_001f_AX

DigitalInput_I1 negiert mit INIT auf DigitalOutput_Q1

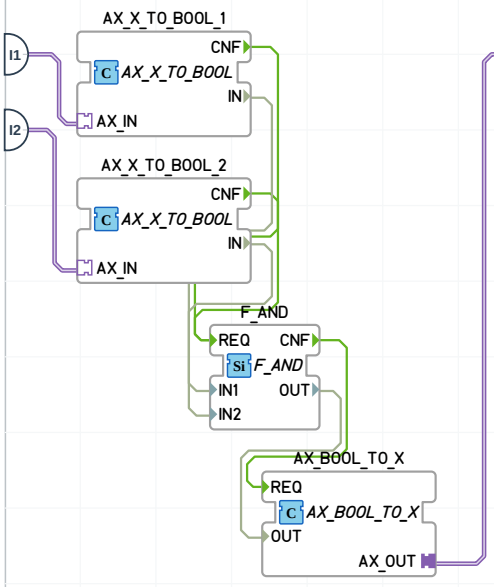


Uebung_001g_AX

DigitalInput_I1 negiert mit INIT und Delay auf DigitalOutput_Q1

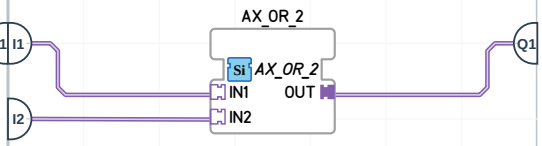
Uebung_002_AX

DigitalInput_I1 auf DigitalOutput_Q1/_Q2, mit Plug and Socket



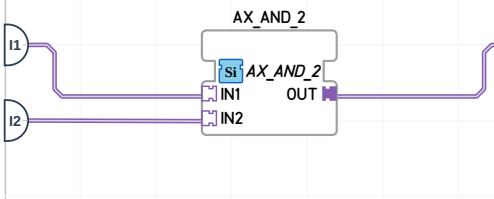
Uebung_002a2_AX

DigitalInput_I1/_I2 mit F_AND (bitwise) auf DigitalOutput_Q1



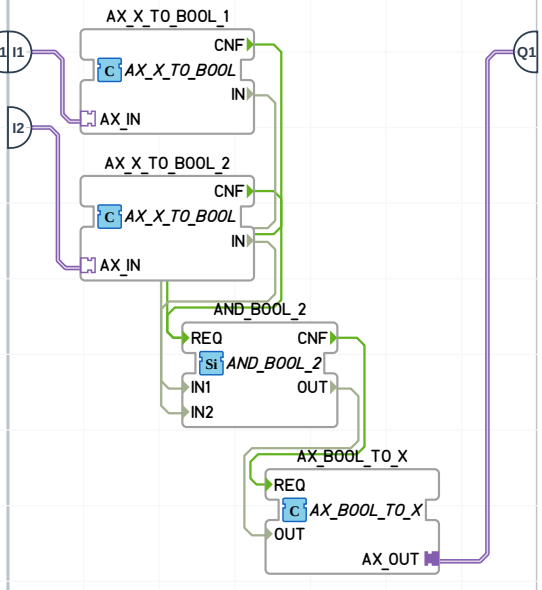
Uebung_002a3_AX

DigitalInput_I1/_I2 mit OR auf DigitalOutput_Q1, mit Plug and Socket



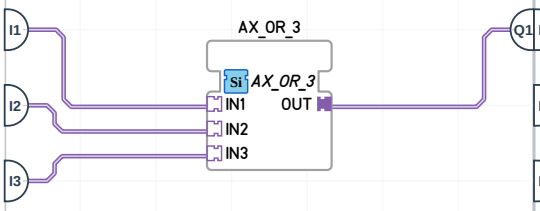
Uebung_002a4_AX

DigitalInput_I1/_I2 mit AX_AND_2 auf DigitalOutput_Q1



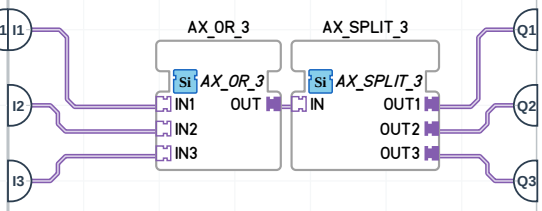
Uebung_002a4b_AX

DigitalInput_I1/_I2 mit AND_BOOL und Negate auf DigitalOutput_Q1



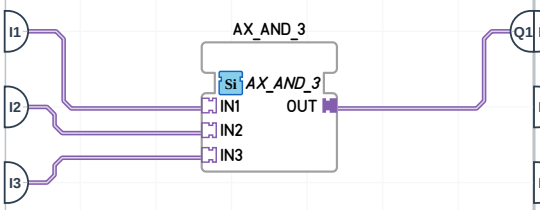
Uebung_002a5_AX

DigitalInput_I1-3 mit OR auf DigitalOutput_Q1, mit Plug and Socket



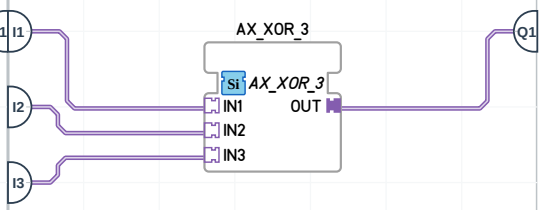
Uebung_002a5b_AX

DigitalInput_I1-3 mit OR auf DigitalOutput_Q1-3, mit Plug and Socket



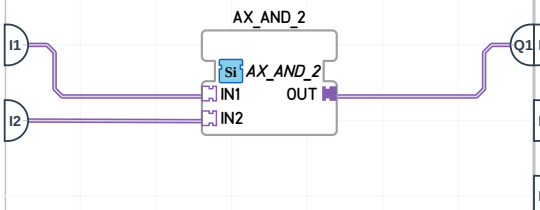
Uebung_002a6_AX

DigitalInput_I1-3 mit AND auf DigitalOutput_Q1, mit Plug and Socket



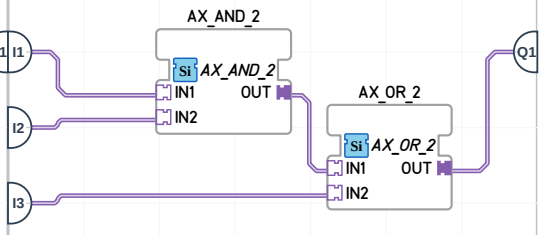
Uebung_002a7_AX

DigitalInput_I1-3 mit XOR auf DigitalOutput_Q1, mit Plug and Socket



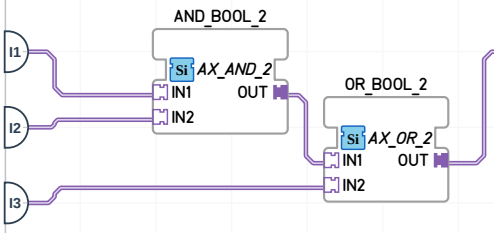
Uebung_002a_AX

DigitalInput_I1/_I2 mit AND auf DigitalOutput_Q1, mit Plug and Socket



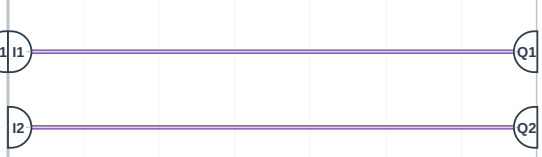
Uebung_002b2_AX

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1, mit AX Adapter



Uebung_002b3_AX

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1



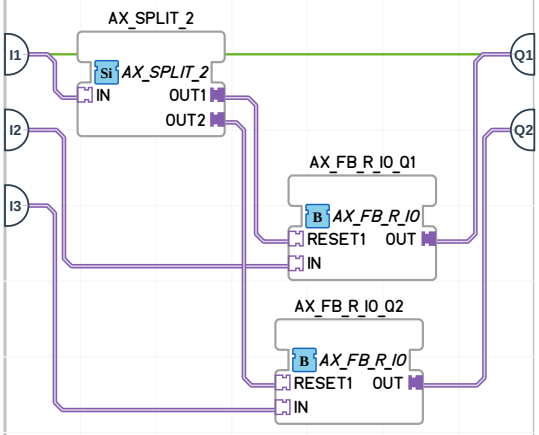
Uebung_003_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – flach mit Adapter



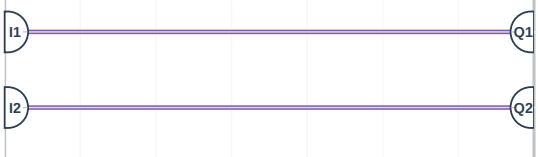
Uebung_003a0_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit untyped Subapp



Uebung_003a2b_AX

2x R_I0 mit IXA



Uebung_003a_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit typed Subapp



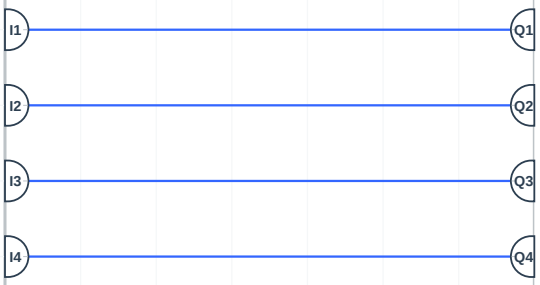
Uebung_003b2_AX

Funk 16 Tasten auf DataPanel 16



Uebung_003b3_AX

Funk 16 Tasten auf DataPanel 16



Uebung_003b_AX

DigitalInput_I1-2-3-4 auf DigitalOutput_Q1-2-3-4

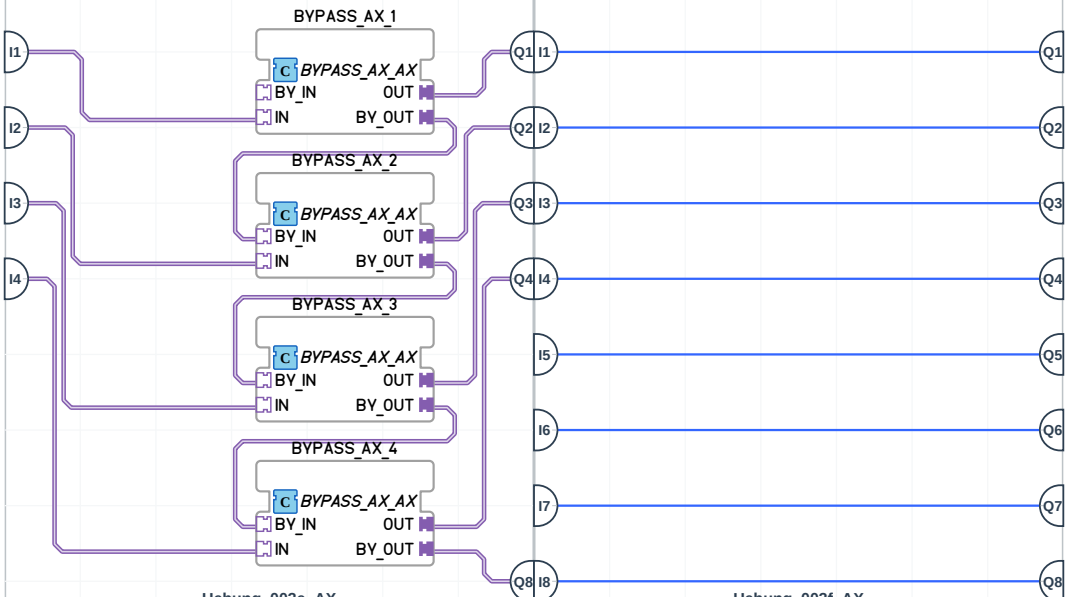
Uebung_003c_AX

DigitalInput_I1-2-3-4 auf AUX



Uebung_003d_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2

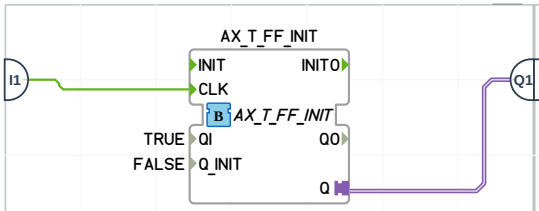


Uebung_003e_AX

DigitalInput_I1-4 auf DigitalOutput_Q1-4 – flach mit Adapter mit Bypass (Druckumlaufventil)

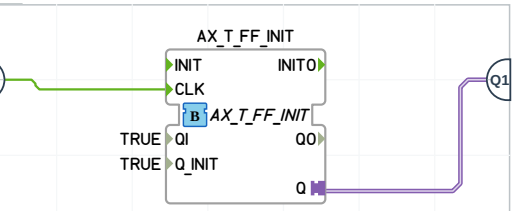
Uebung_003f_AX

DigitalInput_I1-8 auf DigitalOutput_Q1-8



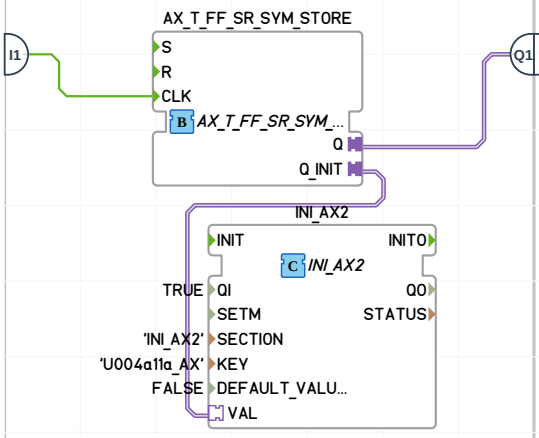
Uebung_004a10a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf FALSE



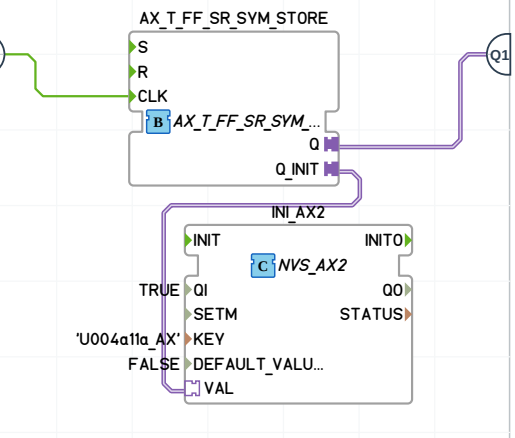
Uebung_004a10b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf TRUE



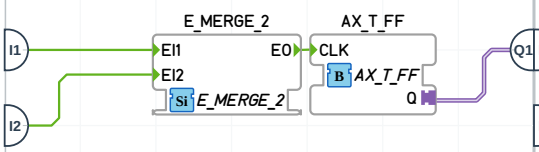
Uebung_004a11a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (INI)



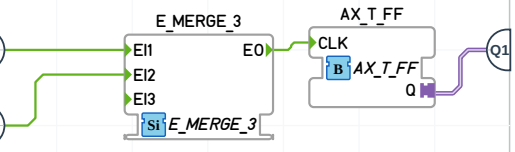
Uebung_004a11b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (NVS)



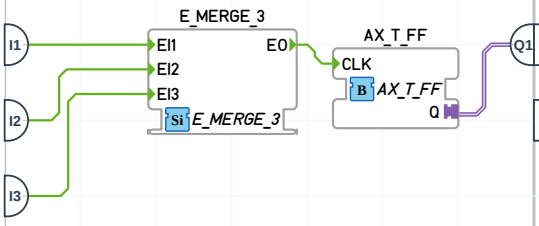
Uebung_004a2_2_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_2



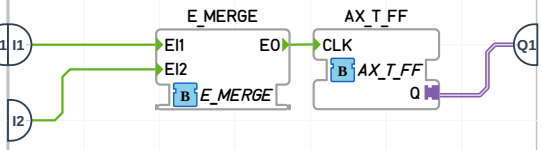
Uebung_004a2_3_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_3



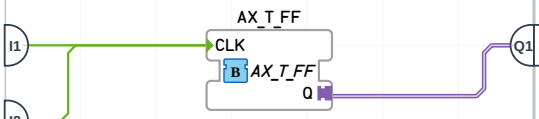
Uebung_004a2_3b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_3 (3 Taster)



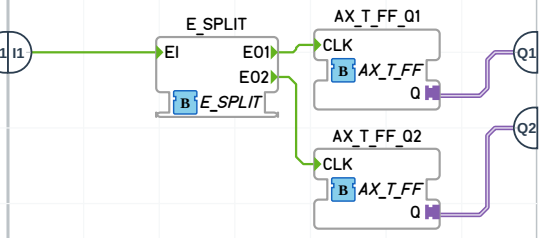
Uebung_004a2_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE



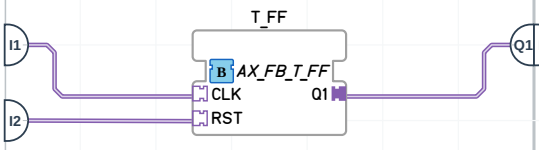
Uebung_004a3_AX

wie Uebung_004a2 aber ohne E_MERGE



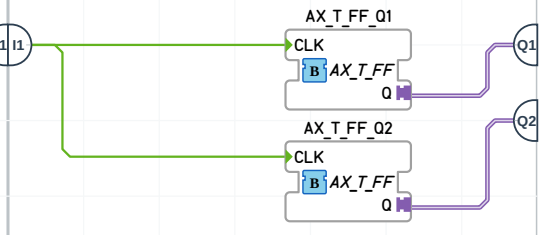
Uebung_004a4_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_SPLIT



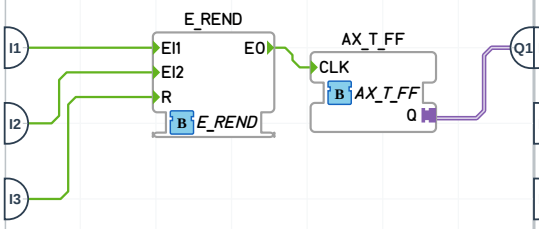
Uebung_004a4_AX_T

Exercise for AX_FB_T_FF (Toggle Flip-Flop with Adapter)



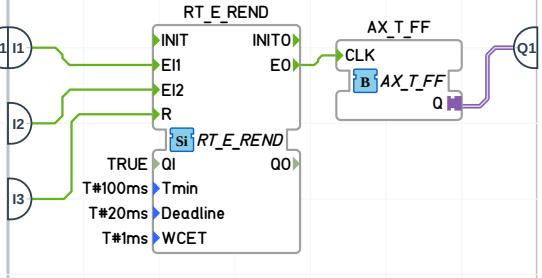
Uebung_004a5_AX

wie Uebung_004a4 aber ohne E_SPLIT



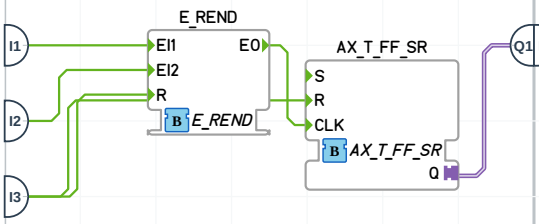
Uebung_004a6_AX

Toggle Flip-Flop mit IE und E_REND (Rendezvous)



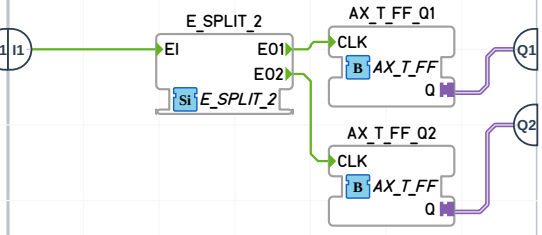
Uebung_004a6a_AX

Toggle Flip-Flop mit IE und E_REND (Rendezvous)



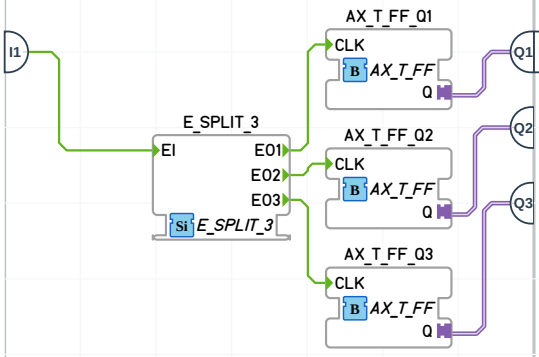
Uebung_004a7_AX

SR und T-Flip-Flop mit IE und E_REND (Rendezvous)



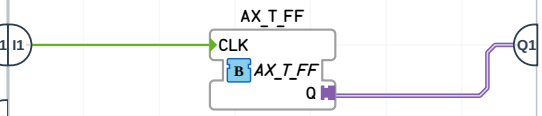
Uebung_004a8_AX

mit E_SPLIT_2



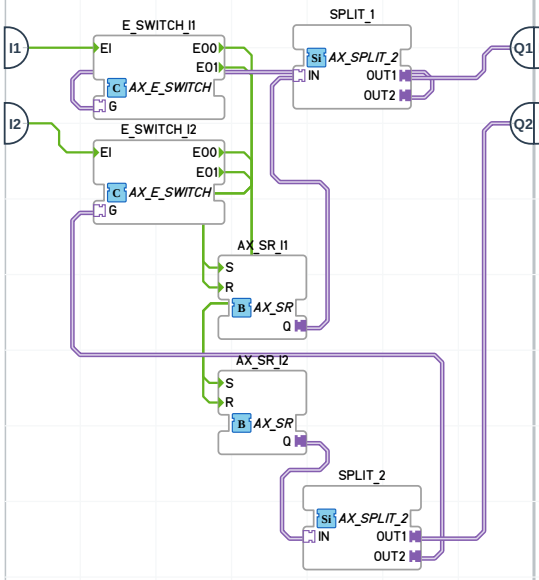
Uebung_004a9_AX

mit E_SPLIT_3



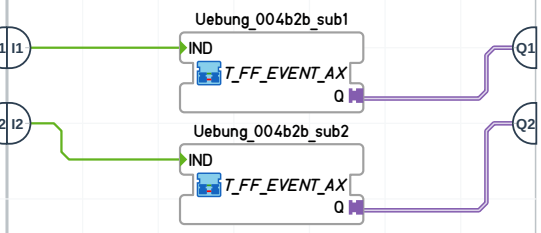
Uebung_004a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK



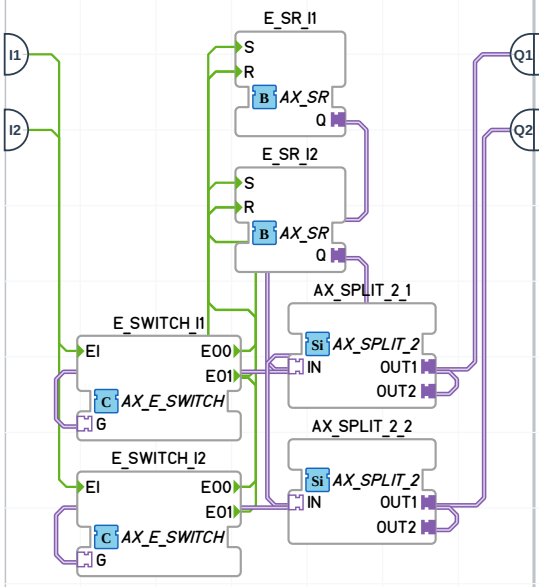
Uebung_004b2_AX

Toggle Flip-Flop mit IE / Split / doppelt (aus AX_E_SWITCH und AX_SR aufgebaut)



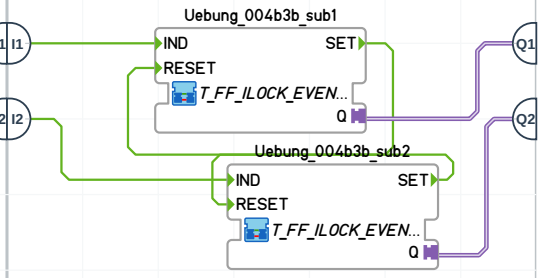
Uebung_004b2b_AX

Zwei unabhängige Toggle-Flip-Flops unter Verwendung von Sub-Applikationen



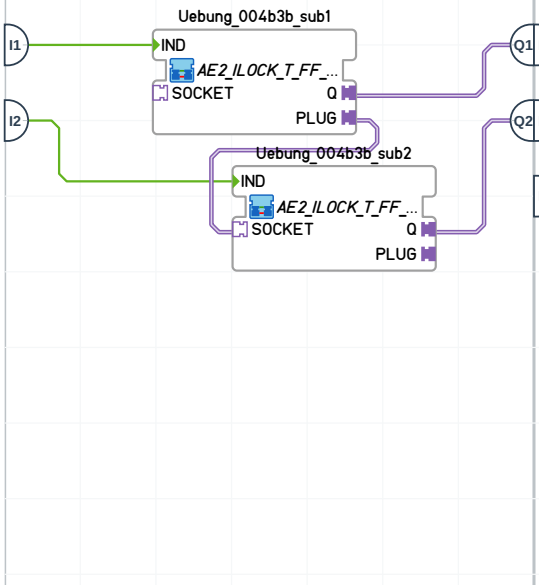
Uebung_004b3_AX

Toggle Flip-Flop mit IE / Split / Verriegelt



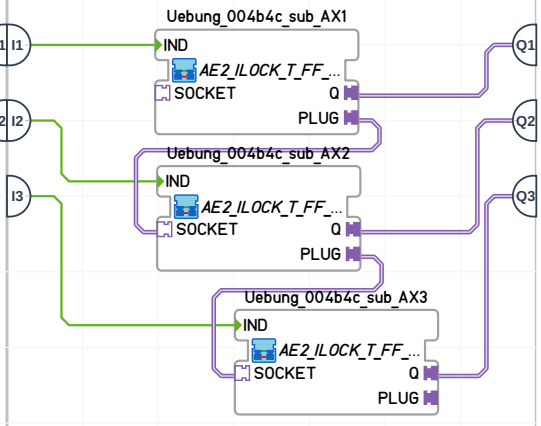
Uebung_004b3b_AX

Zwei gegenseitig verriegelte Toggle-Flip-Flops (Radio-Button-Logik) via Event-Verdrahtung



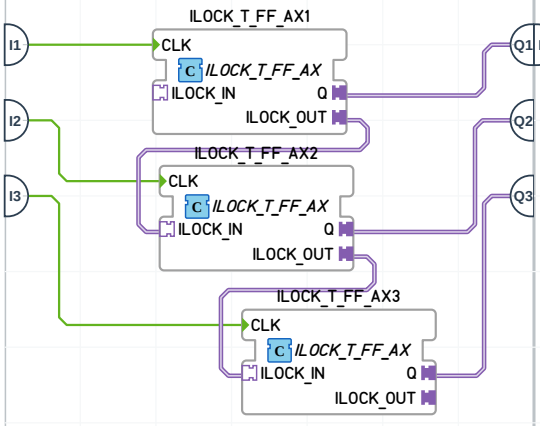
Uebung_004b3c_AX

Zwei gegenseitig verriegelte Toggle-Flip-Flops unter Verwendung von bidirektionalen Adaptern



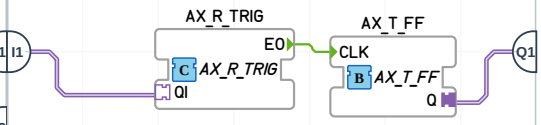
Uebung_004b4c_AX

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter



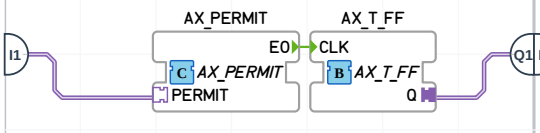
Uebung_004b4d_AX

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter mit ILOCK_T_FF_AX



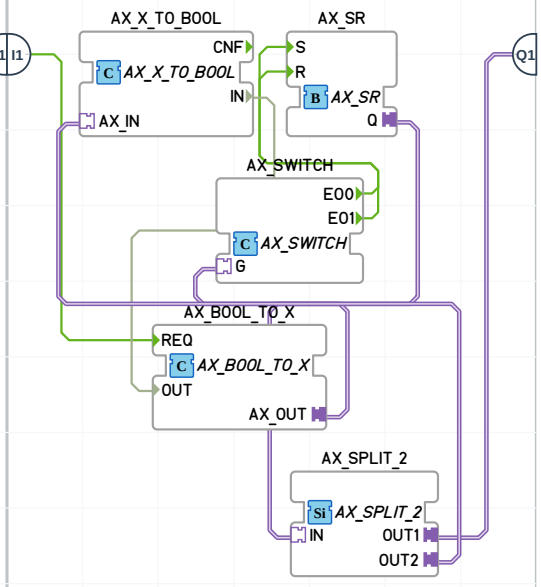
Uebung_004b5_AX

Toggle Flip-Flop mit IXA / AX_R_TRIG + AX_T_FF



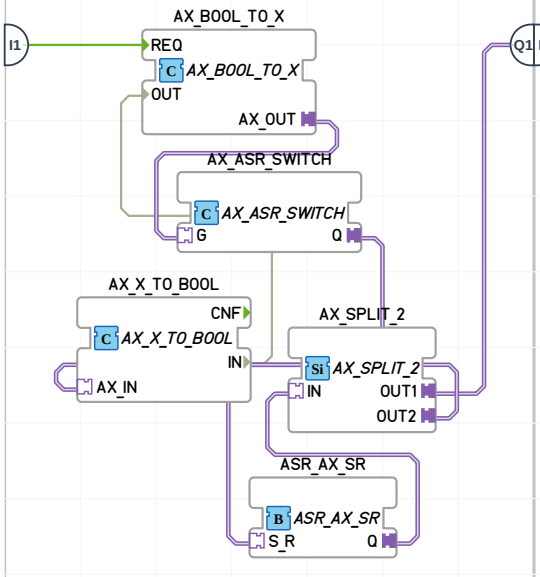
Uebung_004b6_AX

Toggle Flip-Flop mit IXA / AX_PERMIT + AX_T_FF



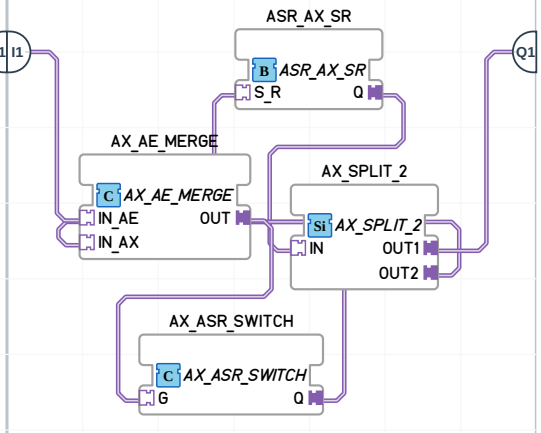
Uebung_004b_AX

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



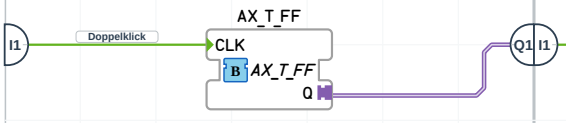
Uebung_004b_AX_ASR

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



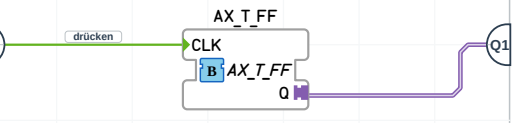
Uebung_004b_AX_ASR_X

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



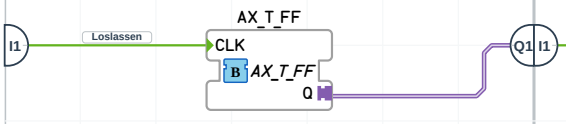
Uebung_004c1_AX

Toggle Flip-Flop mit IE mit BUTTON_DOUBLE_CLICK



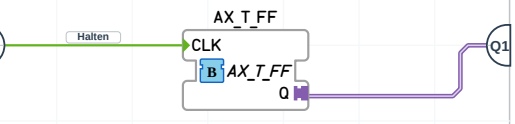
Uebung_004c2_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_START



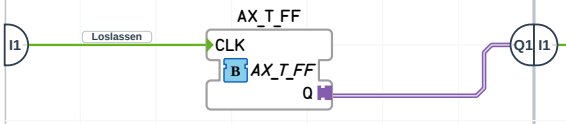
Uebung_004c3_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_UP



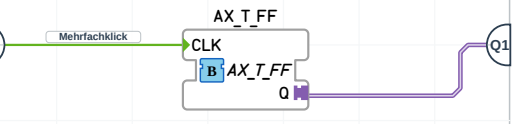
Uebung_004c4_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_HOLD



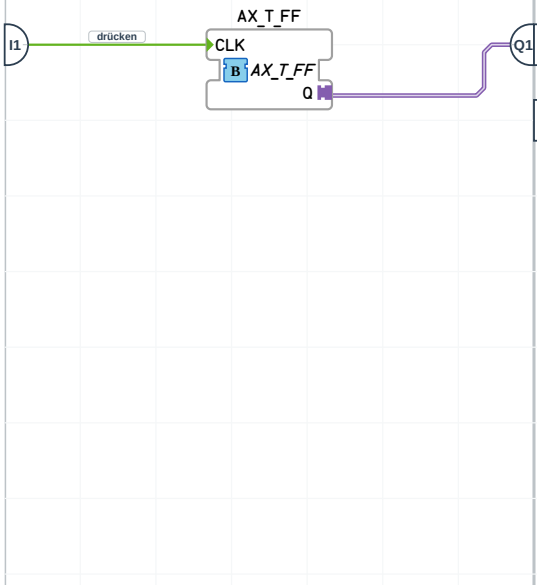
Uebung_004c5_AX

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END



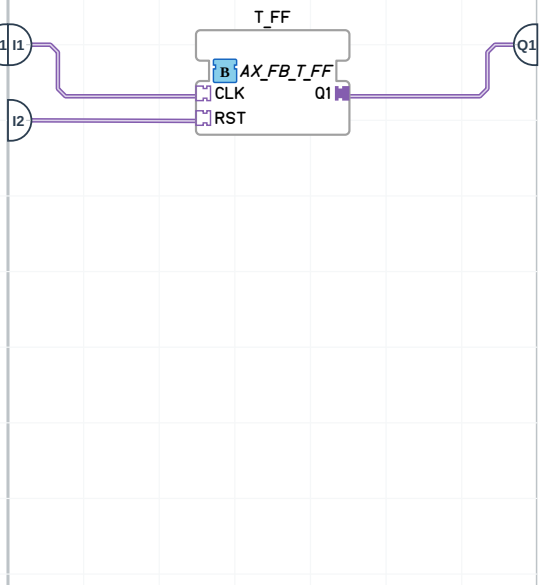
Uebung_004c6_AX

Toggle Flip-Flop mit IE mit BUTTON_MULTIPLE_CLICK
(3-fach Klick)



Uebung_004c7_AX

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END



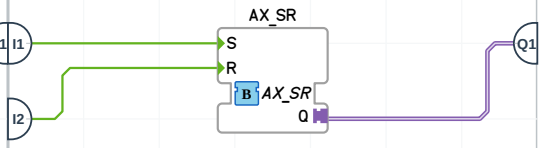
Uebung_004d_T_AX

Exercise for FB_T_FF (Toggle Flip-Flop)



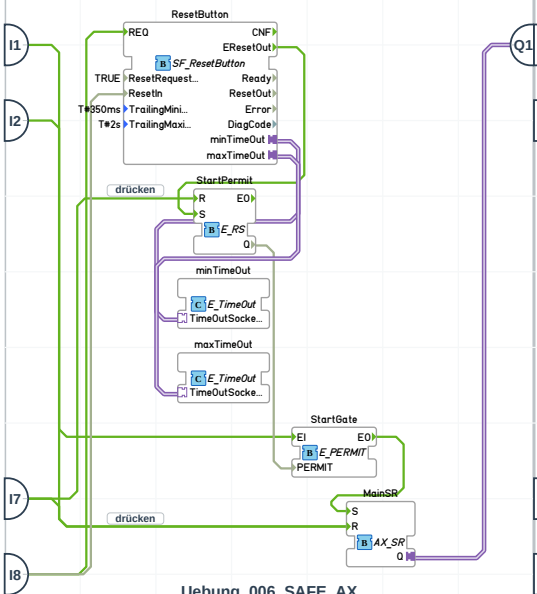
Uebung_005_AX

Toggle Flip-Flop mit IXA



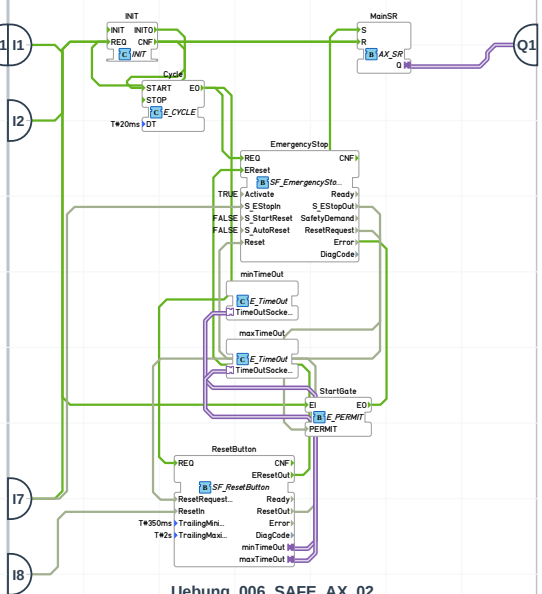
Uebung_006_AX

SR-Flip-Flop mit 2x IE



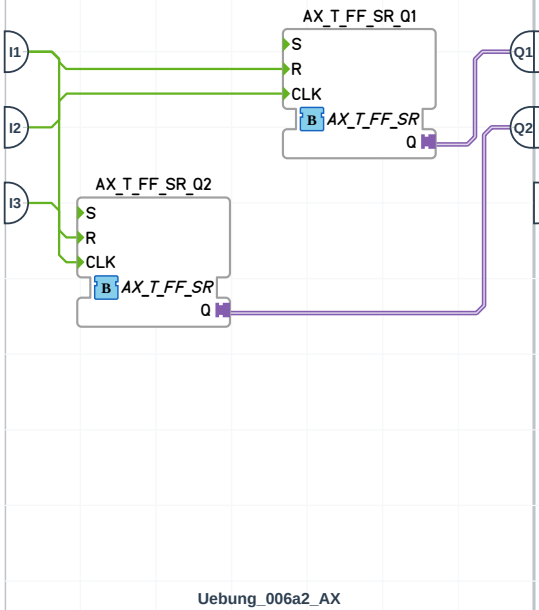
Uebung_006_SAFE_AX

SAFE-Variante: Wiederanlaufsperre fuer Uebung_006_AX (SF_ResetButton, I7/I8)



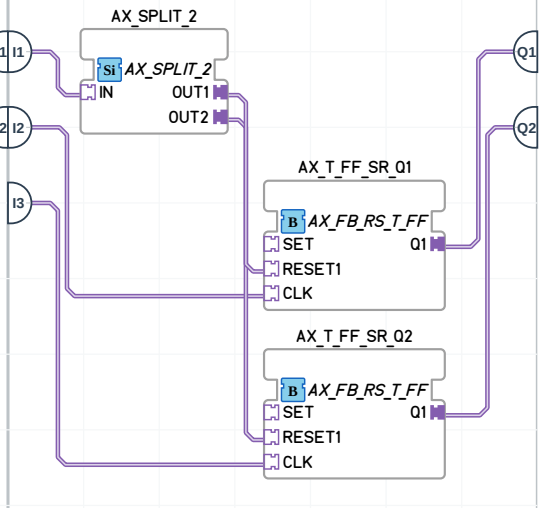
Uebung_006_SAFE_AX_02

Wie Uebung_006_SAFE_AX, aber mit SF_EmergencyStop statt StartPermit/StartGate



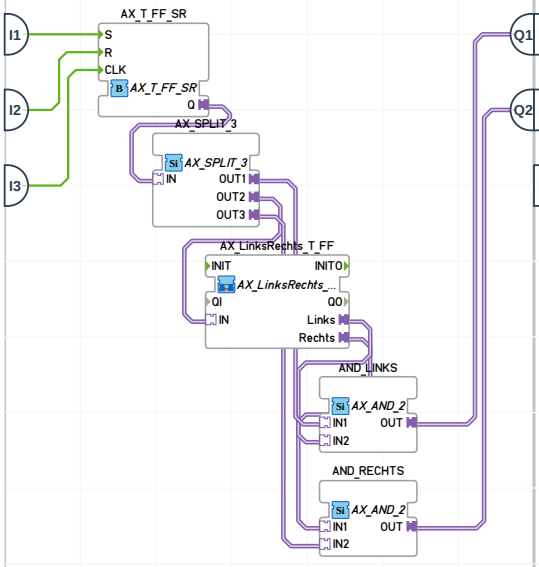
Uebung_006a2_AX

2x SR und T-Flip-Flop mit IE



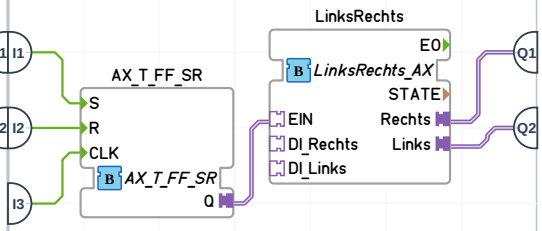
Uebung_006a2b_AX

2x SR und T-Flip-Flop mit IXA



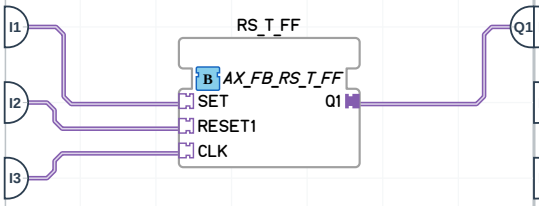
Uebung_006a3_AX

SR und T-Flip-Flop mit 3x IE; links/rechtslauf



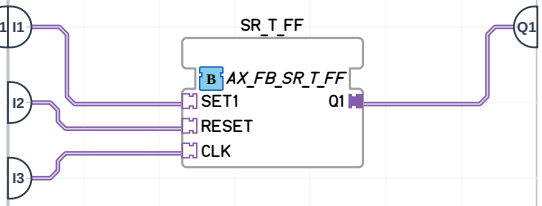
Uebung_006a4_AX

SR und T-Flip-Flop mit 3x IE



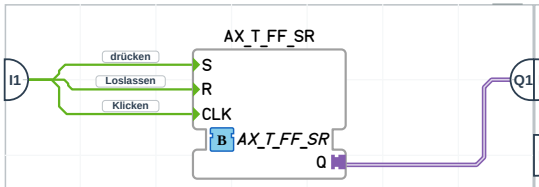
Uebung_006a4_AX_RS

Exercise for AX_FB_RS_T_FF (Reset-Dominant Toggle Flip-Flop with Adapter)



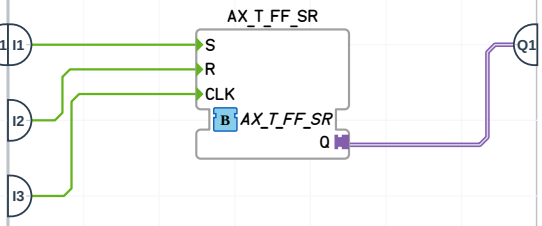
Uebung_006a4_AX_SR

Exercise for AX_FB_SR_T_FF (Set-Dominant Toggle Flip-Flop with Adapter)



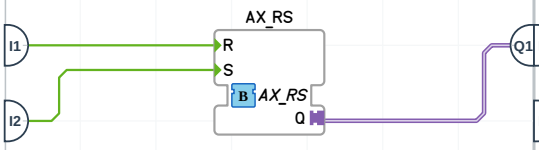
Uebung_006a5_AX

SR und T-Flip-Flop als Rastend/Tastend Implementierung



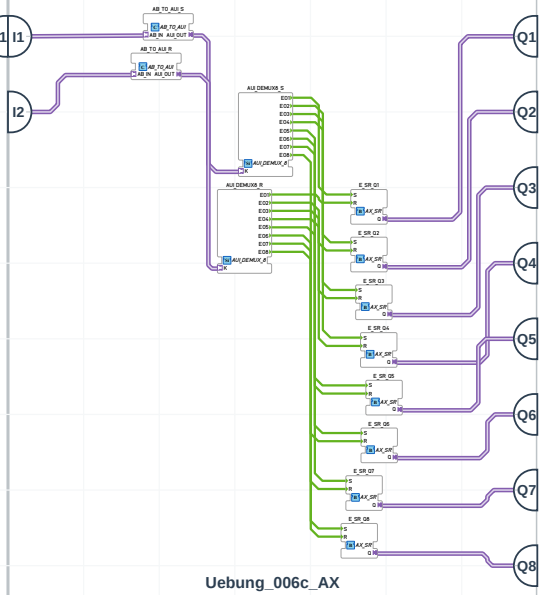
Uebung_006a_AX

SR und T-Flip-Flop mit 3x IE



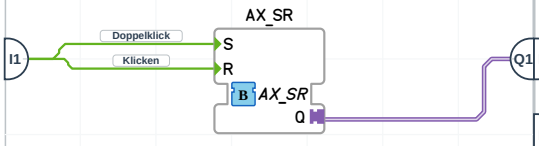
Uebung_006b_AX

RS-Flip-Flop mit 2x IE



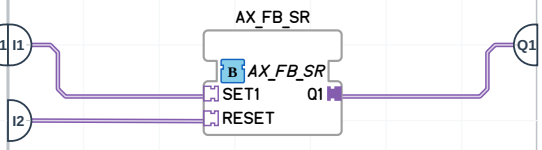
Uebung_006c_AX

SR-Flip-Flop mit IB auf DI_REPEAT



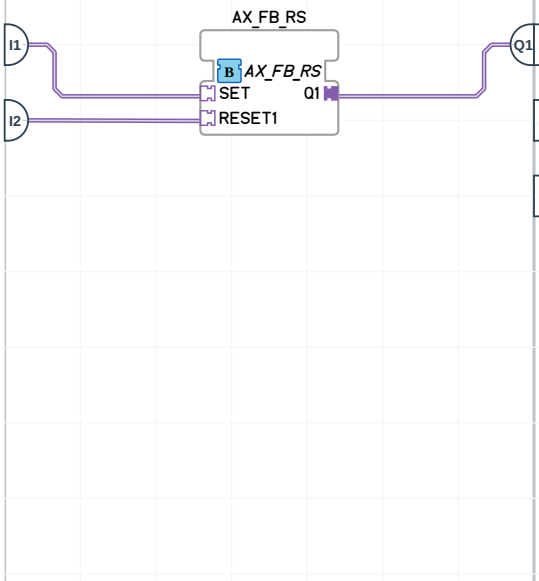
Uebung_006d_AX

SR-Flip-Flop mit 2x IE und Doppelklick.



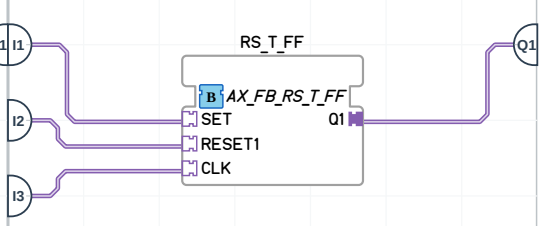
Uebung_006e1_AX

SR-Flip-Flop mit 2x IXA



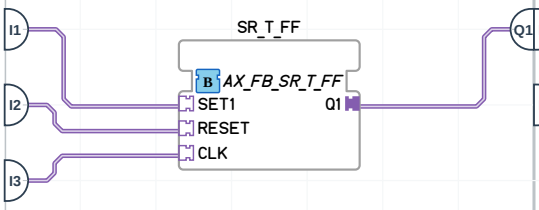
Uebung_006e2_AX

RS-Flip-Flop mit 2x IXA



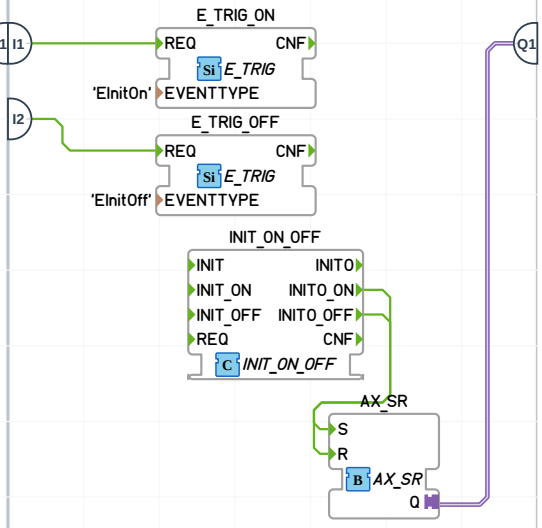
Uebung_006f_RS_AX

Exercise for FB_RS_T_FF (Reset-Dominant Toggle Flip-Flop)



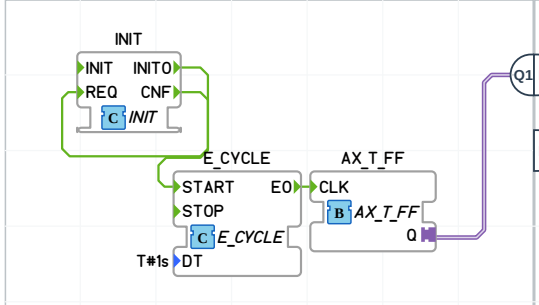
Uebung_006f_SR_AX

Exercise for FB_SR_T_FF (Set-Dominant Toggle Flip-Flop)



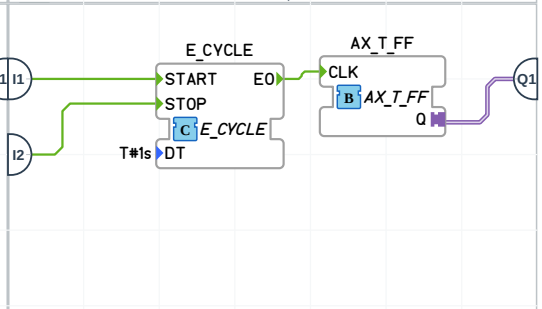
Uebung_006g_AX

SR-Flip-Flop via INIT_ON_OFF, ausgeloest von 2x E_TRIG (EInitOn/EInitOff Broadcast auf unverbundene Events)



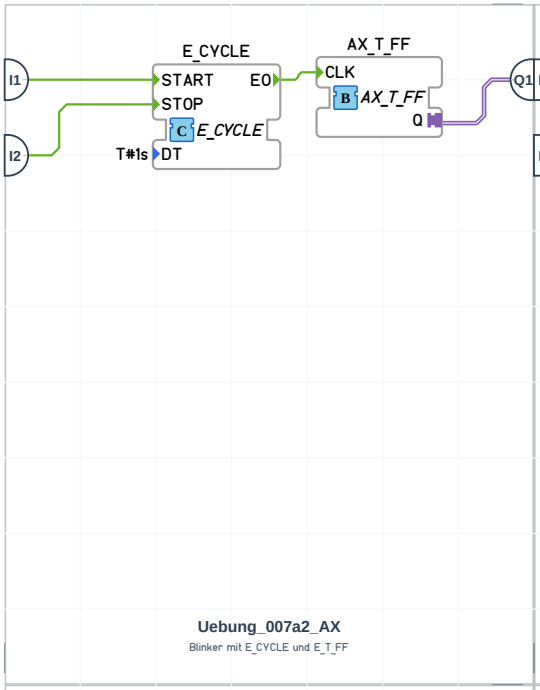
Uebung_007_AX

Blinker mit E_CYCLE und E_T_FF



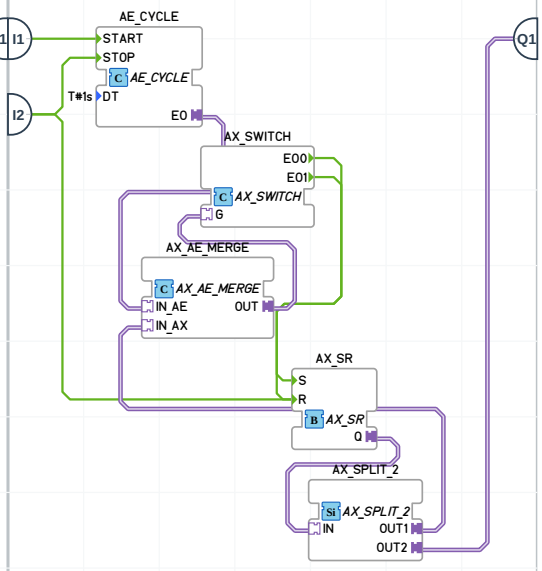
Uebung_007a1_AX

Blinker mit E_CYCLE und E_T_FF



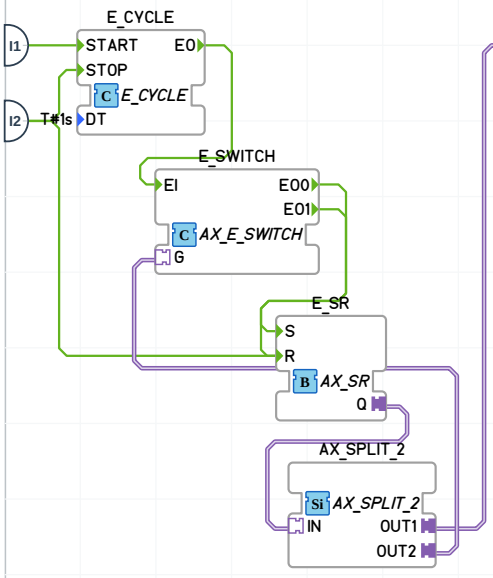
Uebung_007a2_AX

Blinker mit E_CYCLE und E_T_FF



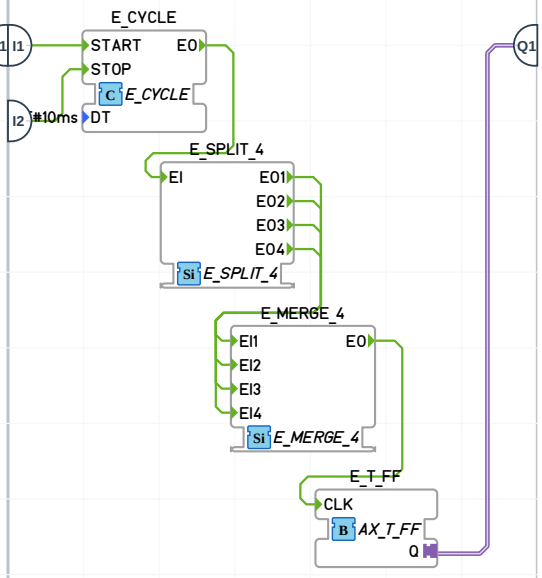
Uebung_007a3_AE

Blinker mit E_CYCLE, E_SWITCH und E_SR



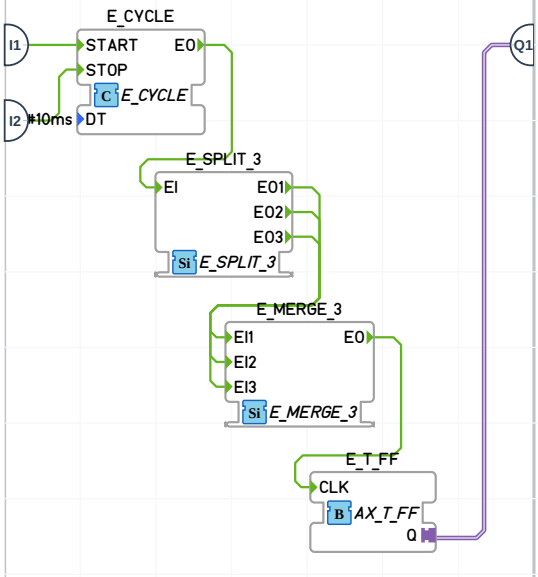
Uebung_007a3_AX

Blinker mit E_CYCLE, E_SWITCH und E_SR



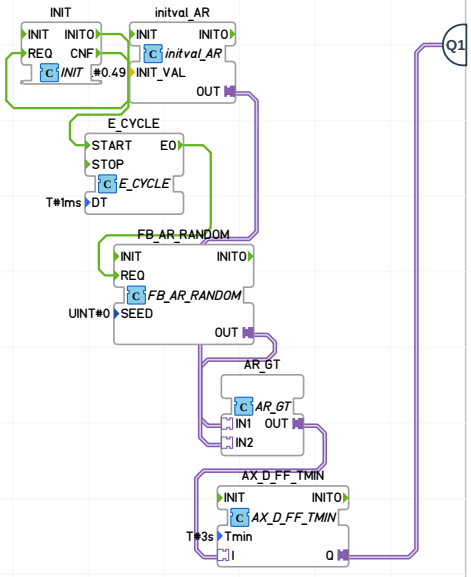
Uebung_007b_AX

Blinker mit E_CYCLE und E_T_FF



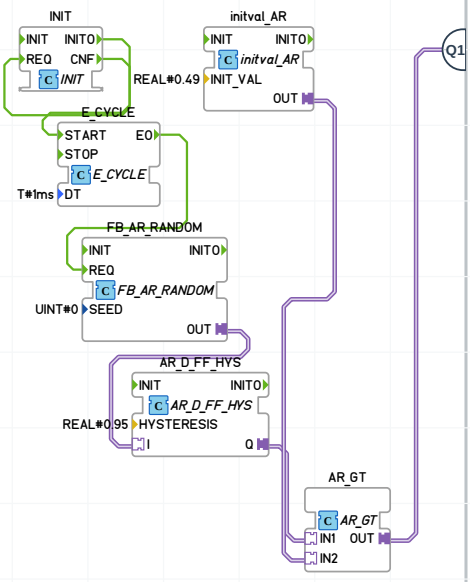
Uebung_007c_AX

Blinker mit E_CYCLE und E_T_FF



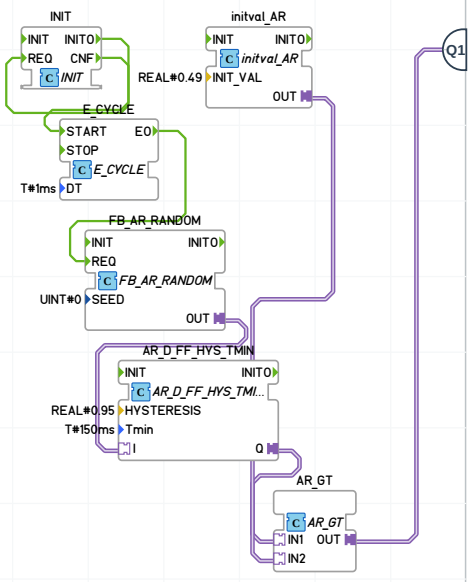
Uebung_007d2_AX

Blinker mit E_CYCLE, FB_AR_RANDOM, AR_GT und AX_D_FF_TMIN



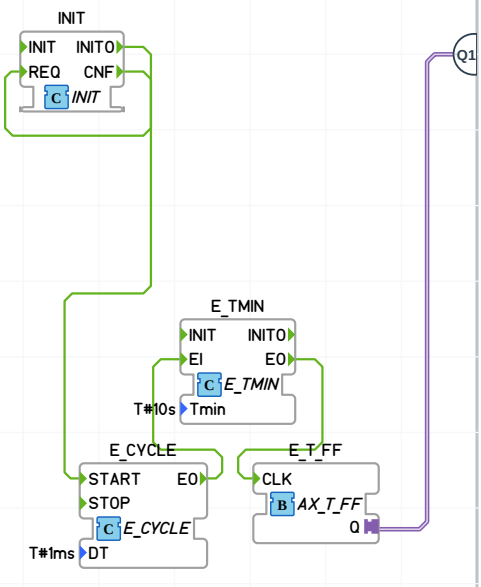
Uebung_007d3_AX

Blinker mit E_CYCLE, FB_AR_RANDOM, AR_D_FF_HYS und AR_GT



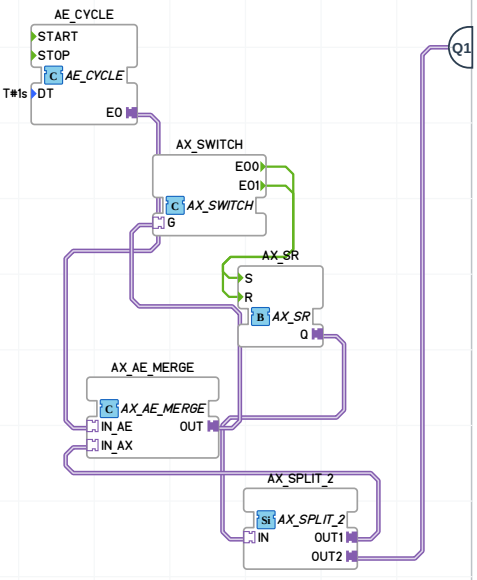
Uebung_007d4_AX

Random Generator mit FB_AR_RANDOM, AR_D_FF_HYS_TMIN, initval_AR und AR_GT auf DigitalOutput_Q1



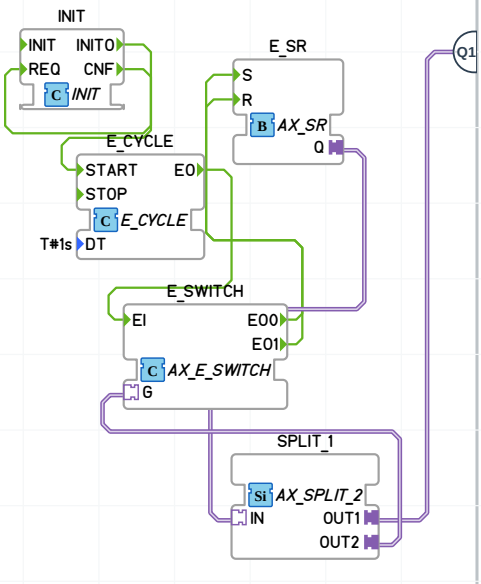
Uebung_007d_AX

Blinker mit E_CYCLE und E_T_FF



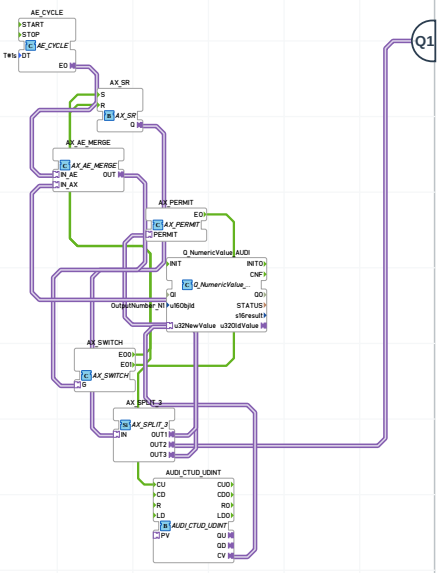
Uebung_008_AE

Blinker mit E_CYCLE, E_SWITCH und E_SR



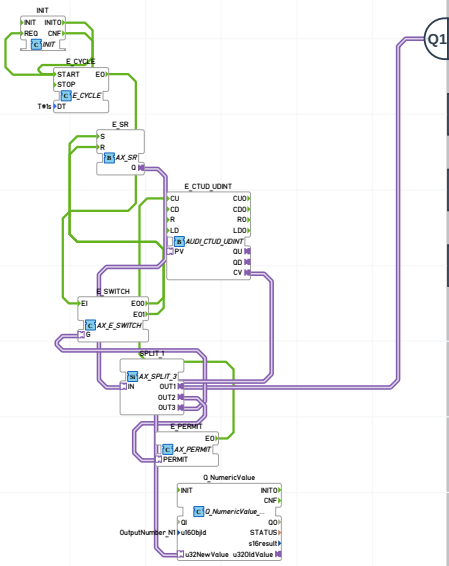
Uebung_008_AX

Blinker mit E_CYCLE, E_SWITCH und E_SR



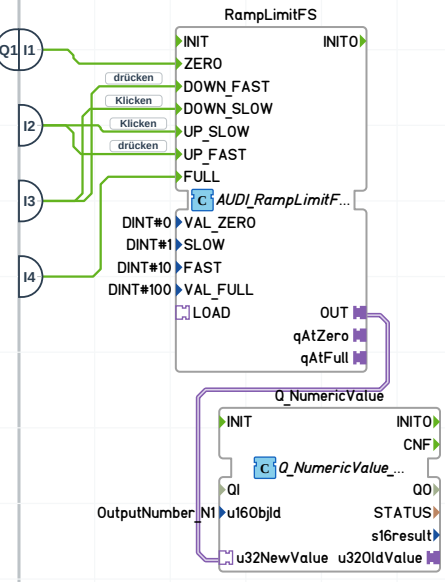
Uebung_009_AUDI

Ticker (AX-Adapter basiert)



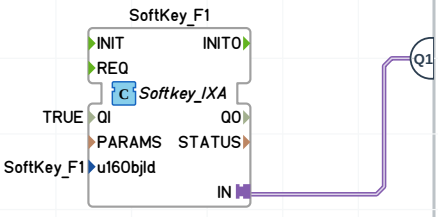
Uebung_009_AX

Ticker



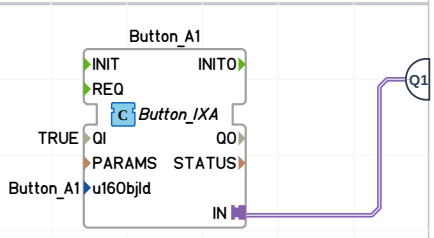
Uebung_009a_AX

RampLimitFS mit AUDI_RampLimitFS Wrapper



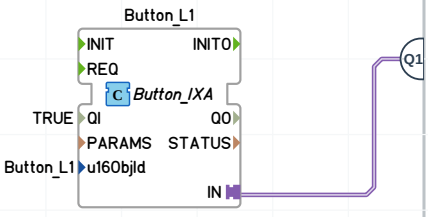
Uebung_010_AX

SoftKey_F1 auf DigitalOutput_Q1



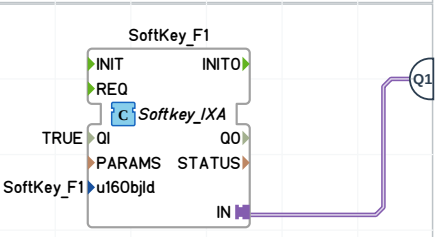
Uebung_010a2_AX

Button_A1 auf DigitalOutput_Q1



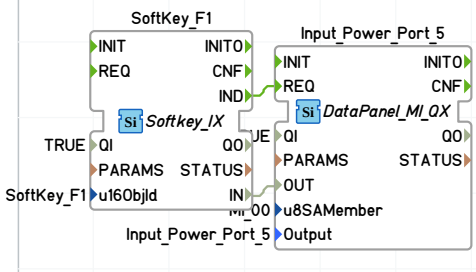
Uebung_010a3_AX

Button_L1 (Latching) auf DigitalOutput_Q1



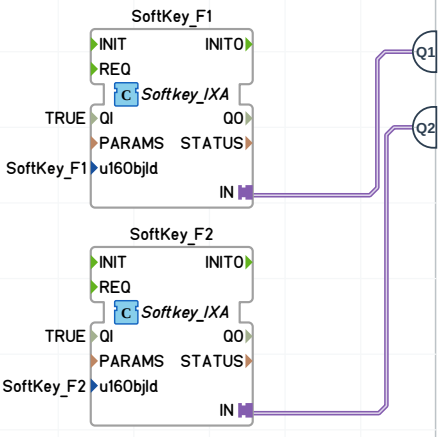
Uebung_010a4_AX

SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



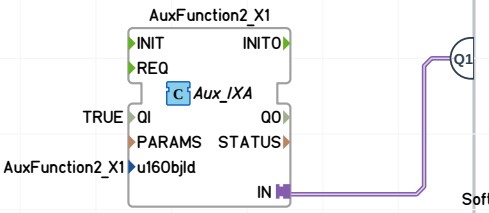
Uebung_010a5_AX

SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



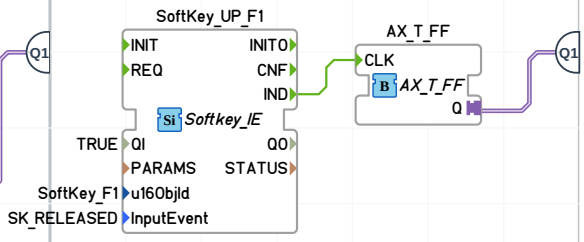
Uebung_010a_AX

SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2



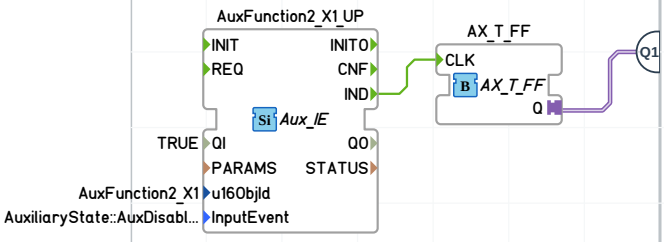
Uebung_010b1_AX

AuxFunction2_X1 auf DigitalOutput_Q1



Uebung_010b2_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED



Uebung_010b3_AX

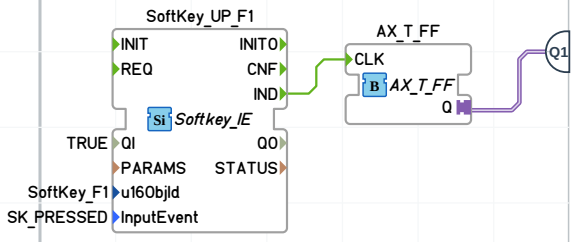
Toggle Flip-Flop mit IE AuxFunction2_X1

Uebung_010b4_AX

SoftKey_F1-8 auf DigitalOutput_Q1-8

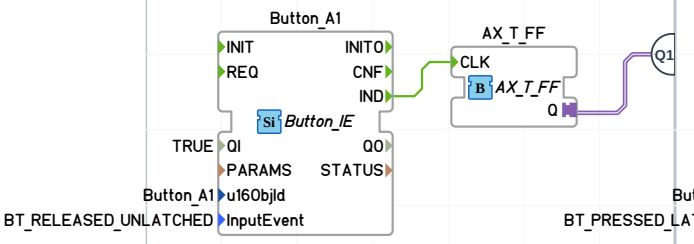
Uebung_010b5_AX

SoftKey_F1-10 auf DigitalOutput_Q1-10



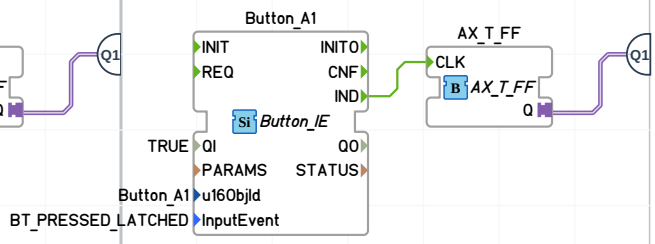
Uebung_010b6_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_PRESSED



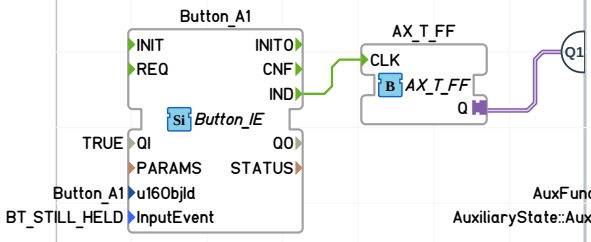
Uebung_010b7_AX

Toggle Flip-Flop mit IE Button_A1 BT_RELEASED_UNLATCHED



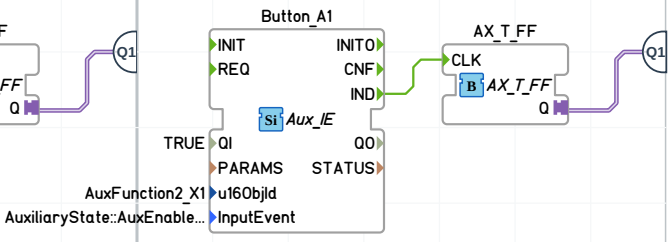
Uebung_010b8_AX

Toggle Flip-Flop mit IE Button_A1 BT_PRESSED_LATCHED



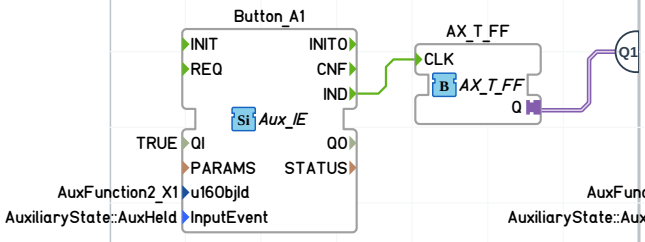
Uebung_010b9_AX

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



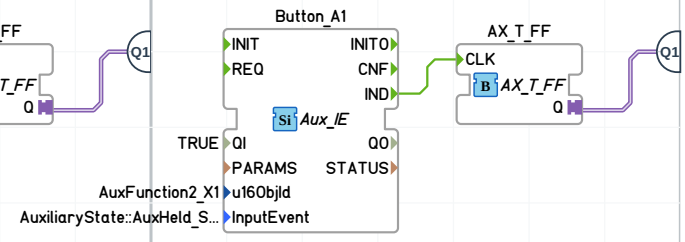
Uebung_010bA2_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxEnabled



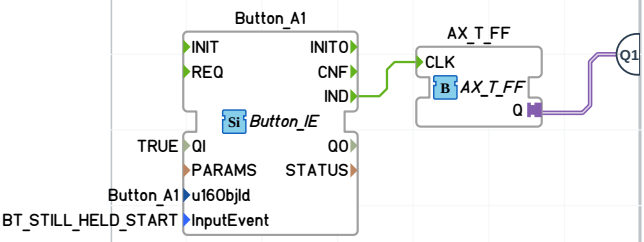
Uebung_010bA3_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld



Uebung_010bA4_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld_START



Uebung_010bA_AX

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



Uebung_010c2_AX

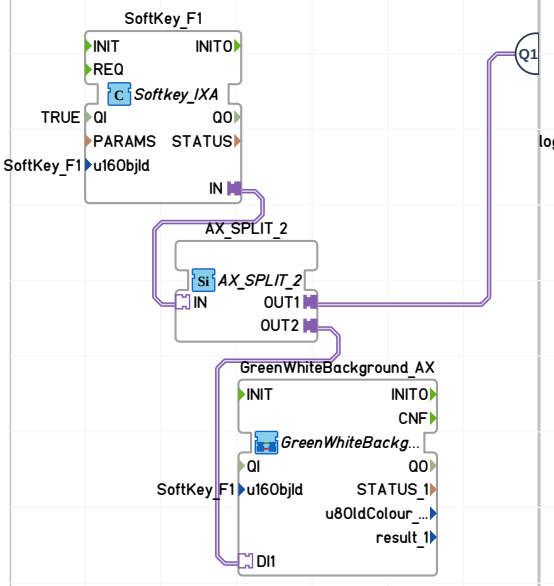
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Subapp

Uebung_010c3_AX

SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Typed Subapp

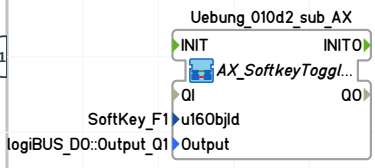
Uebung_010c4_AX

SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2 mit GreenWhiteBackground mit Typed Subapp



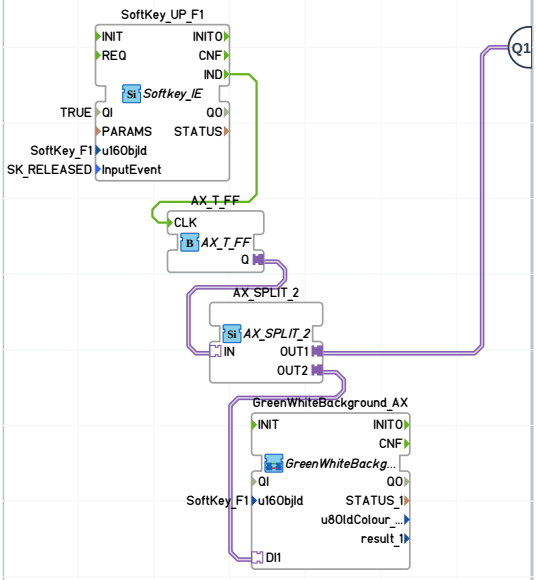
Uebung_010c_AX

SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground



Uebung_010d2_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED mit GreenWhiteBackground mit Subapp

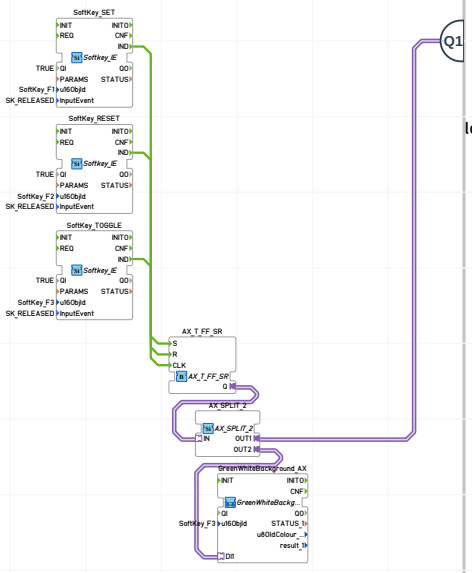


Uebung_010d_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED mit GreenWhiteBackground

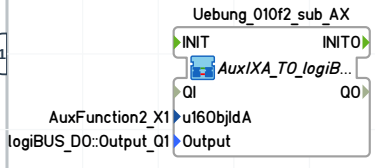
Uebung_010e2_AX

SR und T-Flip-Flop mit 3x SoftKey (Softkey_IE, SK_RELEASED) mit GreenWhiteBackground am Toggle-SoftKey mit Subapp



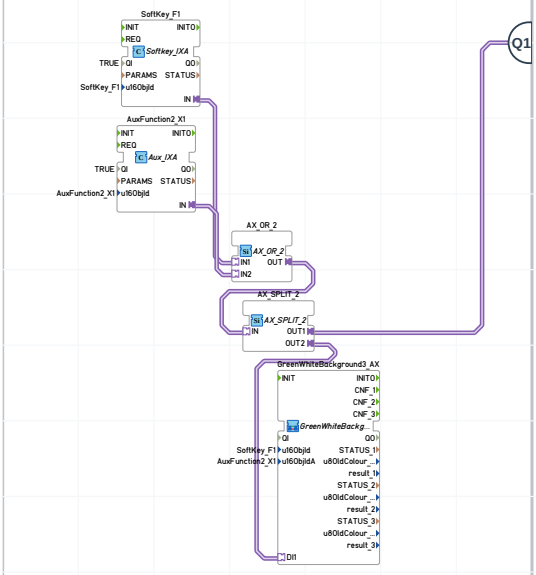
Uebung_010e_AX

SR und T-Flip-Flop mit 3x SoftKey (Softkey_IE, SK_RELEASED) mit GreenWhiteBackground am Toggle-SoftKey



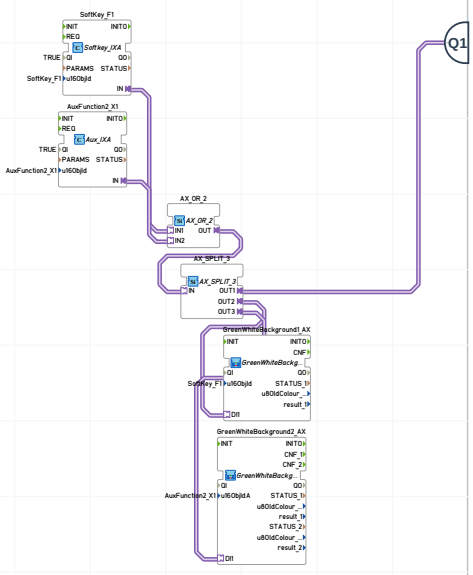
Uebung_010f2_AX

AuxFunction2_X1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Subapp



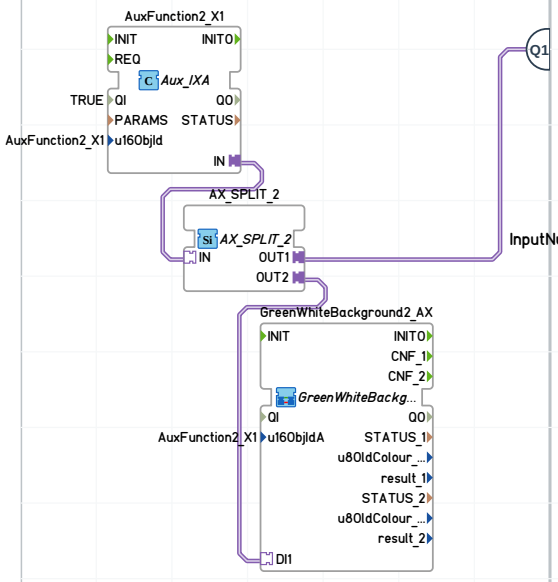
Uebung_010f3_AX

SoftKey_F1 ODER AuxFunction2_X1 auf DigitalOutput_Q1
mit GreenWhiteBackground (SoftKey UND Aux)



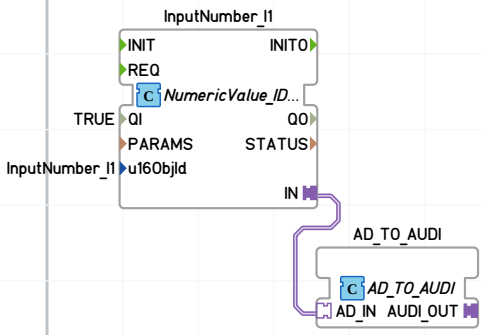
Uebung_010f4_AX

SCHLECHTER STIL (Punktabzug!): SoftKey_F1 ODER
AuxFunction2_X1 auf DigitalOutput_Q1 mit 2x
GreenWhiteBackground statt GreenWhiteBackground3_AX



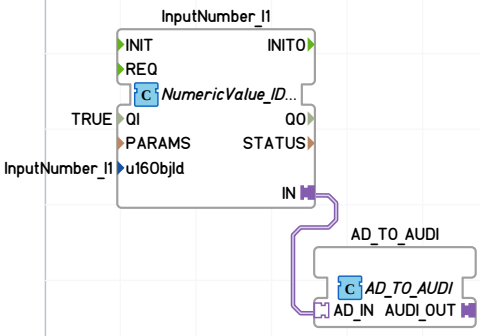
Uebung_010f_AX

AuxFunction2_X1 auf DigitalOutput_Q1 mit
GreenWhiteBackground



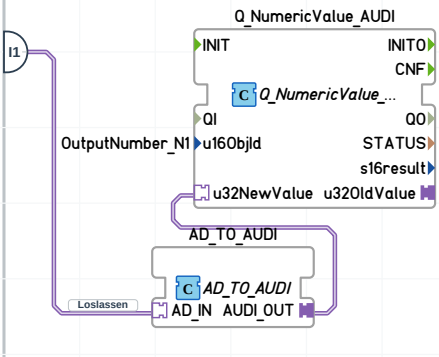
Uebung_011_AUDI

Numeric Value Input



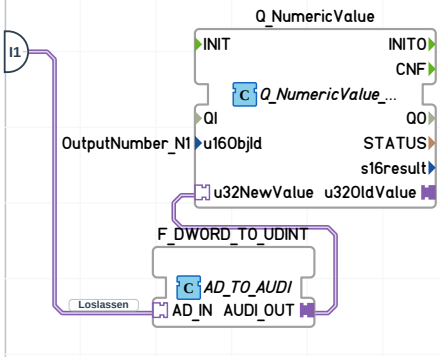
Uebung_011_AX

Numeric Value Input



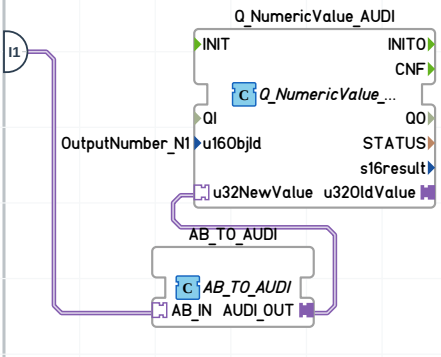
Uebung_011a2_AUDI

Numeric Value Output und BUTTON_PRESS_REPEAT_DONE



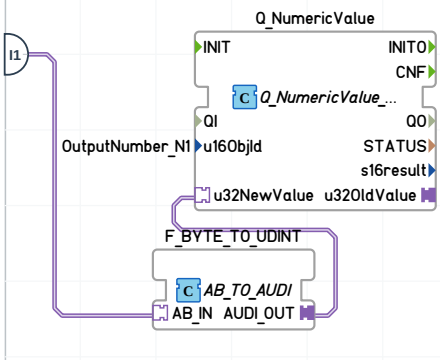
Uebung_011a2_AX

Numeric Value Output und BUTTON_PRESS_REPEAT_DONE



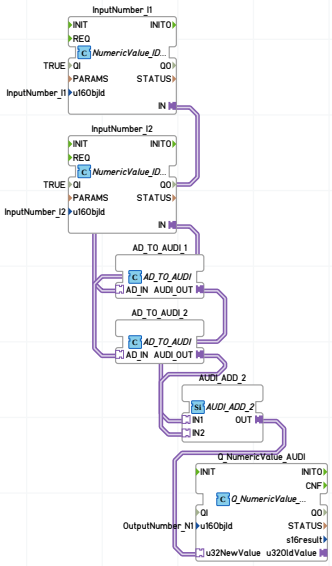
Uebung_011a_AUDI

Numeric Value Output und BUTTON_PRESS_REPEAT_DONE



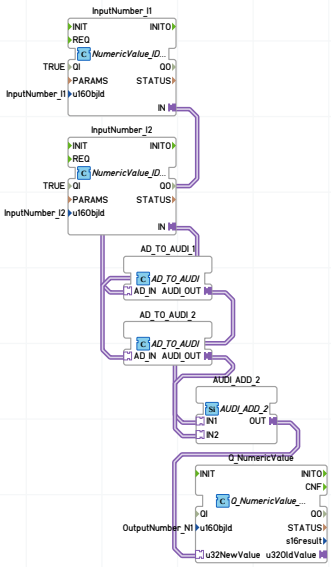
Uebung_011a_AX

Numeric Value Output und BUTTON_PRESS_REPEAT_DONE



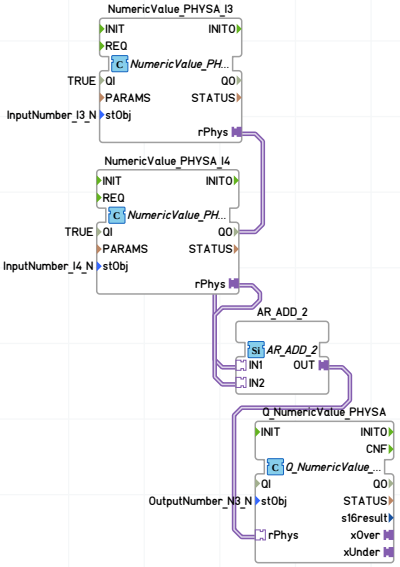
Uebung_011b1_AUDI

Numeric Value Input ADD



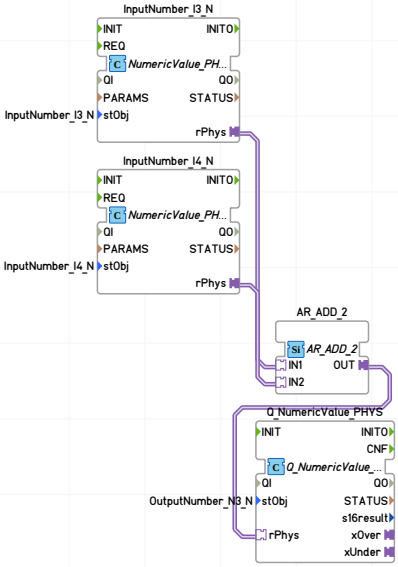
Uebung_011b1_AX

Numeric Value Input ADD



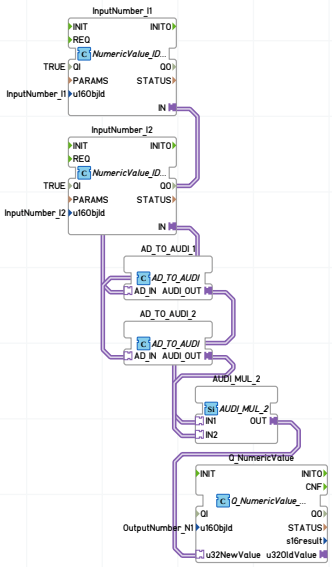
Uebung_011b1_PHYSA

Numeric Value Input ADD with Physical Values



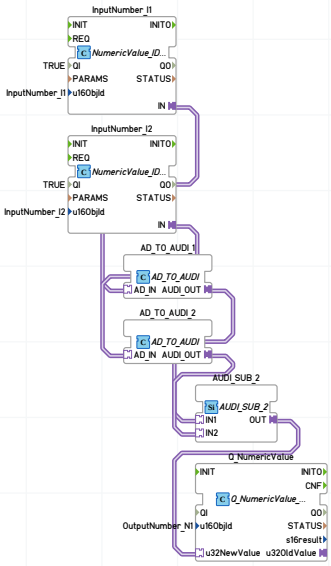
Uebung_011b1_PHYS_AX

Numeric Value Input ADD (PHYS)



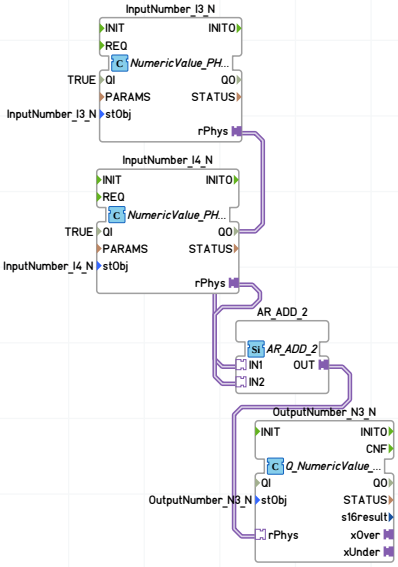
Uebung_011b2_AX

Numeric Value Input MUL



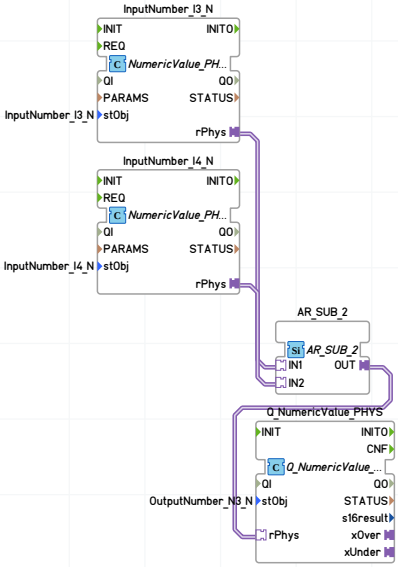
Uebung_011b3_AX

Numeric Value Input SUB



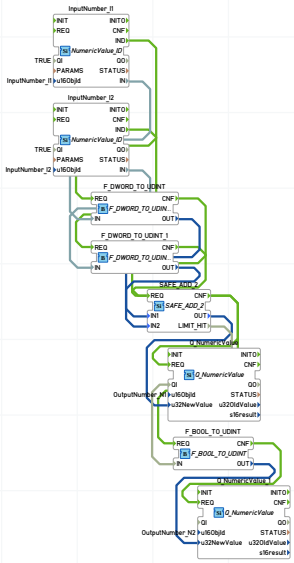
Uebung_011b3_PHYSA

Numeric Value Input ADD with Physical Values



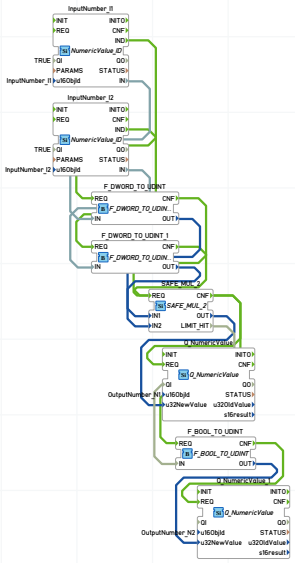
Uebung_011b3_PHYS_AX

Numeric Value Input SUB (PHYS)



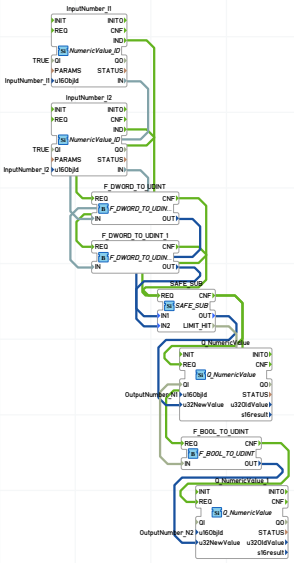
Uebung_011b4_AX

Numeric Value Input SAFE_ADD



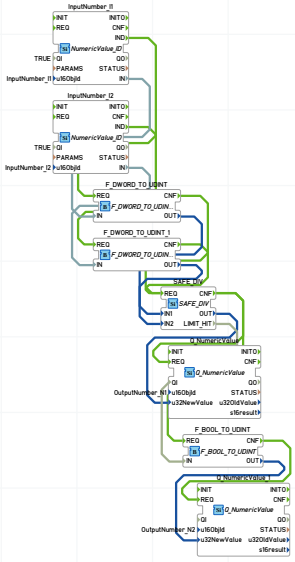
Uebung_011b5_AX

Numeric Value Input SAFE_MUL



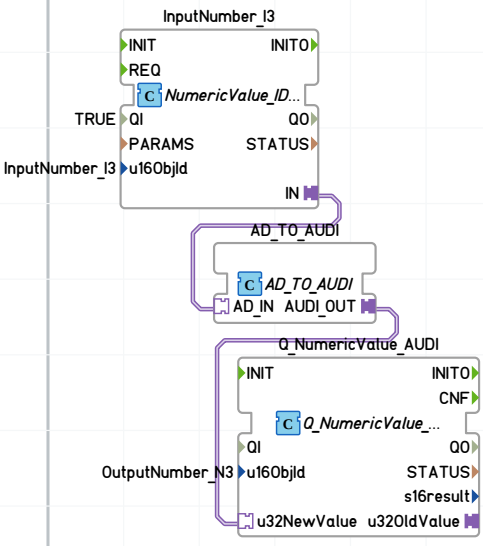
Uebung_011b6_AX

Numeric Value Input SAFE_SUB



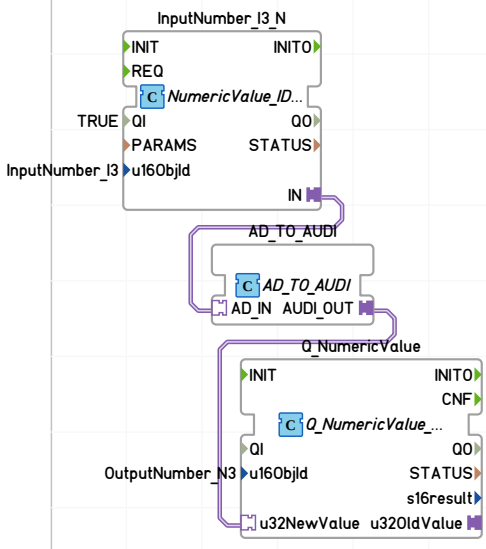
Uebung_011b7_AX

Numeric Value Input SAFE_DIV



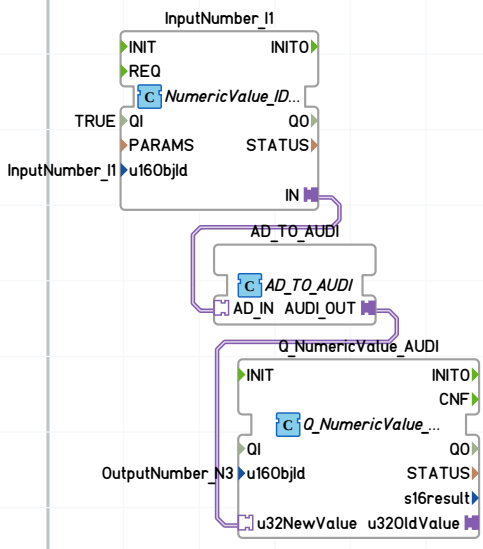
Uebung_011c_AUDI

Numeric Value Input I3 Durchschleifen auf N3



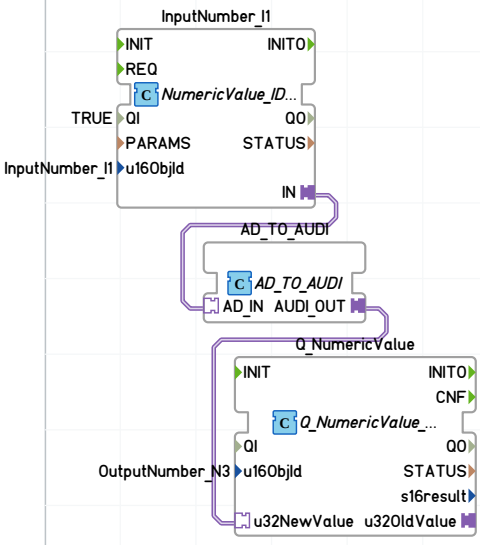
Uebung_011c_AX

Numeric Value Input I3 Durchschleifen auf N3



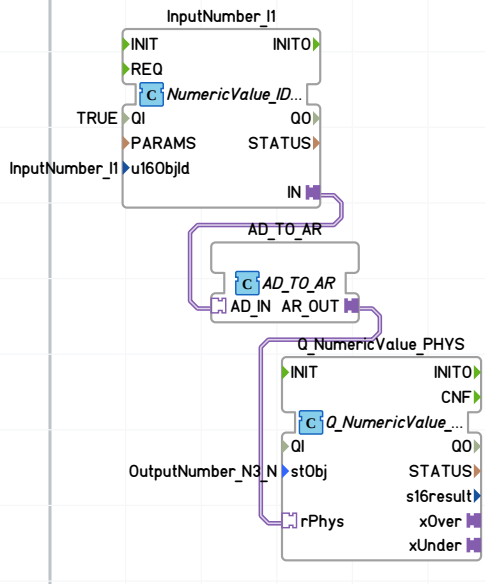
Uebung_011d_AUDI

Numeric Value Input I1 Durchschleifen auf N3
(Offset/Scale Effekt)



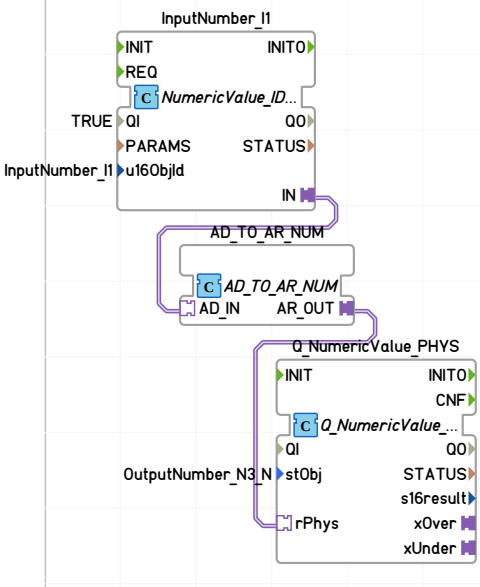
Uebung_011d_AX

Numeric Value Input I1 Durchschleifen auf N3 (Offset/Scale Effekt)



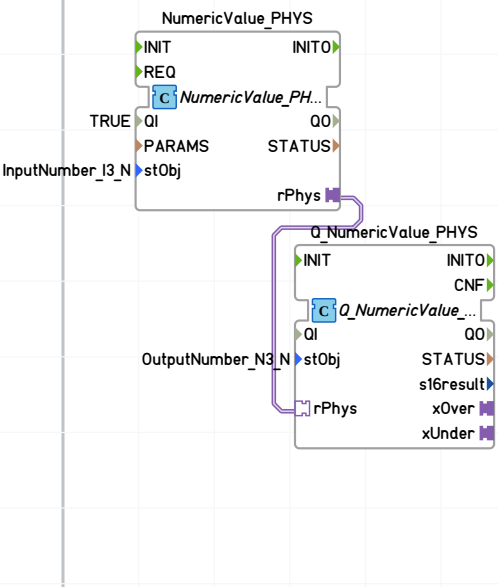
Uebung_011e_MIXA

Numeric Value Input I1 Durchschleifen auf N3 (Software Scale via NumericObjectPool_S) falsch gemischt!



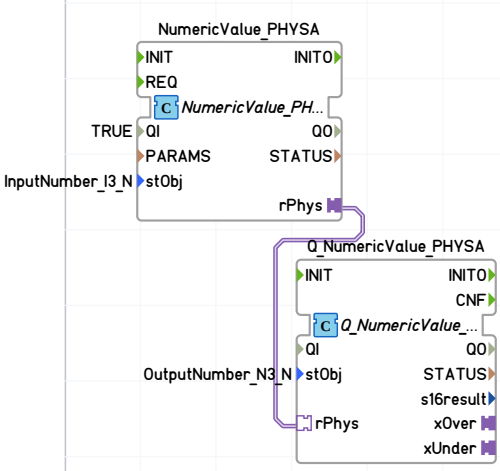
Uebung_011e_MIX_AX

Numeric Value Input I1 Durchschleifen auf N3 (Software Scale via NumericObjectPool_S) falsch gemischt!



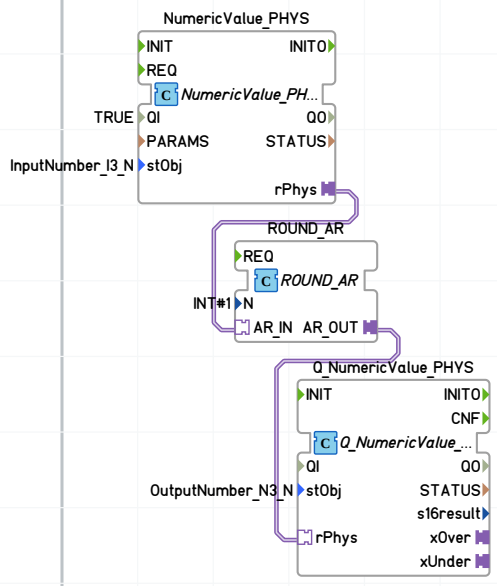
Uebung_011f_AX

Numeric Value Input I3 Durchschleifen auf N3 (Input und Output PHYS via NumericObjectPool_S)



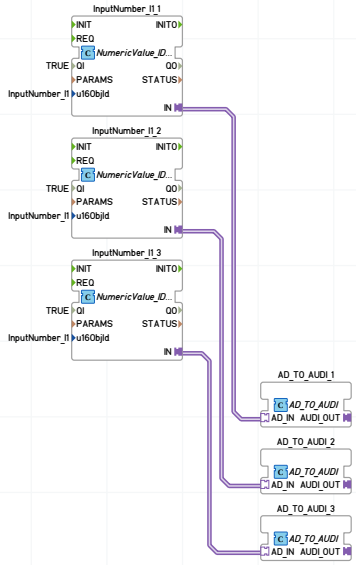
Uebung_011f_PHYSA

Numeric Value Input I1 Durchschleifen auf N3 mit PHYSA (Physikalische Werte)



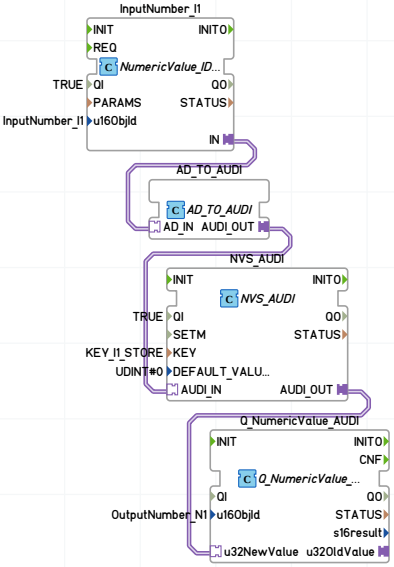
Uebung_011f_ROUND_AR_AX

Wie Uebung_011f_AX, aber mit OSCAT ROUND_AR dazwischen (rundet auf N Nachkommastellen)



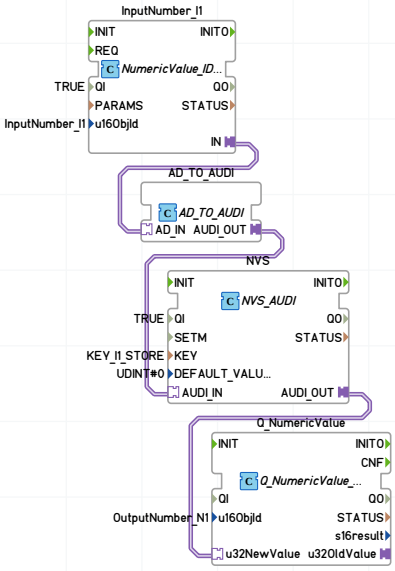
Uebung_011g1_AX

Numeric Value Input -- multiple I00bservers



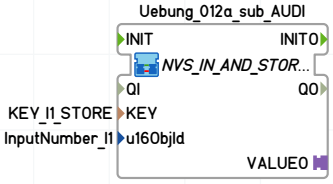
Uebung_012_AUDI

Numeric Value Input und Speichern



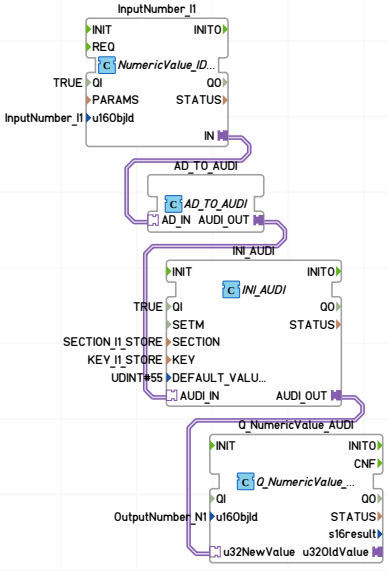
Uebung_012_AX

Numeric Value Input und Speichern NVS



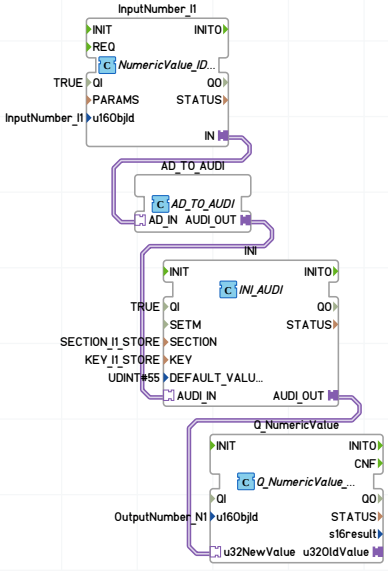
Uebung_012a_AX

Numeric Value Input und Speichern (Adapter Version)



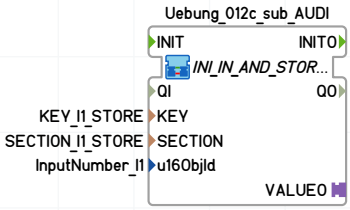
Uebung_012b_AUDI

Numeric Value Input und Speichern



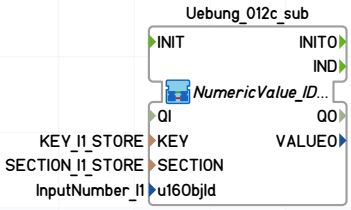
Uebung_012b_AX

Numeric Value Input und Speichern INI



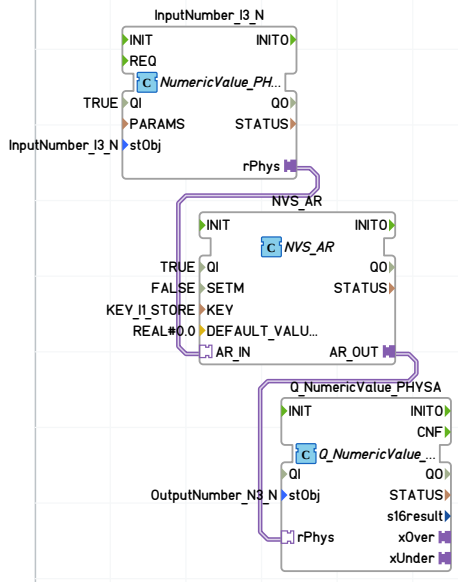
Uebung_012c_AUDI

Numeric Value Input und Speichern INI mit Subapp (Adapter Version)



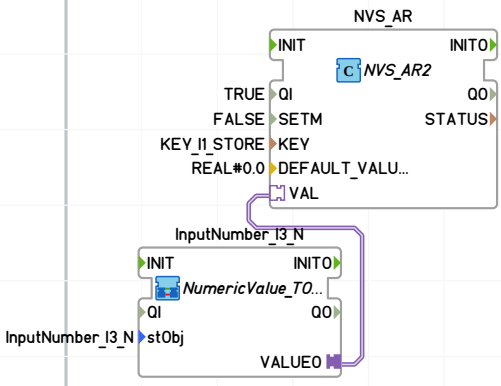
Uebung_012c_AX

Numeric Value Input und Speichern INI mit Subapp



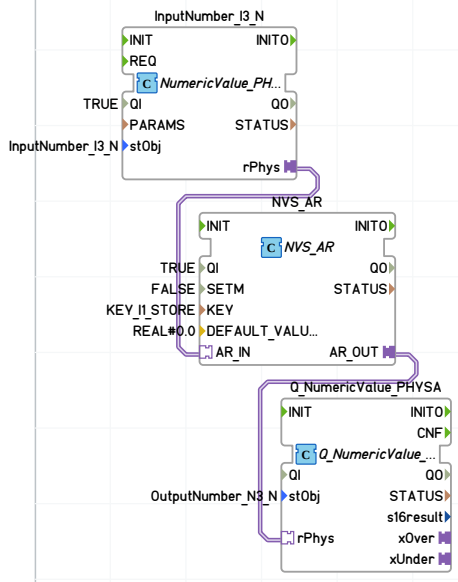
Uebung_012d_AR

Numeric Value Input PHYS und Speichern NVS (AR Adapter, plain)



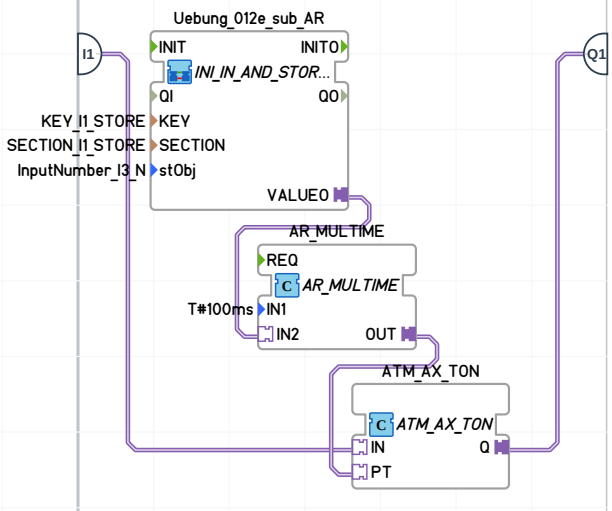
Uebung_012d_AR2

Numeric Value Input PHYS und Speichern NVS via AR2 (bidirektional) – NumericValue_TO_AR2 direkt an ein NVS_AR2 angeschlossen



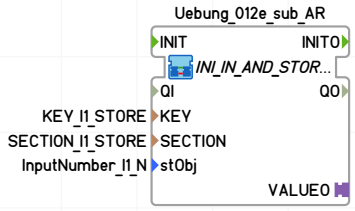
Uebung_012d_AX

Numeric Value Input PHYS und Speichern NVS



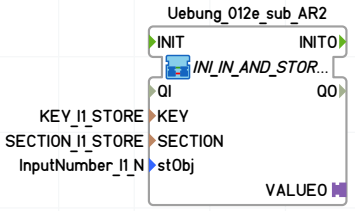
Uebung_012e2_AR

Numeric Value Input PHYS und Speichern INI mit Subapp (AR Adapter)



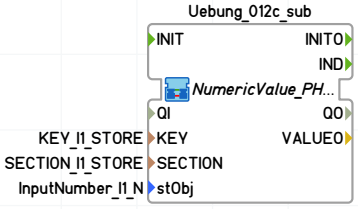
Uebung_012e_AR

Numeric Value Input PHYS und Speichern INI mit Subapp (AR Adapter)



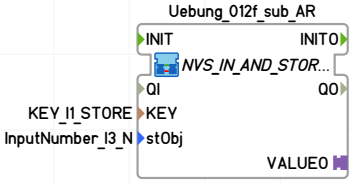
Uebung_012e_AR2

Numeric Value Input PHYS und Speichern INI mit Subapp (AR2 Adapter, echter in sich geschlossener Baustein)



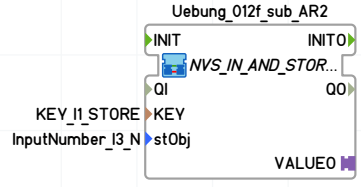
Uebung_012e_AX

Numeric Value Input PHYS und Speichern INI mit Subapp



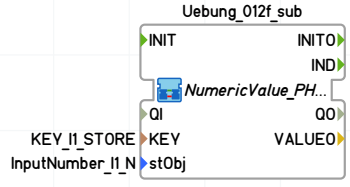
Uebung_012f_AR

Numeric Value Input PHYS und Speichern NVS mit Subapp (AR Adapter)



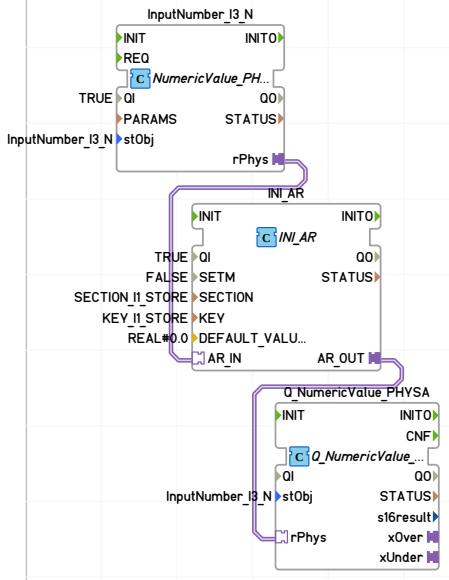
Uebung_012f_AR2

Numeric Value Input PHYS und Speichern NVS mit Subapp (AR2 Adapter, echter in sich geschlossener Baustein)



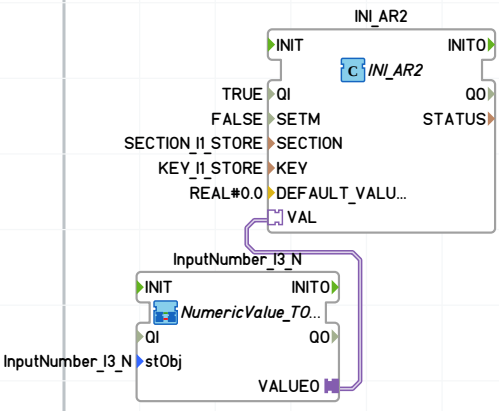
Uebung_012f_AX

Numeric Value Input PHYS und Speichern NVS mit Subapp



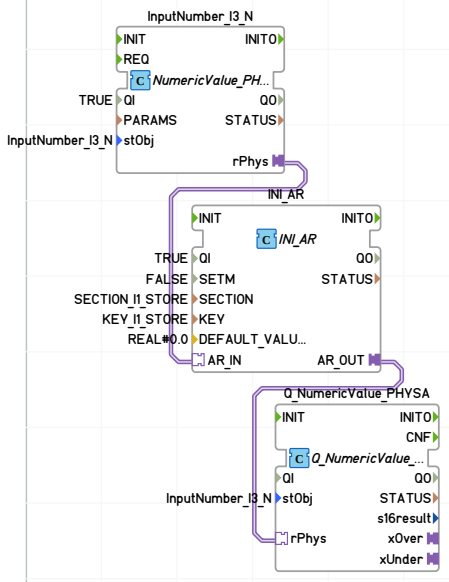
Uebung_012g_AR

Numeric Value Input PHYS und Speichern INI (AR Adapter, plain)



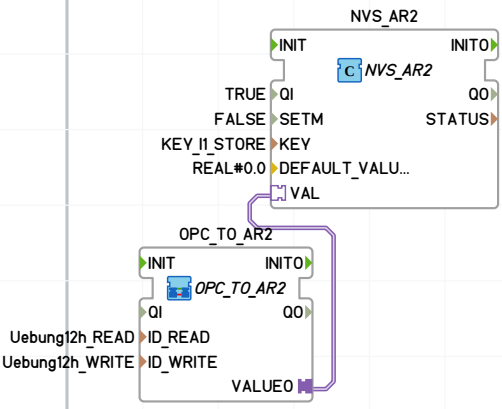
Uebung_012g_AR2

Numeric Value Input PHYS und Speichern INI via AR2 (bidirektional) – NumericValue_TO_AR2 direkt an ein INI_AR2 angeschlossen



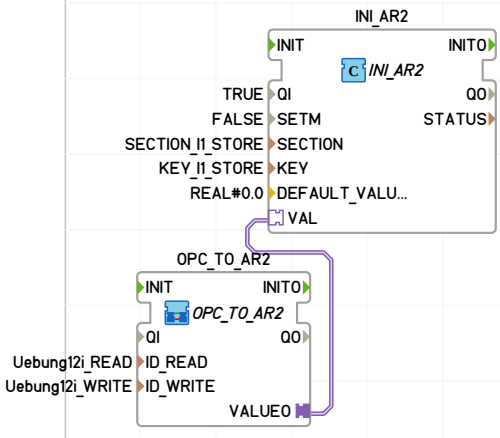
Uebung_012g_AX

Numeric Value Input PHYS und Speichern INI



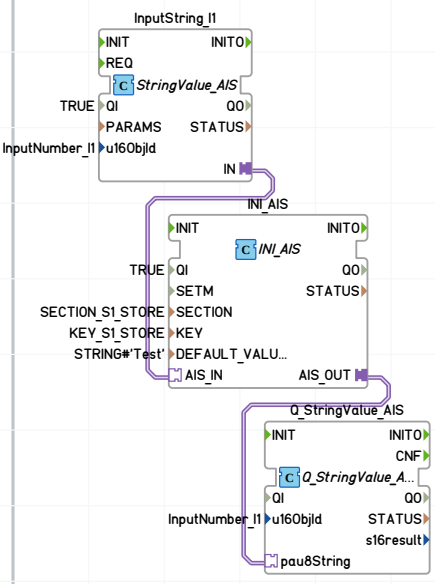
Uebung_012h_AR2

Nur OPC-UA (kein VT-Feld) und Speichern NVS via AR2 – OPC_TO_AR2 direkt an ein NVS_AR2 angeschlossen



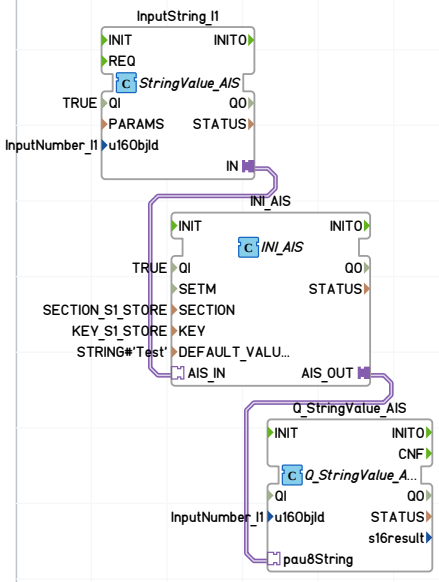
Uebung_012i_AR2

Nur OPC-UA (kein VT-Feld) und Speichern INI via AR2 – OPC_TO_AR2 direkt an ein INI_AR2 angeschlossen



Uebung_012j_AIS

String Input und Speichern



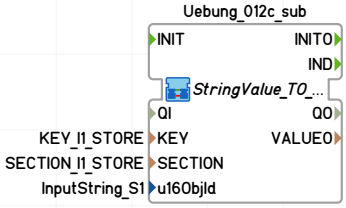
Uebung_012j_AX

String Input und Speichern INI



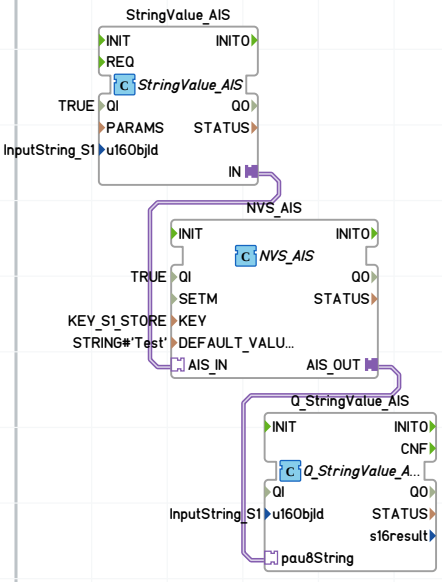
Uebung_012k_AIS

String Input und Speichern INI mit Subapp (AIS Adapter)



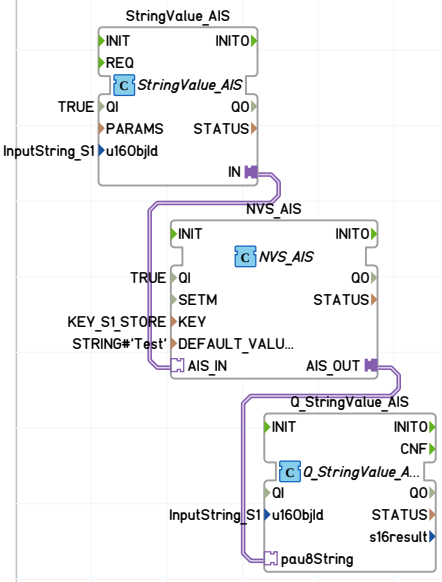
Uebung_012k_AX

String Input und Speichern INI mit Subapp



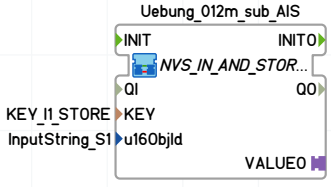
Uebung_012l_AIS

String Input und Speichern NVS (AIS Adapter, plain)



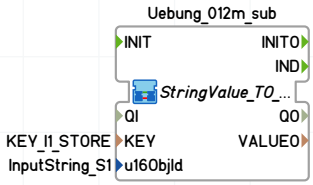
Uebung_012l_AX

String Input und Speichern NVS



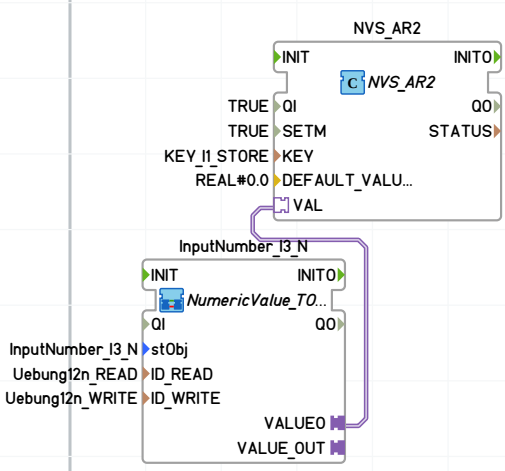
Uebung_012m_AIS

String Input und Speichern NVS mit Subapp (AIS Adapter)



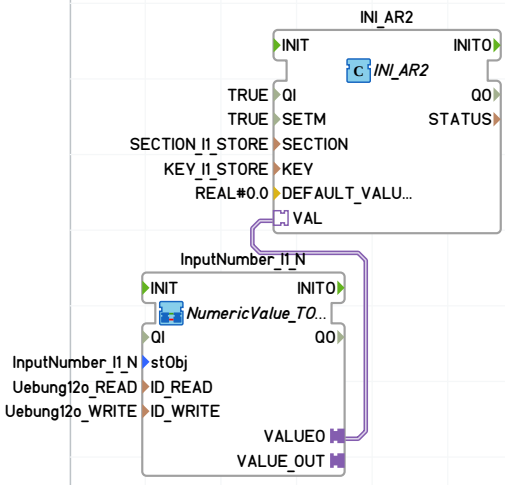
Uebung_012m_AX

String Input und Speichern NVS mit Subapp



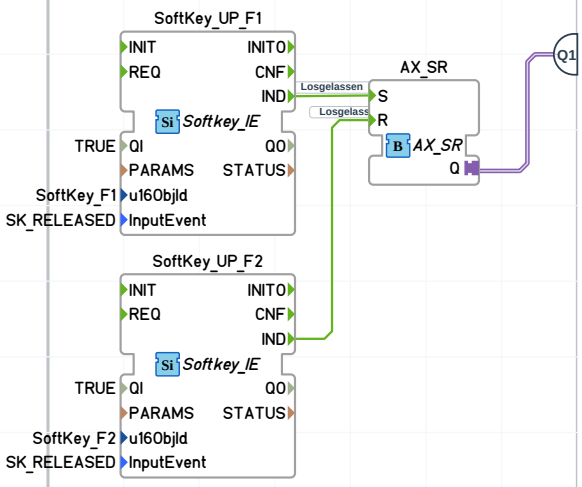
Uebung_012n_AR2

VT-Feld UND OPC-UA (letzter Schreiber gewinnt) und Speichern NVS via AR2 – NumericValue_TO_AR2_OPC direkt an ein NVS_AR2 angeschlossen



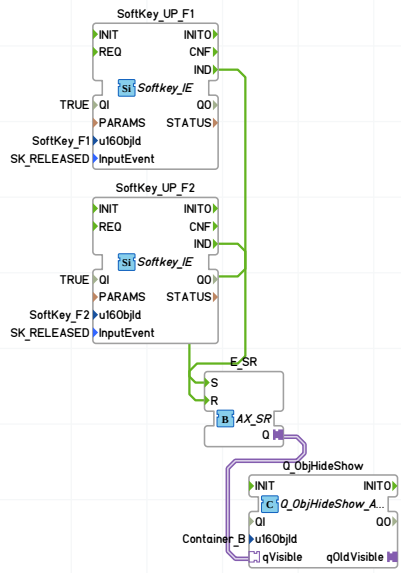
Uebung_012o_AR2

VT-Feld UND OPC-UA (letzter Schreiber gewinnt) und Speichern INI via AR2 – NumericValue_TO_AR2_OPC direkt an ein INI_AR2 angeschlossen



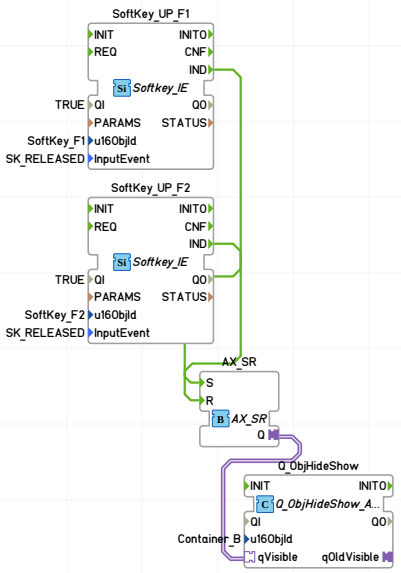
Uebung_013_AX

SR-Flip-Flop (Softkey)



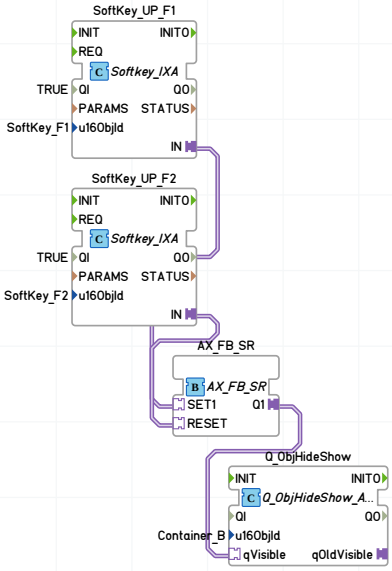
Uebung_014_AX

Container (visible/invisible)



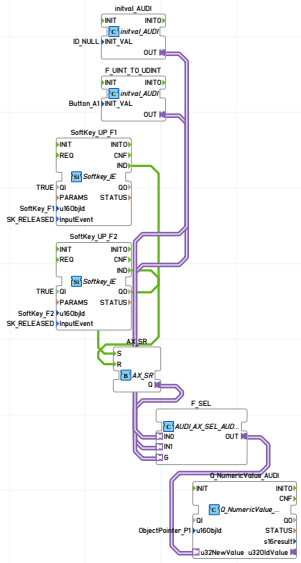
Uebung_014_AXa

Container (visible/invisible)



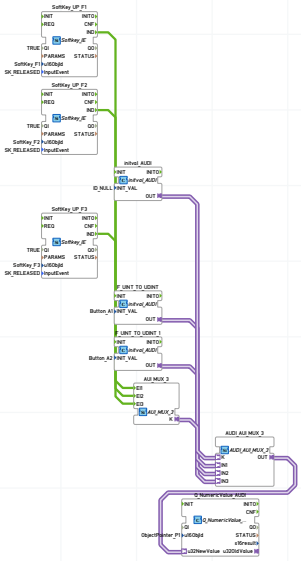
Uebung_014_AXb

Container (visible/invisible)



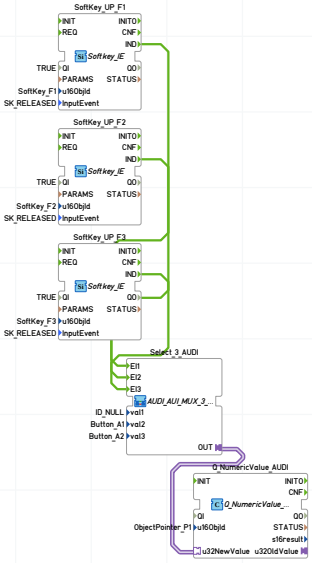
Uebung_015_AX

Object Pointer umschalten



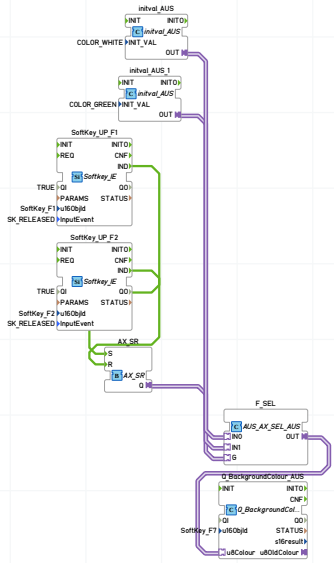
Uebung_015a_AX

Object Pointer umschalten -- 3-fach



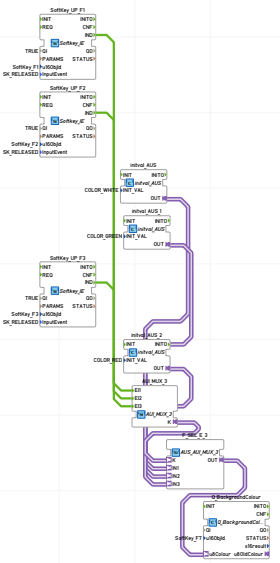
Uebung_015b_AX

Object Pointer umschalten -- 3-fach



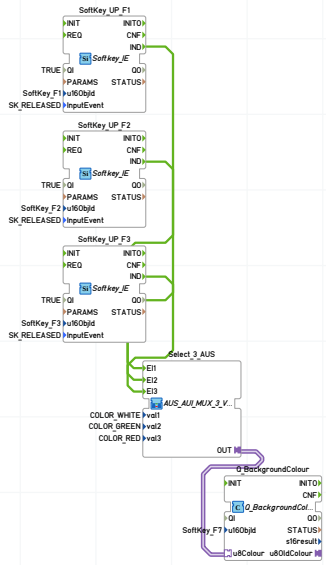
Uebung_016_AX

Background Colour umschalten



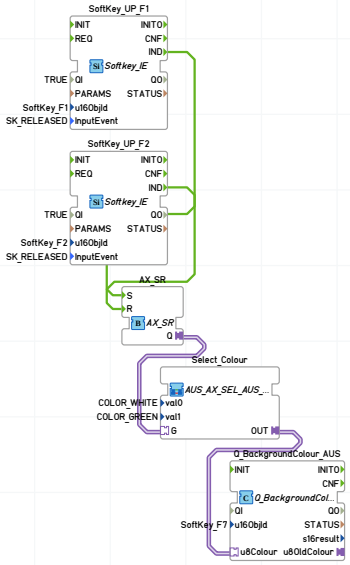
Uebung_016a_AX

Background Colour umschalten -- 3-fach



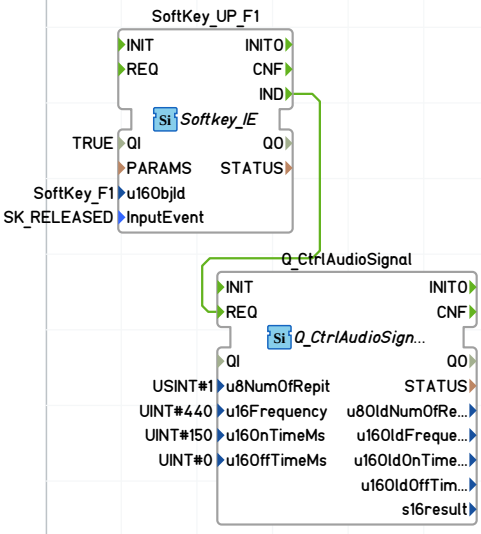
Uebung_016b_AX

Background Colour umschalten -- 3-fach mit SubApp



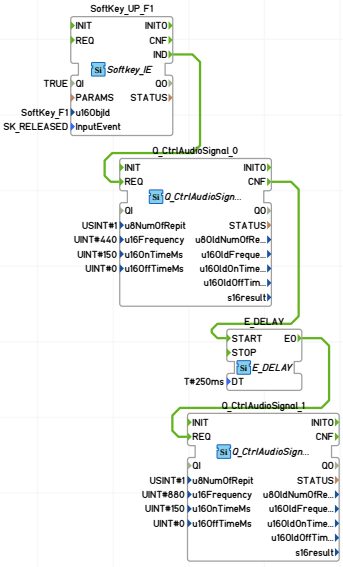
Uebung_016c_AX

Background Colour umschalten -- mit SubApp für SEL und INITVAL



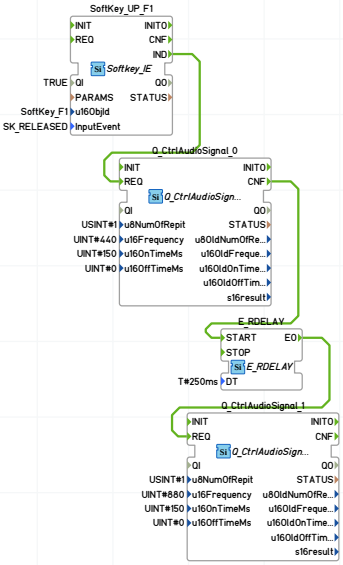
Uebung_017_AX

Control Audio Signal



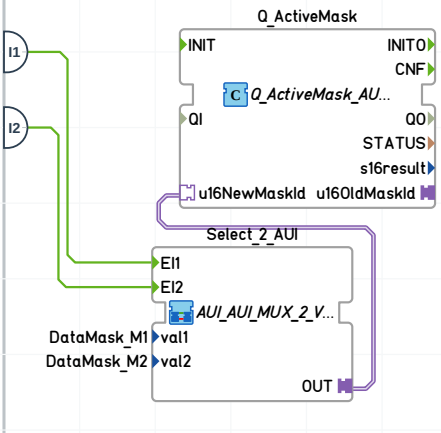
Uebung_018_AX

Control Audio Signal und Delay



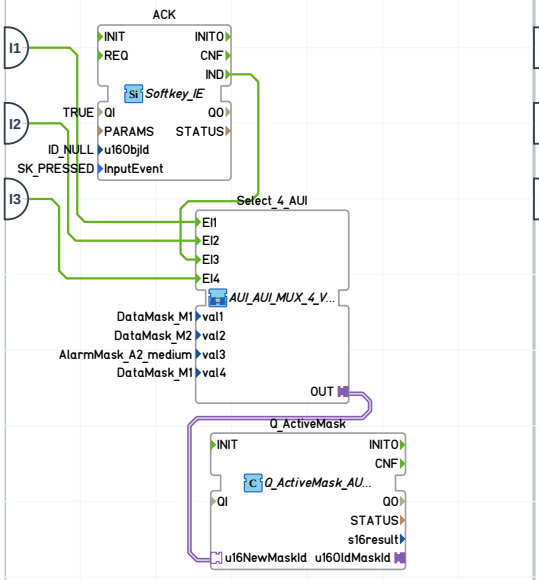
Uebung_018a_AX

Control Audio Signal und Delay



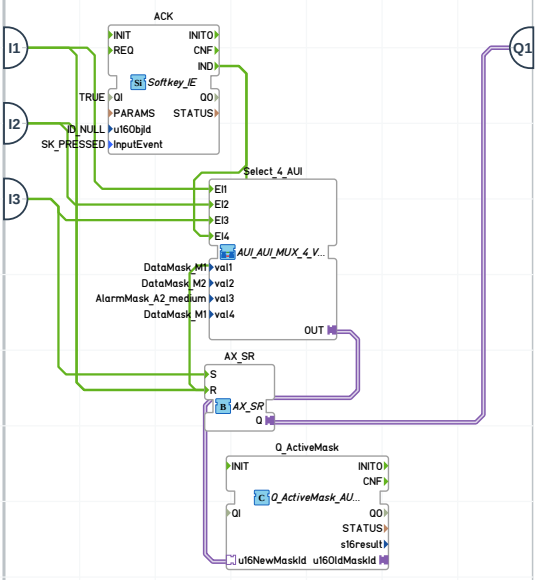
Uebung_019_AX

Umschalten einer Maske



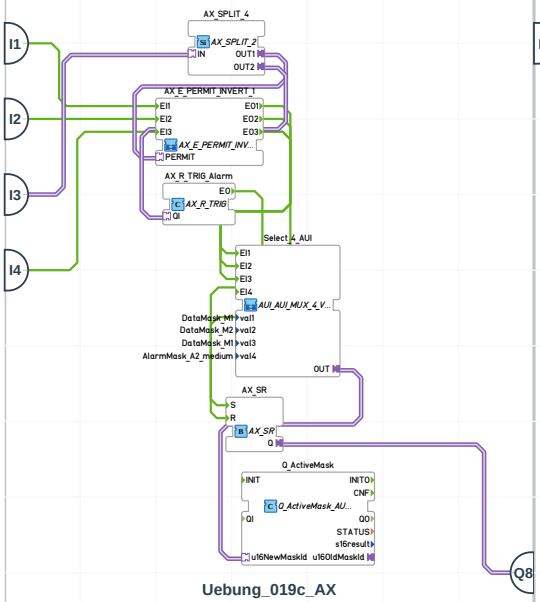
Uebung_019a_AX

Umschalten einer Maske



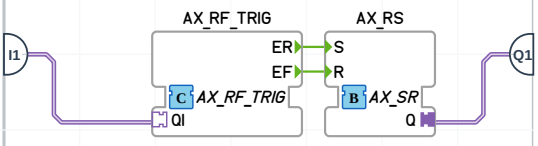
Uebung_019b_AX

Umschalten einer Maske, mit Plug and Socket



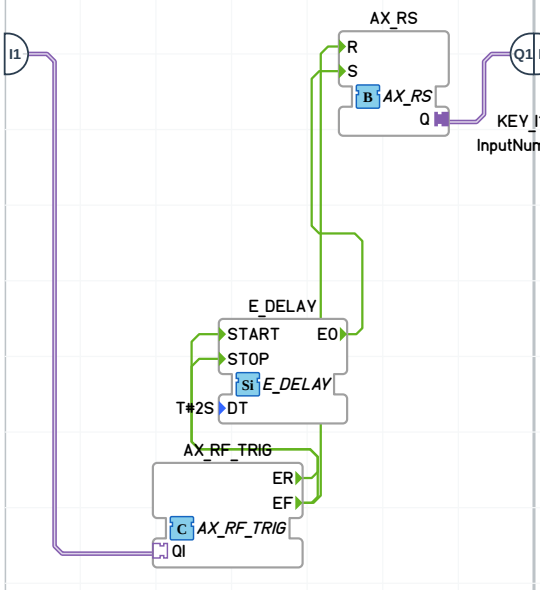
Uebung_019c_AX

Umschalten einer Maske, mit Plug and Socket



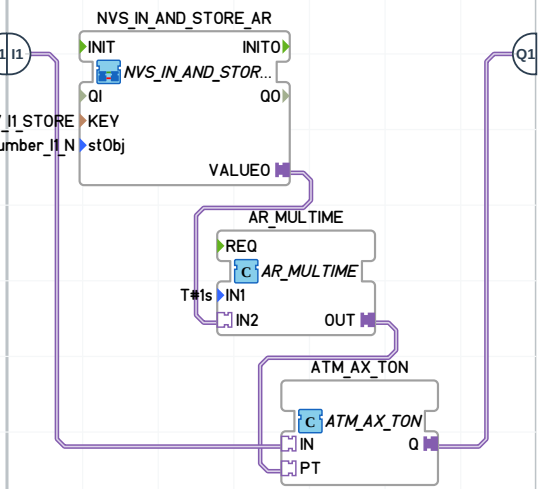
Uebung_020a_AX

DigitalInput_I1 auf DigitalOutput_Q1 via AX_SR (Puffer)



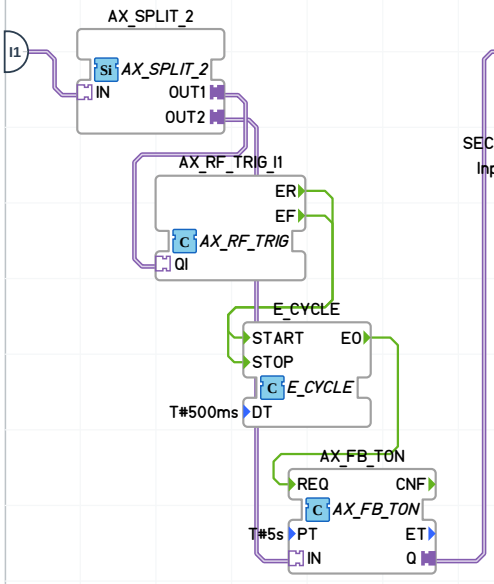
Uebung_020b_AX

DigitalInput_I1 auf DigitalOutput_Q1; TON Einschaltverzögert; aufgelöst



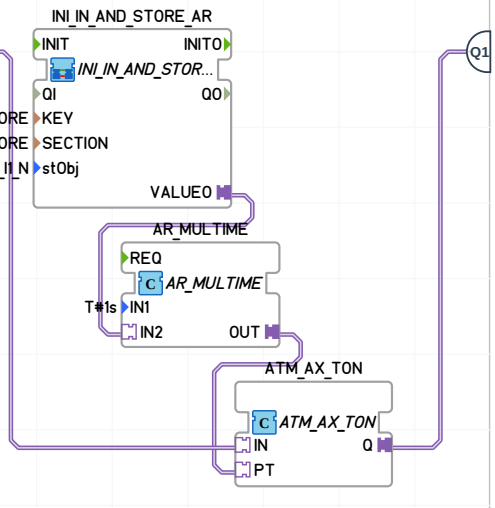
Uebung_020c2_AX

DigitalInput_I1 auf DigitalOutput_Q1; E_TON Einschaltverzögert; Zeit eingeben und speichern, NVS



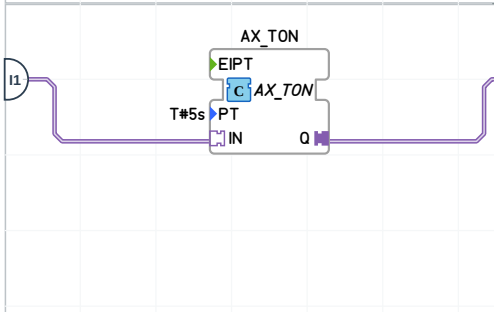
Uebung_020c3_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TON; Einschaltverzögert



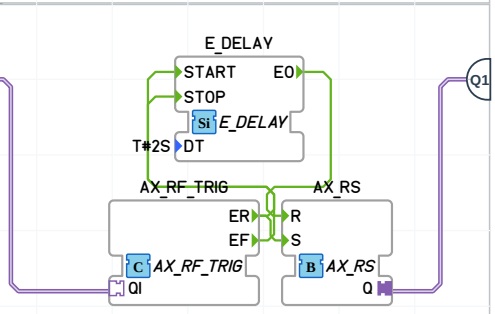
Uebung_020c4_AX

DigitalInput_I1 auf DigitalOutput_Q1; E_TON Einschaltverzögert; Zeit eingeben und speichern, INI



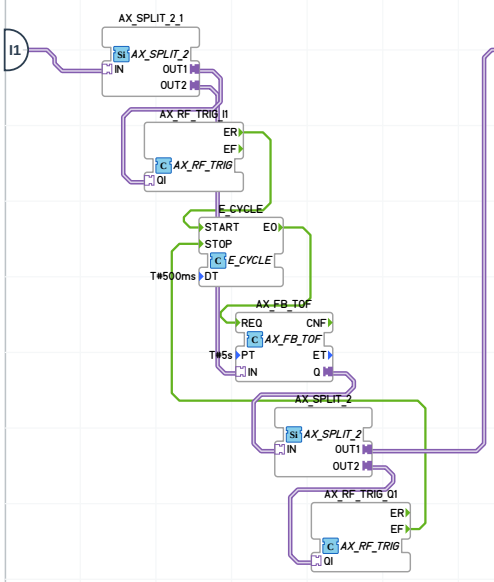
Uebung_020c_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TON; Einschaltverzögert



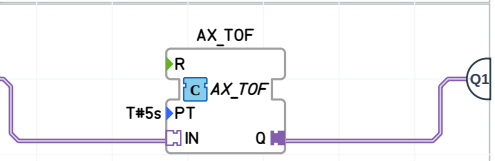
Uebung_020d_AX

DigitalInput_I1 auf DigitalOutput_Q1; TOF Ausschaltverzögert; aufgelöst



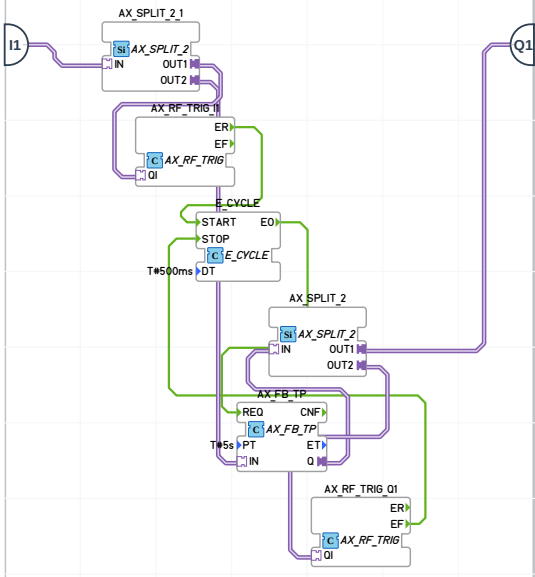
Uebung_020e2_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TOF; Ausschaltverzögert



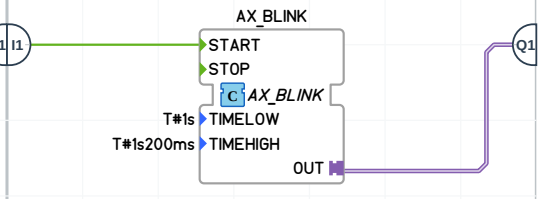
Uebung_020e_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TOF; Ausschaltverzögert



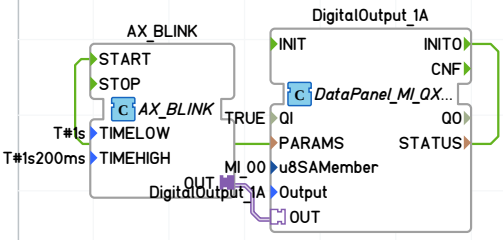
Uebung_020f2_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TP; Impulsformend



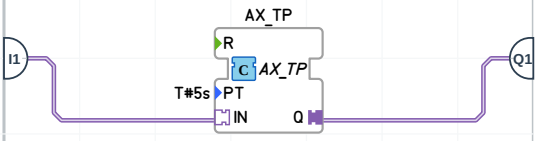
Uebung_020f3_AX

DigitalInput_I1 auf DigitalOutput_Q1; BLINKER



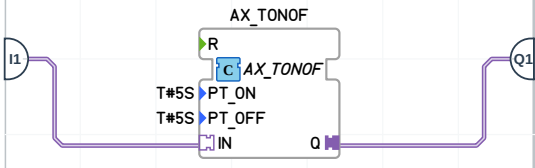
Uebung_020f4_AX

DigitalInput_I1 auf DataPanel_1A; BLINKER



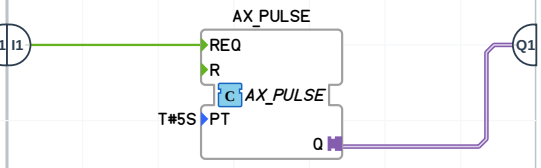
Uebung_020f_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TP; Impulsformend



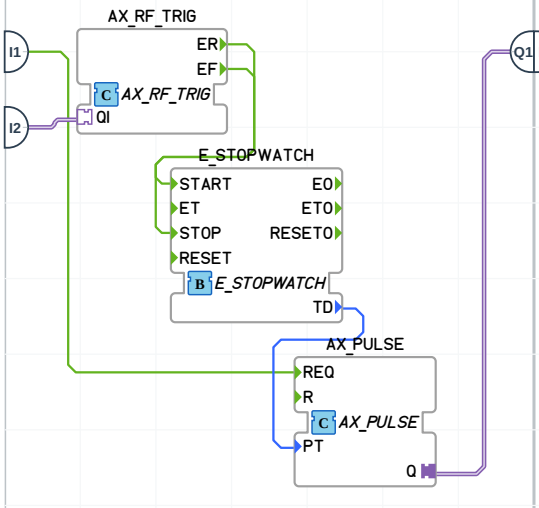
Uebung_020g_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TONOF; Ein-/Ausschaltverzögert



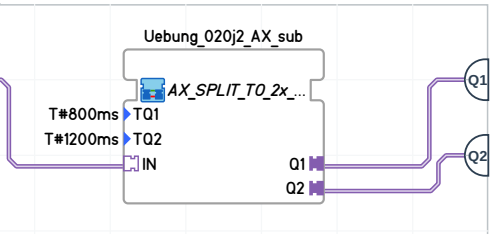
Uebung_020h_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_PULSE; Impulsformend



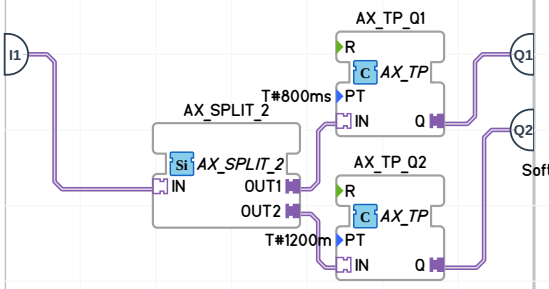
Uebung_020i_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_PULSE; Teach-In



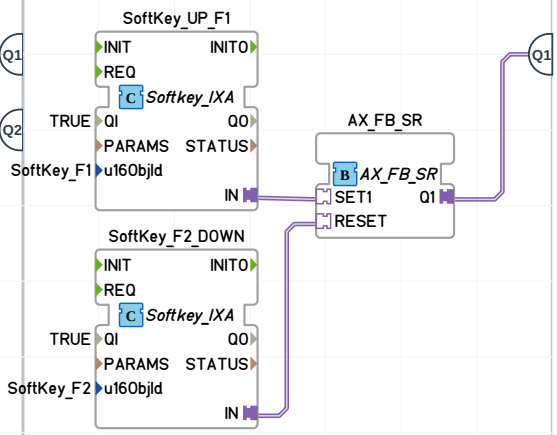
Uebung_020j2_AX

DigitalInput_I1 auf DigitalOutput_Q1/2; 2xAX_TP; Impulsformend mit SUB



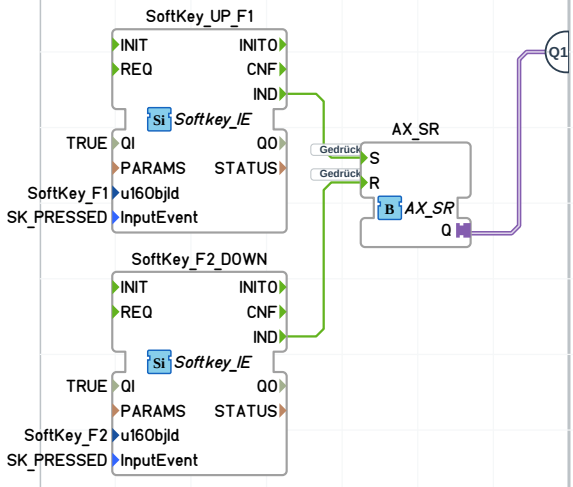
Uebung_020j_AX

DigitalInput_I1 auf DigitalOutput_Q1/2; 2xAX_TP; Impulsformend



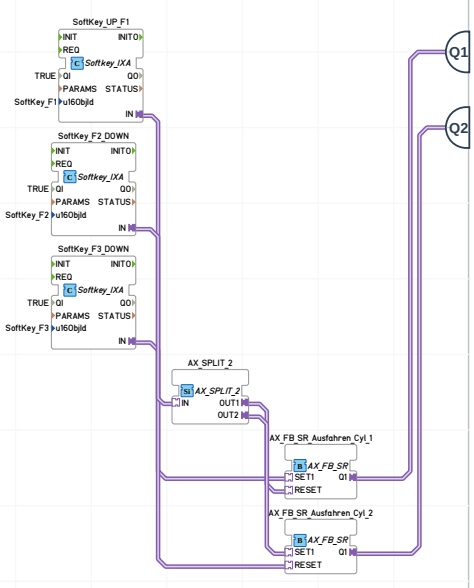
Uebung_021_AX

Spiegelabfolge (1) – AX Variante



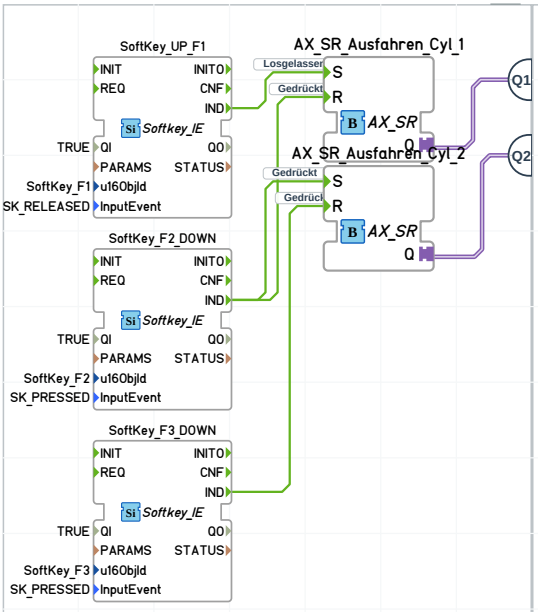
Uebung_021b_AX2

Spiegelabfolge (1) – AX Variante



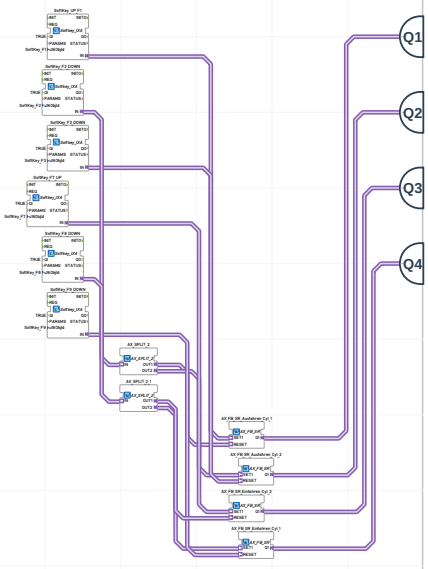
Uebung_022_AX

Spiegelabfolge (2) – AX Variante



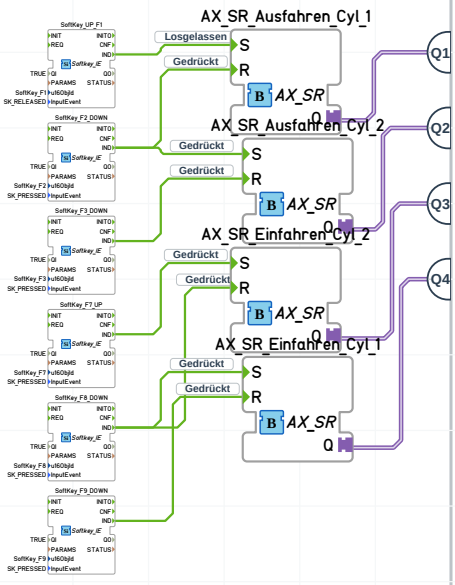
Uebung_022b_AX2

Spiegelabfolge (2) – AX Variante



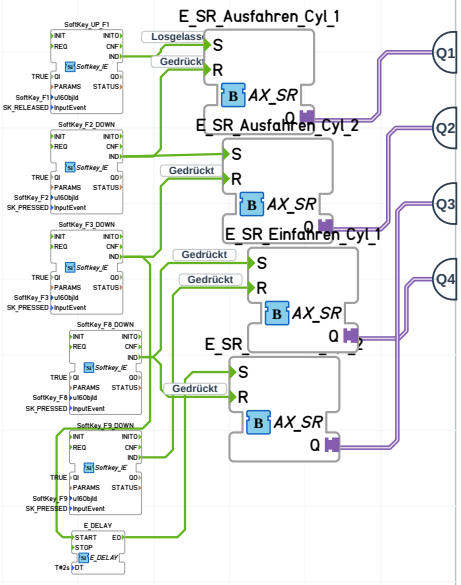
Uebung_023_AX

Spiegelabfolge (3) – AX Variante



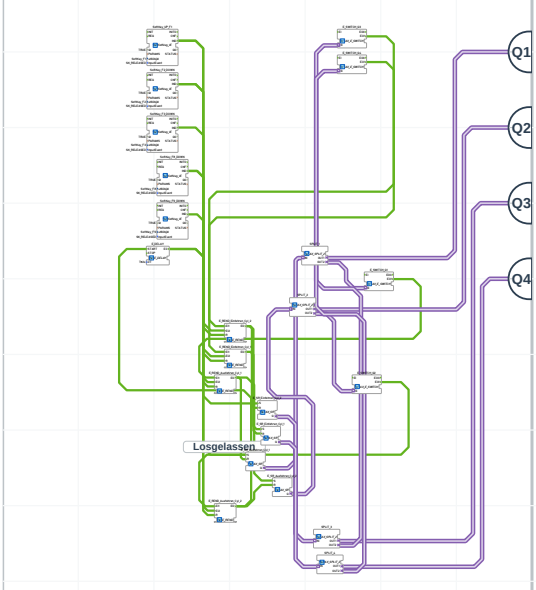
Uebung_023b_AX2

Spiegelabfolge (3) – AX Variante



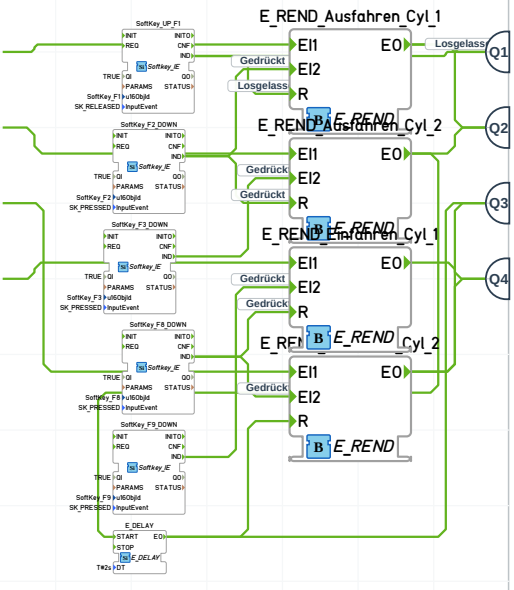
Uebung_024_AX

Spiegelabfolge (4)



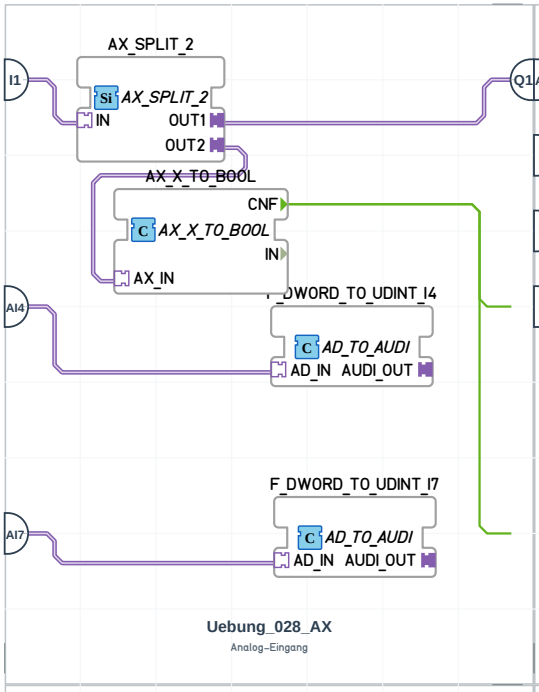
Uebung_025_AX

Spiegelabfolge (5)



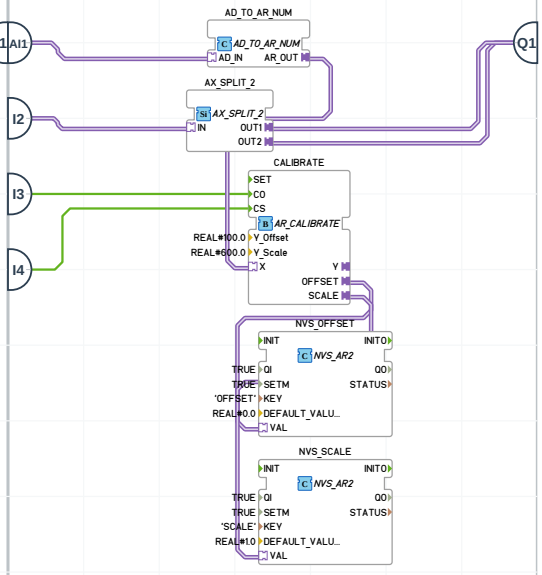
Uebung_026_AX

Spiegelabfolge (6)



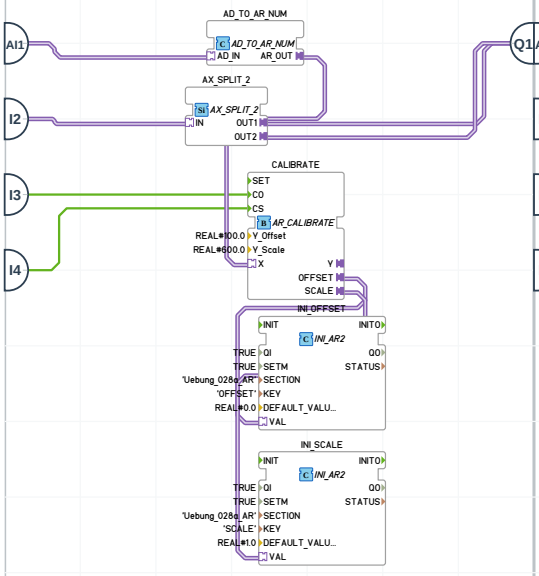
Uebung_028_AX

Analog-Eingang



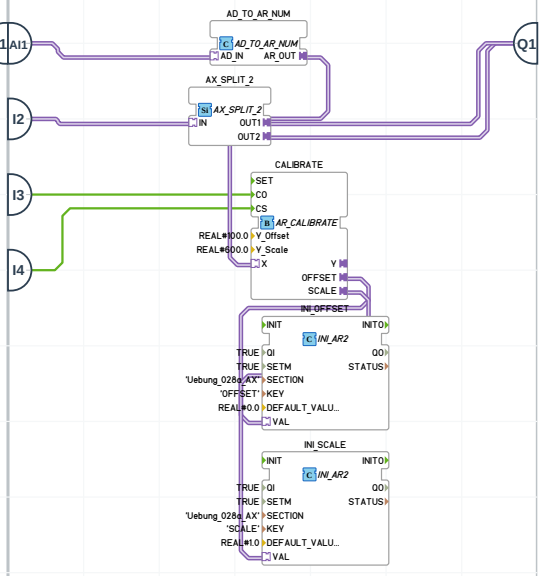
Uebung_028a2_AR

Analog-Eingang Kalibrierung mit Adaptern NVS



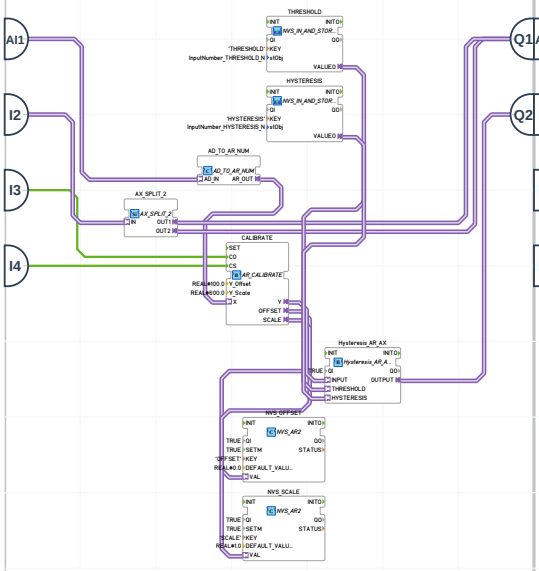
Uebung_028a_AR

Analog-Eingang Kalibrierung mit Adaptern INI



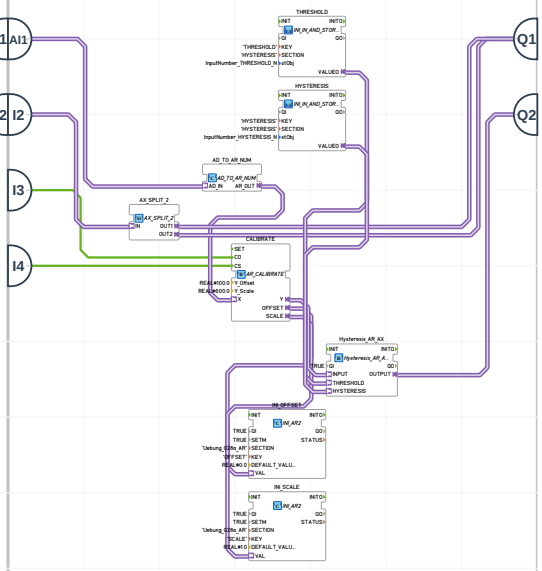
Uebung_028a_AX

Analog-Eingang Kalibrierung mit Adaptern INI



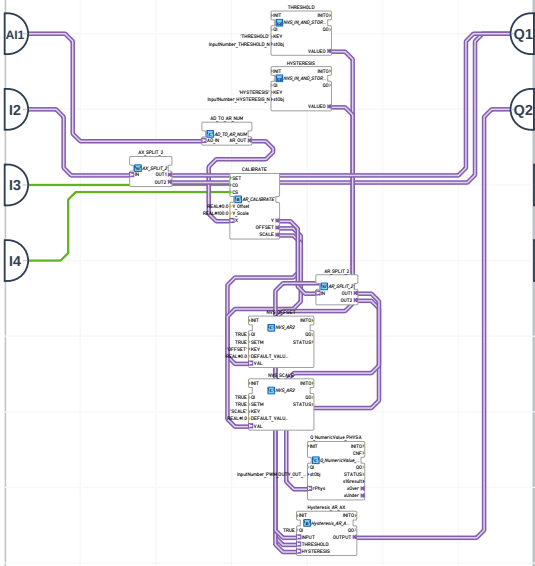
Uebung_028b2_AR

Analog-Eingang Kalibrierung mit Adaptern NVS und Hysterese-Regler am Ausgang



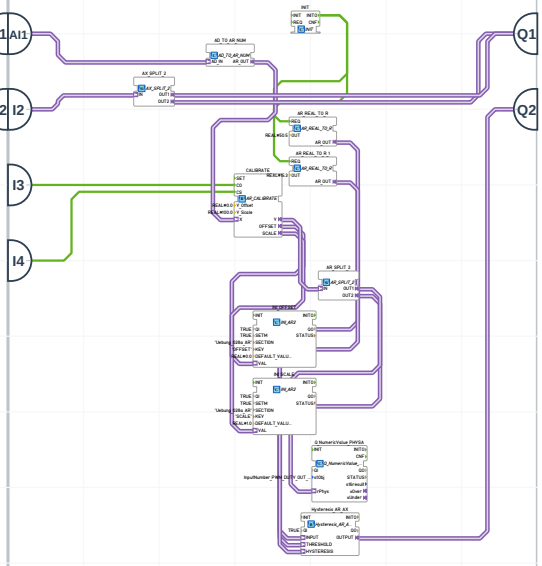
Uebung_028b_AR

Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese-Regler am Ausgang



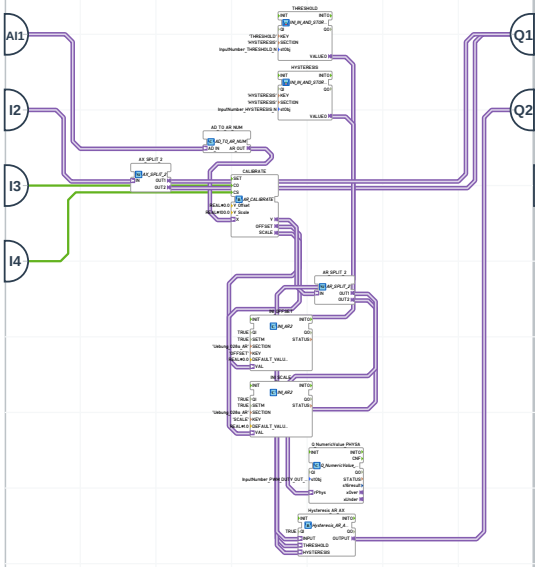
Uebung_028c2_AR

Analog-Eingang Kalibrierung mit Adaptern NVS und Hysterese-Regler am Ausgang und Display Anzeige



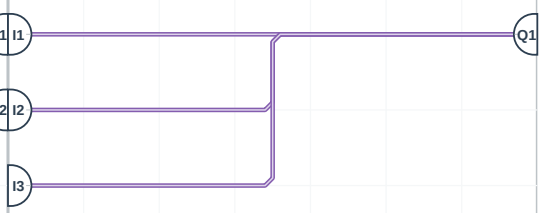
Uebung_028c3_AR

Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese-Regler am Ausgang und Display Anzeige



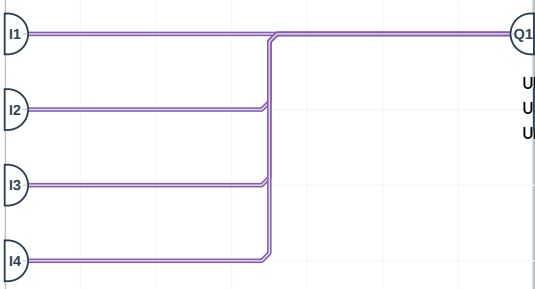
Uebung_028c_AR

Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese-Regler am Ausgang und Display Anzeige



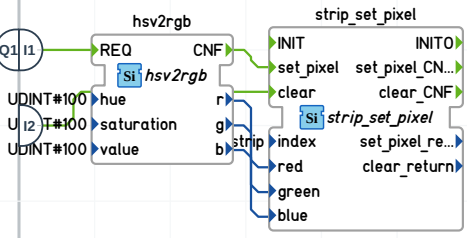
Uebung_029_AX

LED_00 Blinkende LED, mit Plug and Socket



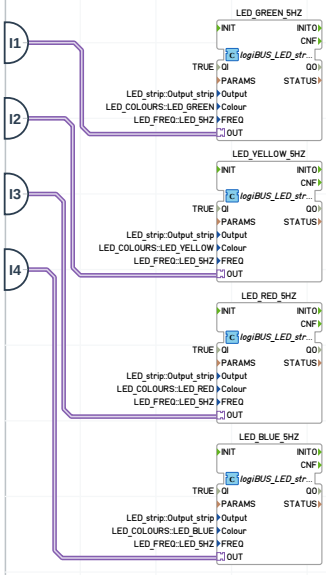
Uebung_030_AX

LED_PWM Blinkende LED, mit Plug and Socket



Uebung_031_AX

LED Strip

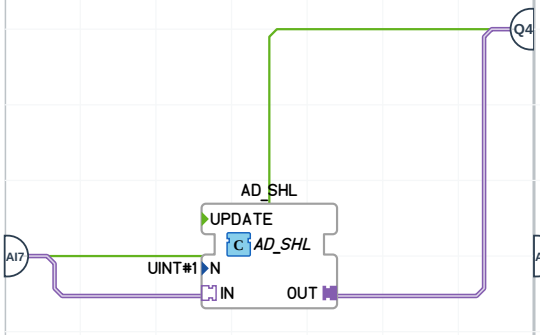


Uebung_032_AX

LED Strip Blinkende LED, mit Plug and Socket

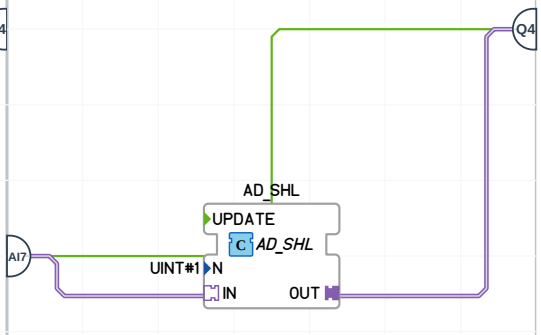
Uebung_033_AX

LED Strip Blinkende LED, mit Plug and Socket



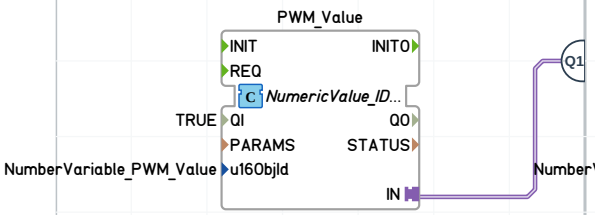
Uebung_034_AD

Analog-Eingang auf PWM Ausgang (Adapter Version)



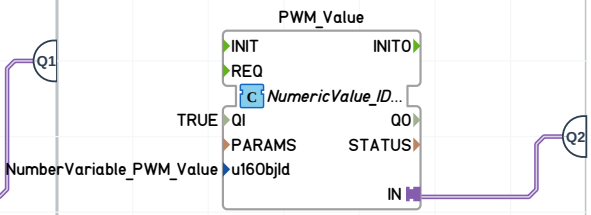
Uebung_034_AX

Analog-Eingang auf PWM Ausgang (Adapter Version)



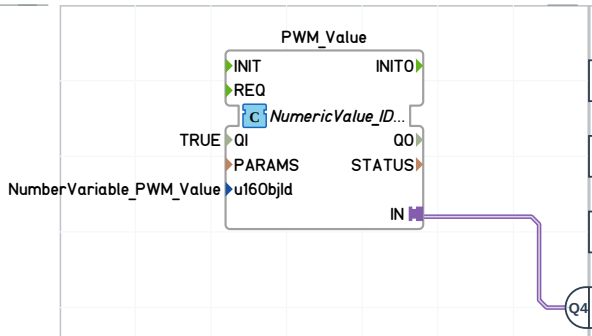
Uebung_034a1_Q1_AX

Numeric Value Input auf PWM Ausgang (Adapter Version)



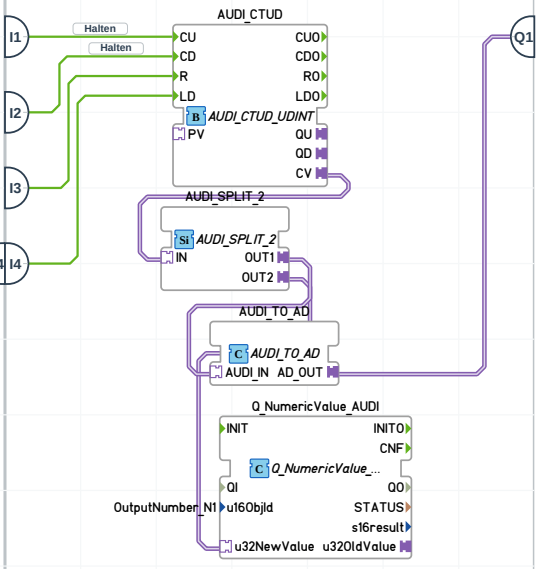
Uebung_034a1_Q2_AX

Numeric Value Input auf PWM Ausgang (Adapter Version)



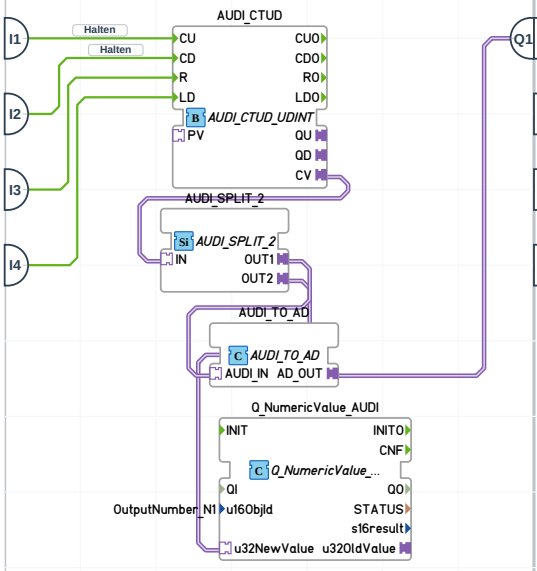
Uebung_034a1_Q4_AX

Numeric Value Input auf PWM Ausgang (Adapter Version)



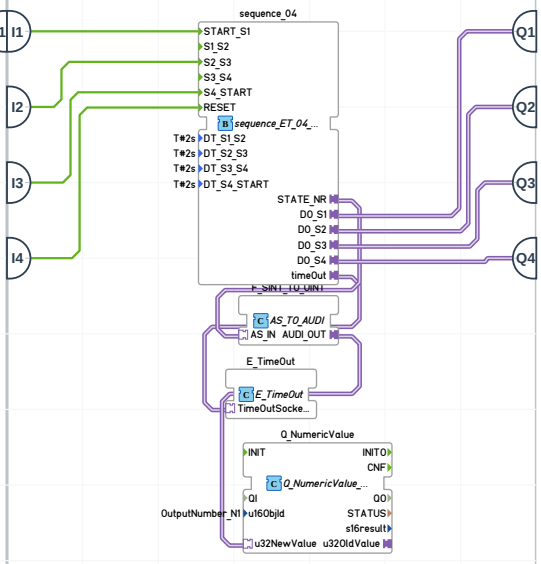
Uebung_034b_AUDI

LONG_PRESS_HOLD-Eingang auf PWM Ausgang (Adapter Version) mit Terminal-Ausgabe



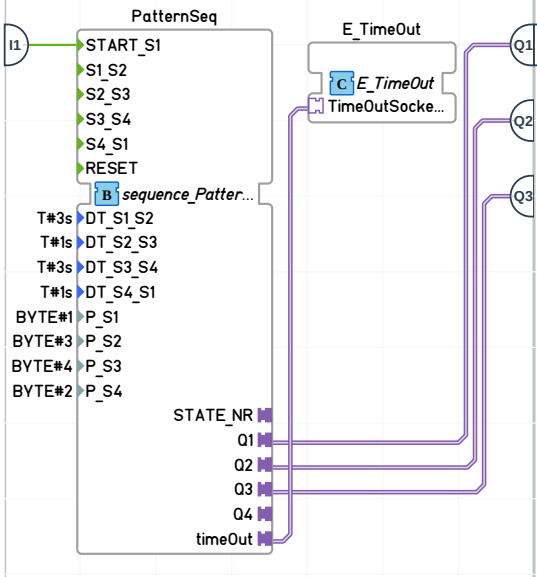
Uebung_034b_AX

LONG_PRESS_HOLD-Eingang auf PWM Ausgang (Adapter Version) mit Terminal-Ausgabe



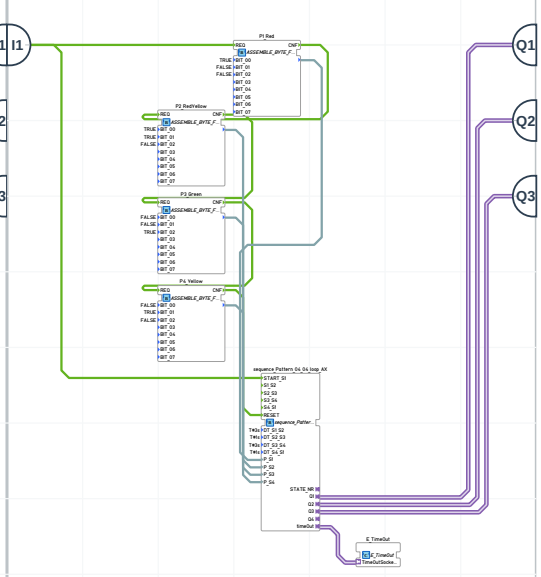
Uebung_035_AX

Spiegelabfolge V2 mit Schrittkette



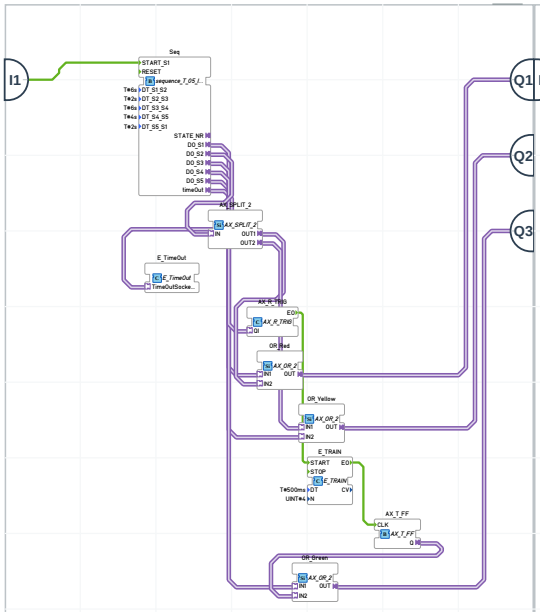
Uebung_035a1_AX

Ampelsteuerung mit Pattern Sequencer (AX)



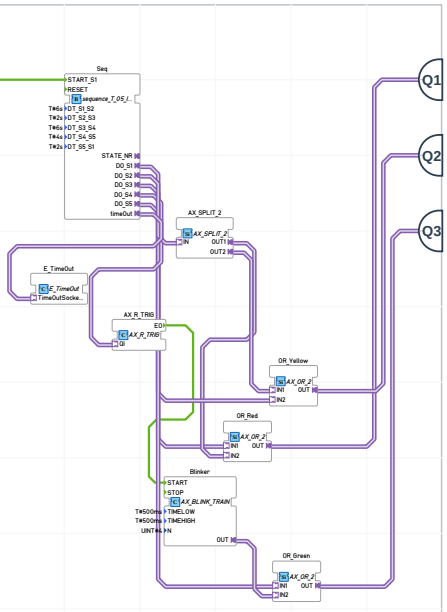
Uebung_035a1b_AX

Ampelsteuerung mit Pattern Sequencer (AX) und Bit-Assembler



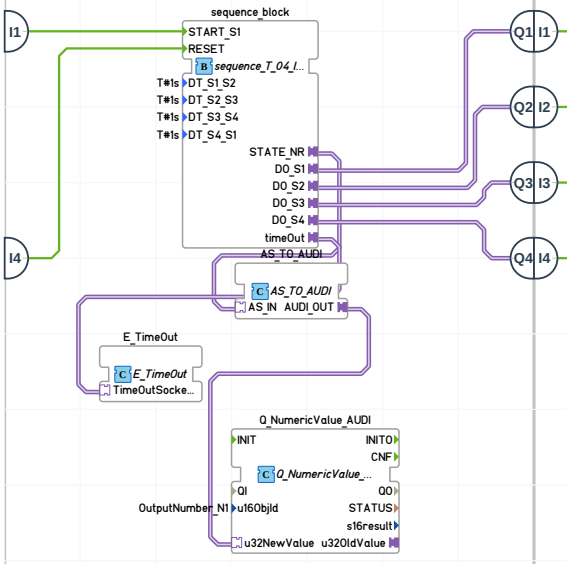
Uebung_035a2_AX

Ampelschaltung Österreich (AX) mit Sequence T05



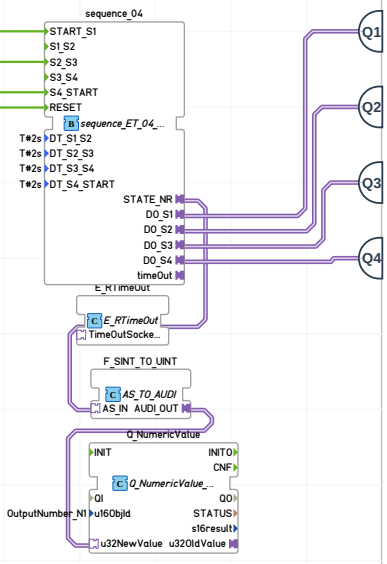
Uebung_035a3_AX

Ampelschaltung Österreich (AX) mit Blinker



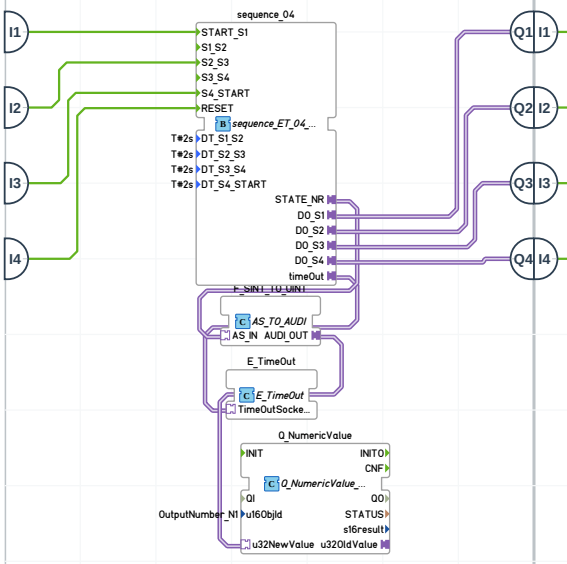
Uebung_035a_AX

Lauflicht 4 reine Zeitsteuerung (Adapter Version)



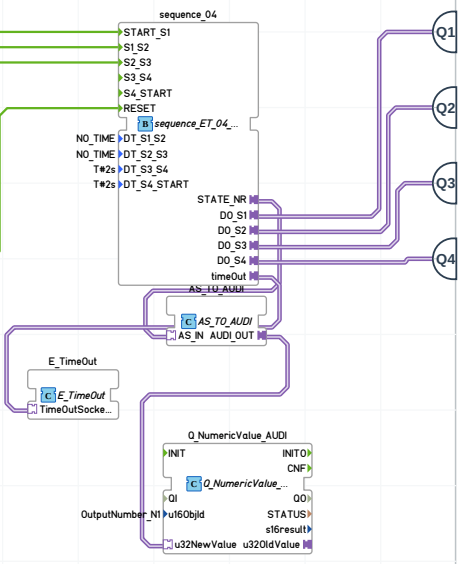
Uebung_035b_AX

Spiegelabfolge V2 mit Schrittkette



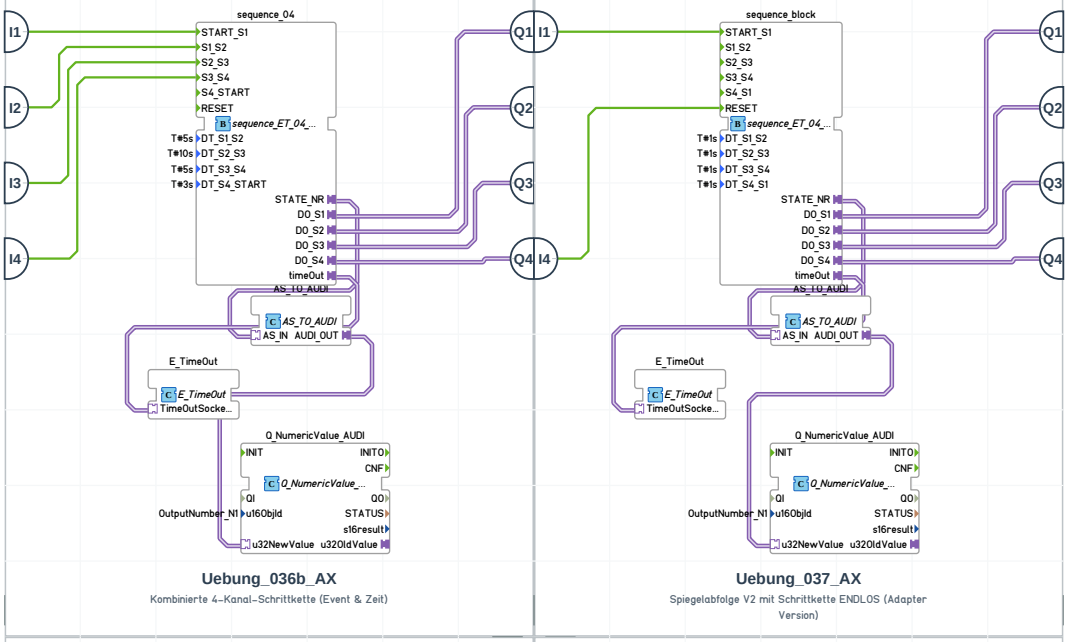
Uebung_035c_AX

Spiegelabfolge V2 mit Schrittkette



Uebung_036_AX

Spiegelabfolge V2 mit Schrittkette

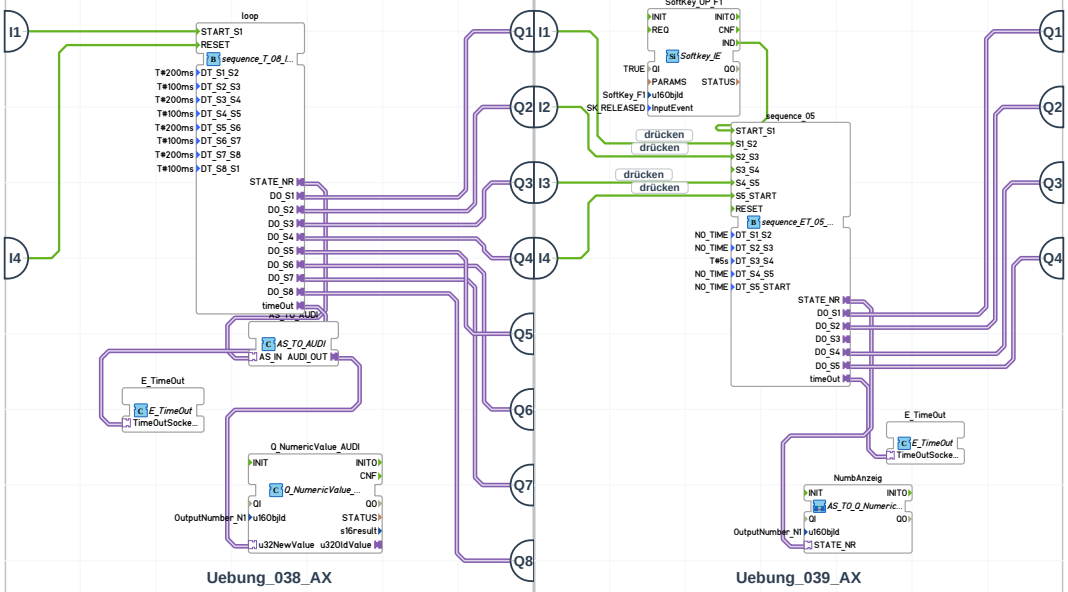


Uebung_036b_AX

Kombinierte 4-Kanal-Schrittkette (Event & Zeit)

Uebung_037_AX

Spiegelabfolge V2 mit Schrittkette ENDLOS (Adapter Version)

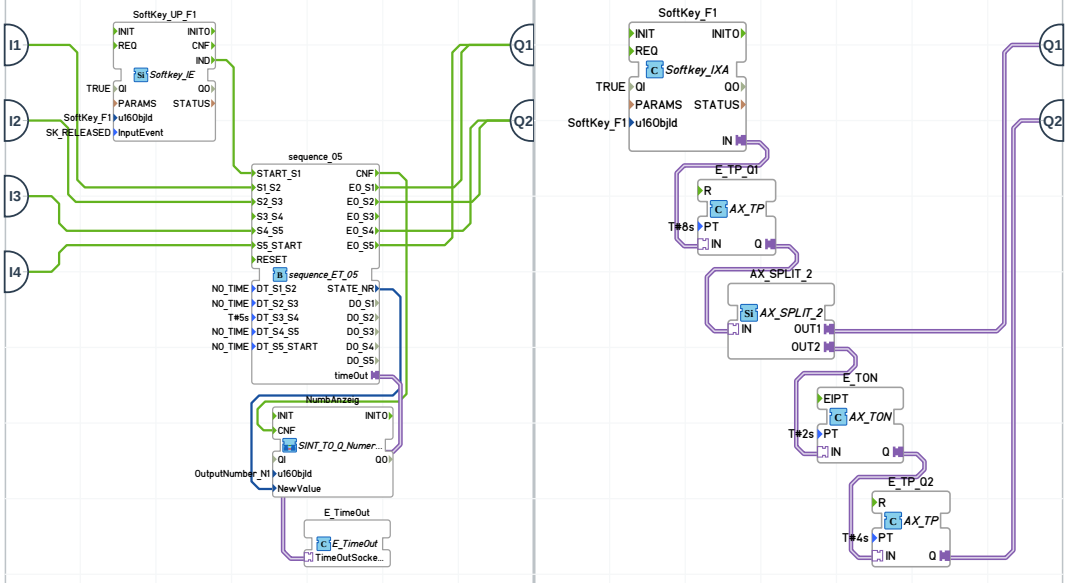


Uebung_038_AX

Lauflicht 8 reine Zeitsteuerung

Uebung_039_AX

Spiegelabfolge V2 mit Schrittkette

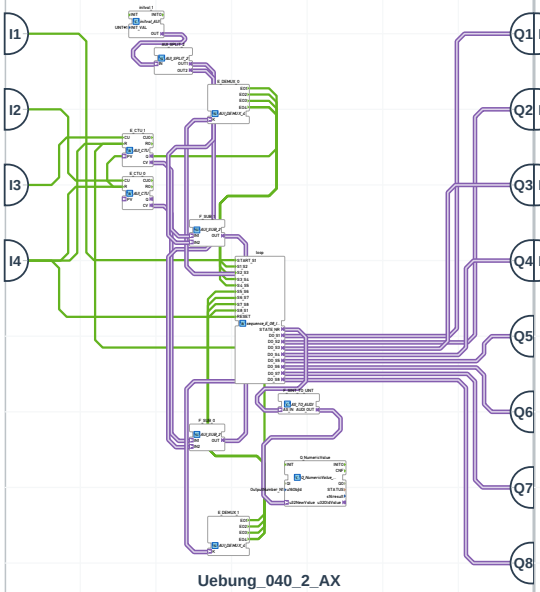


Uebung_039a_AX

Spiegelabfolge V2 mit Schrittkette

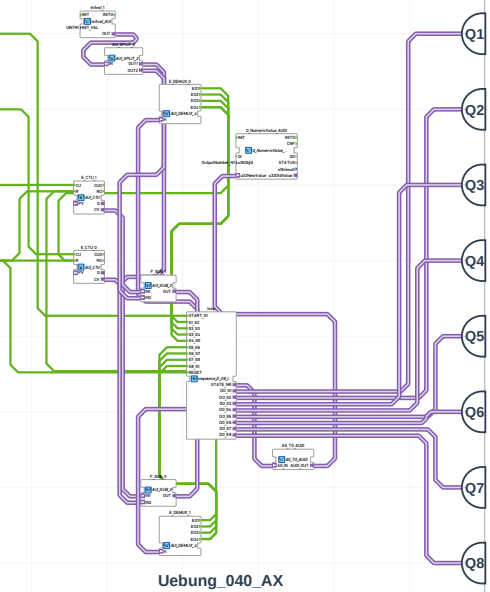
Uebung_039b_AX

Spiegelabfolge V2 mit Schrittkette



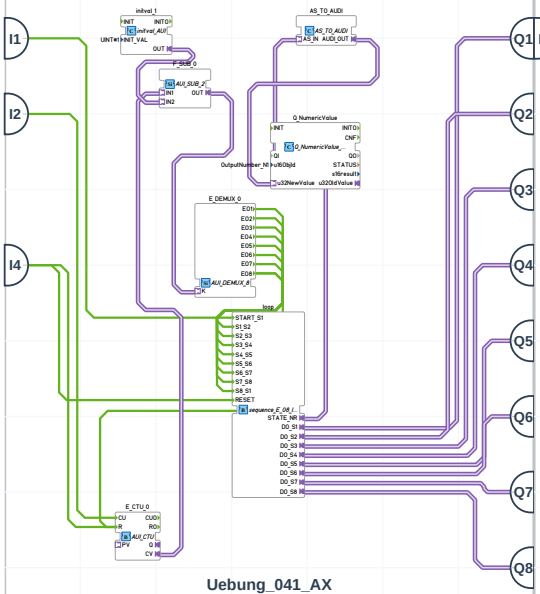
Uebung_040_2_AX

Lauflicht 8



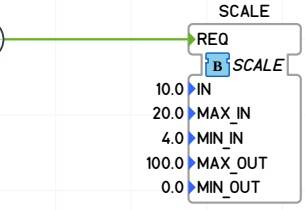
Uebung_040_AX

Lauflicht 8 Event gesteuert



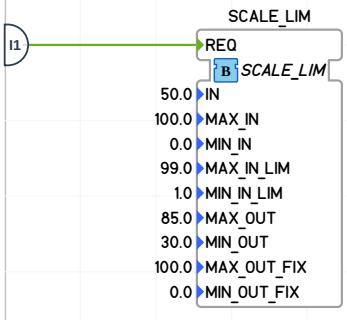
Uebung_041_AX

Lauflicht 8 mit einer Taste, mit Plug and Socket



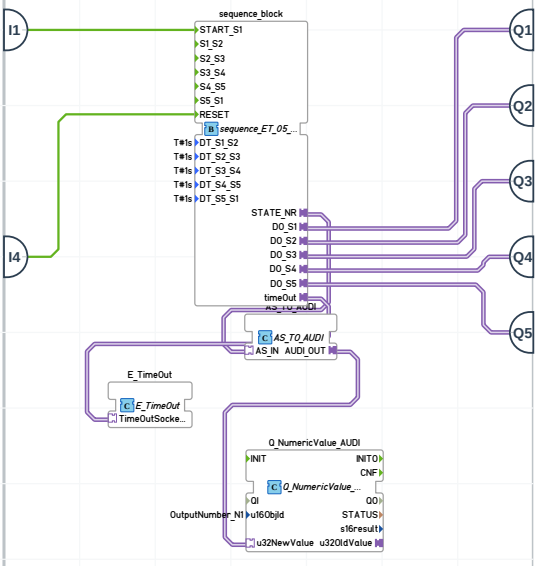
Uebung_042_AX

Scaling Function Block Testing



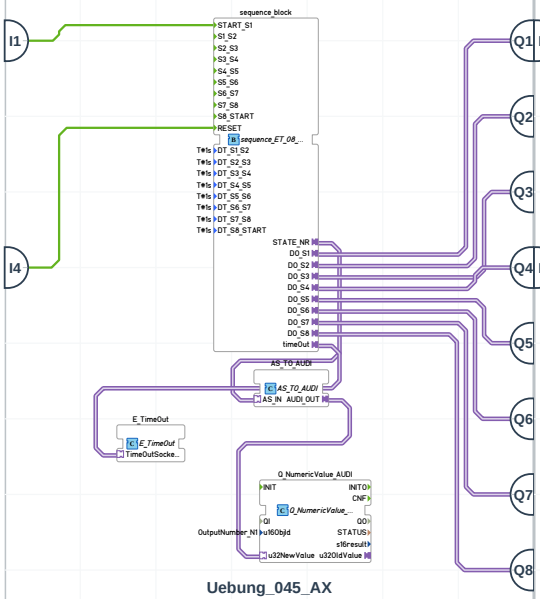
Uebung_043_AX

Scaling Function Block with limits Testing



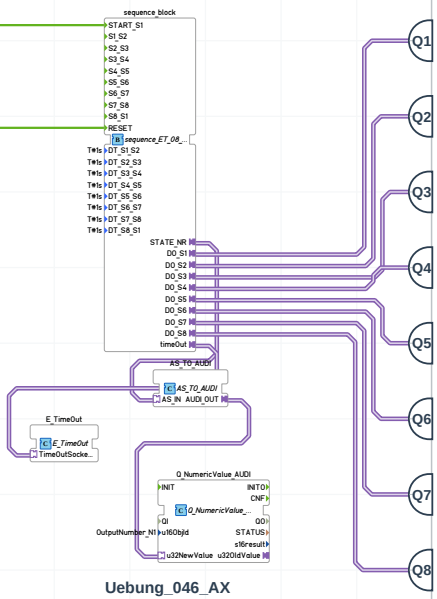
Uebung_044_AX

Spiegelabfolge 5-Kanal mit Schrittkette ENDLOS
(Adapter Version)



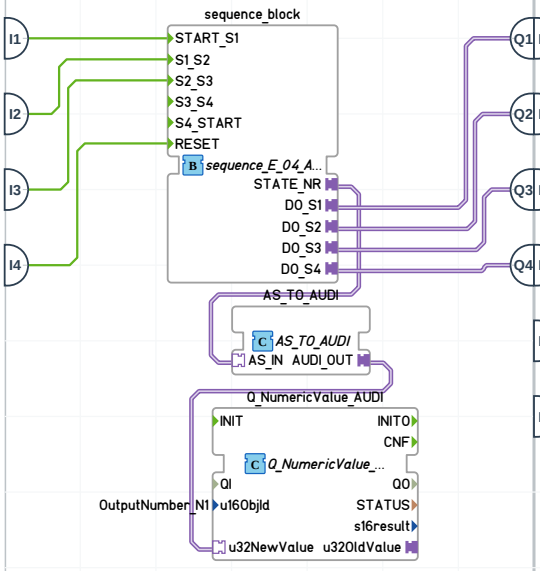
Uebung_045_AX

Ablaufsteuerung 8-Kanal (Event & Zeit) (Adapter Version)



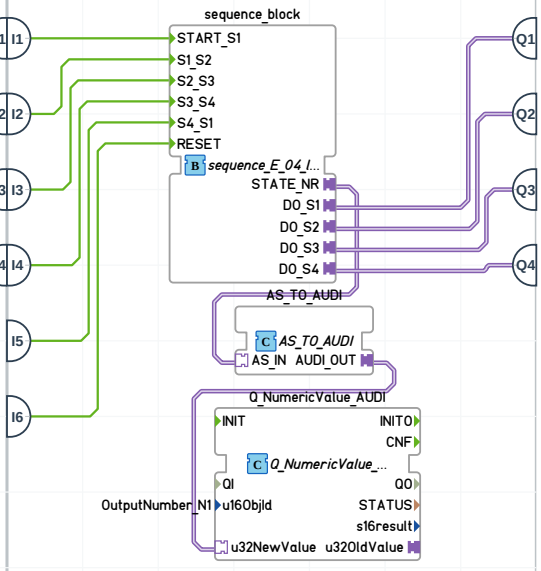
Uebung_046_AX

Ablaufsteuerung 8-Kanal ENDLOS (Event & Zeit) (Adapter Version)



Uebung_047_AX

Reine Event-Schrittkette 4-Kanal (Adapter Version)



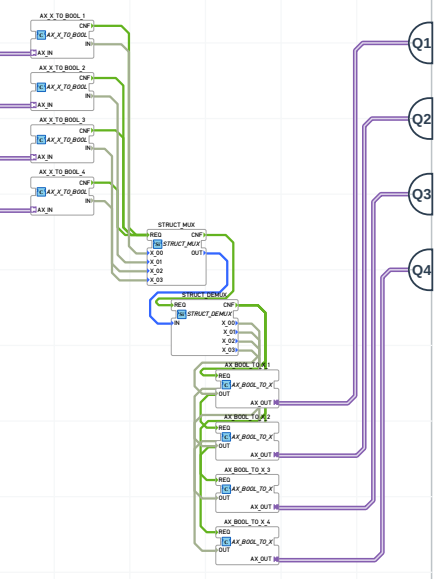
Uebung_048_AX

Reine Event-Schrittkette 4-Kanal ENDLOS (Adapter Version)



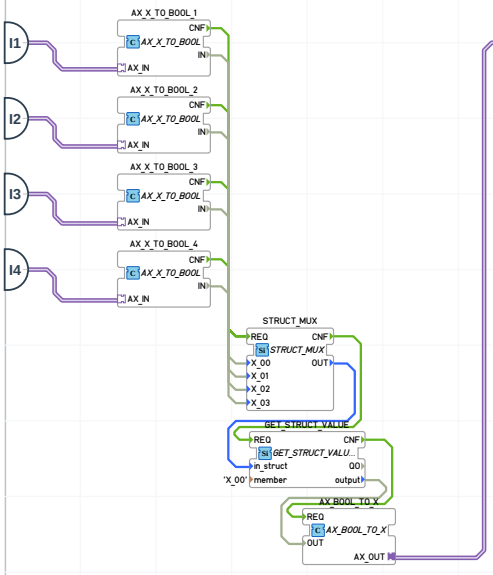
Uebung_049_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket



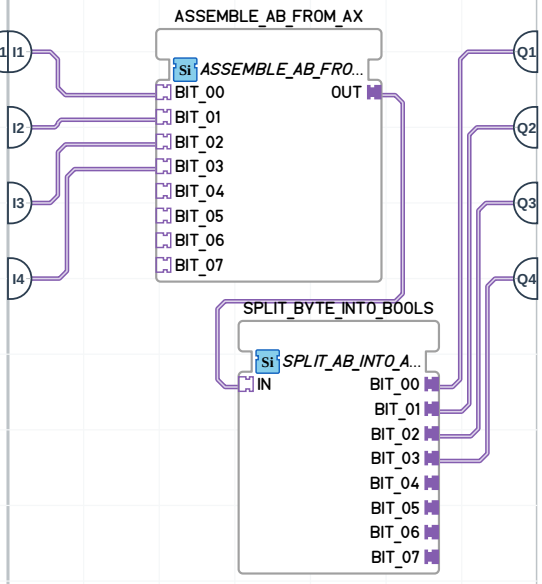
Uebung_051_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket



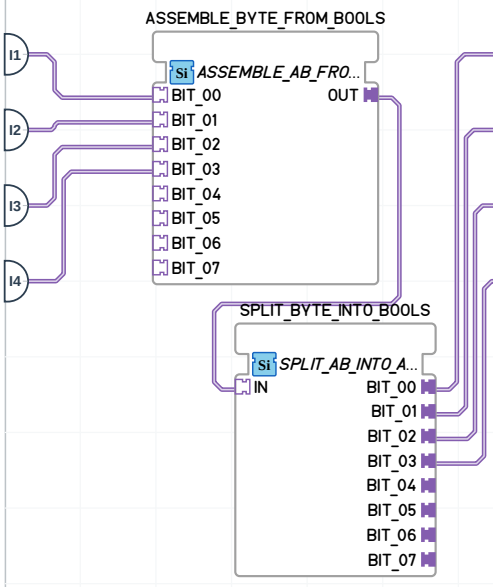
Uebung_052_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket



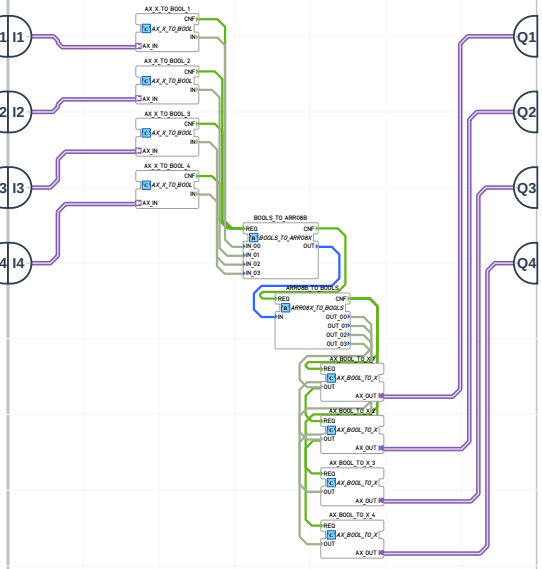
Uebung_053_AB

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4



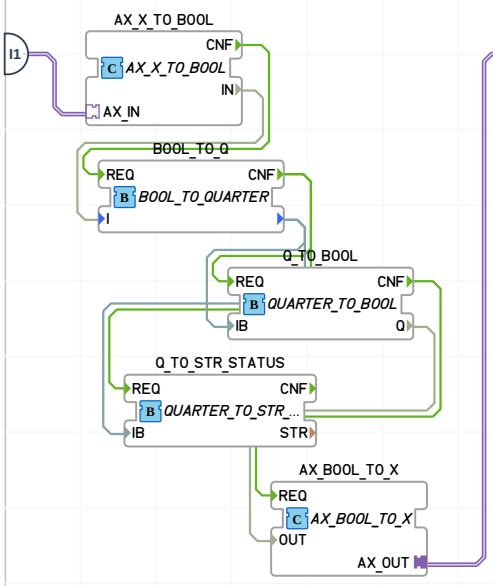
Uebung_053_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4



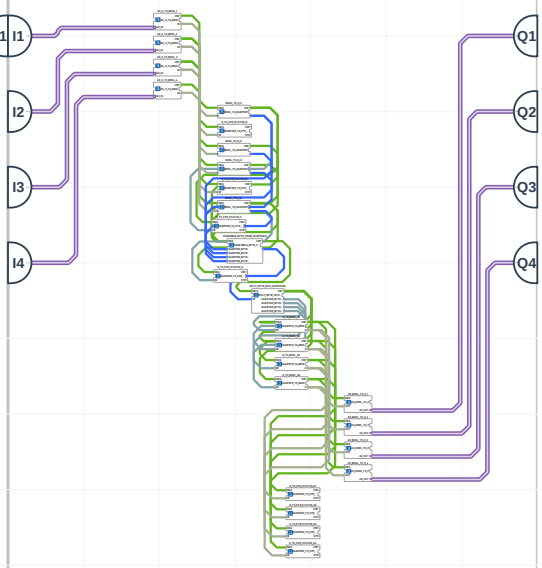
Uebung_054_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket



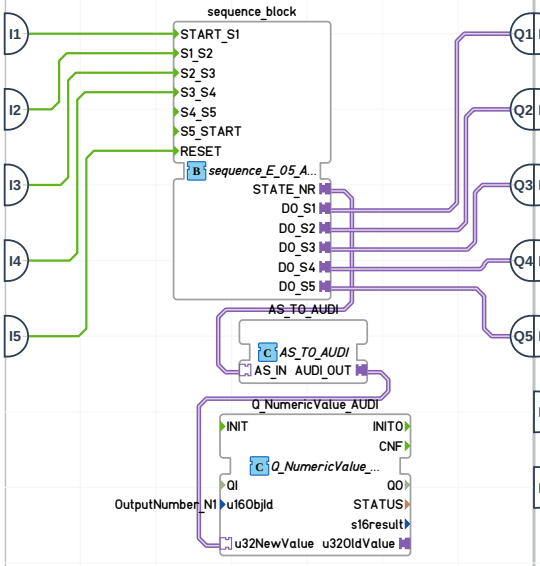
Uebung_055_AX

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



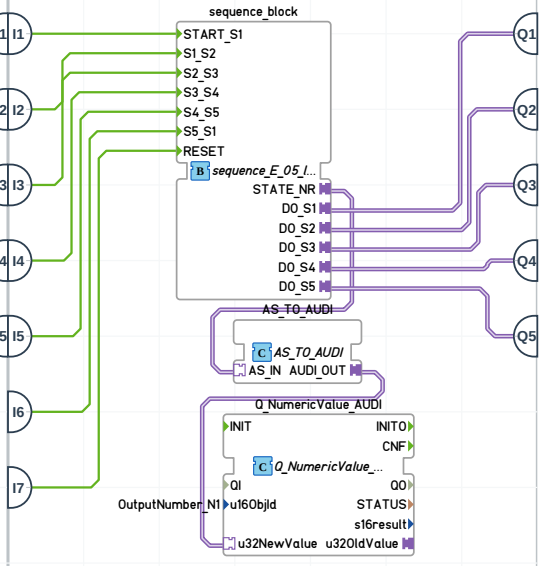
Uebung_056_AX

DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket



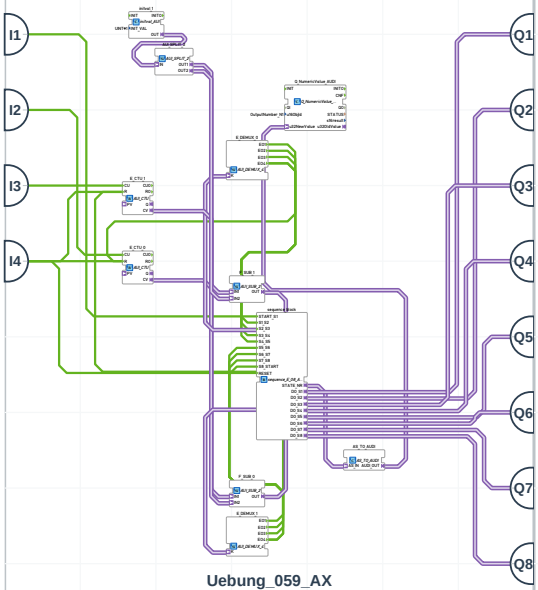
Uebung_057_AX

Reine Event-Schrittkette 5-Kanal (Adapter Version)



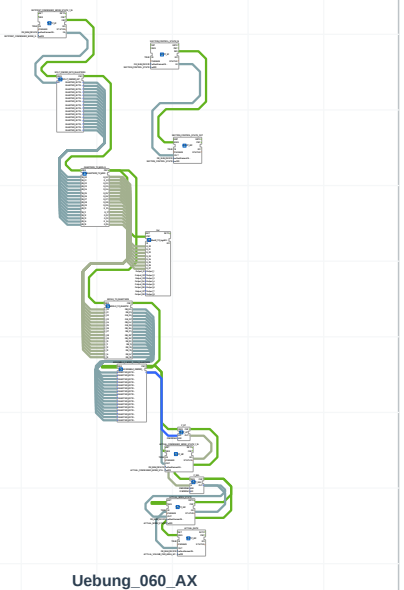
Uebung_058_AX

Reine Event-Schrittkette 5-Kanal ENDLOS (Adapter Version)



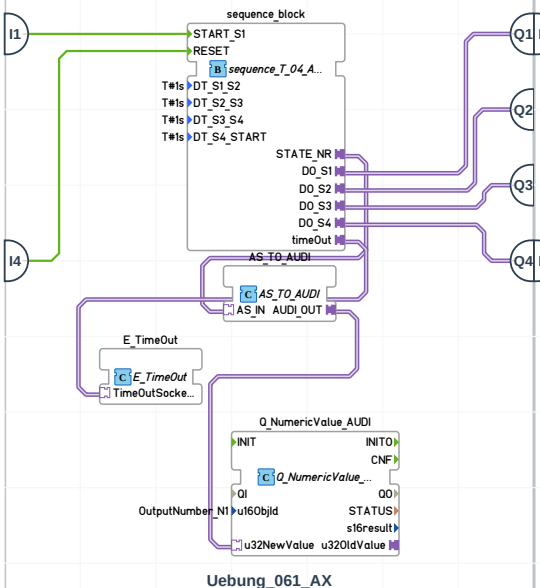
Uebung_059_AX

Reine Event-Schrittkette 8-Kanal (Adapter Version)



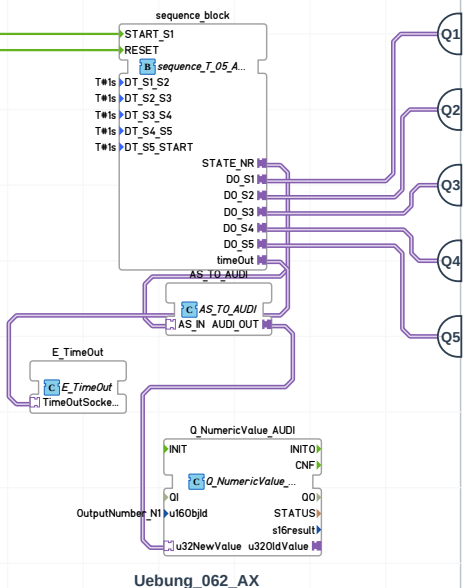
Uebung_060_AX

für TC-SC also Task Controller Section Control, mit Plug and Socket



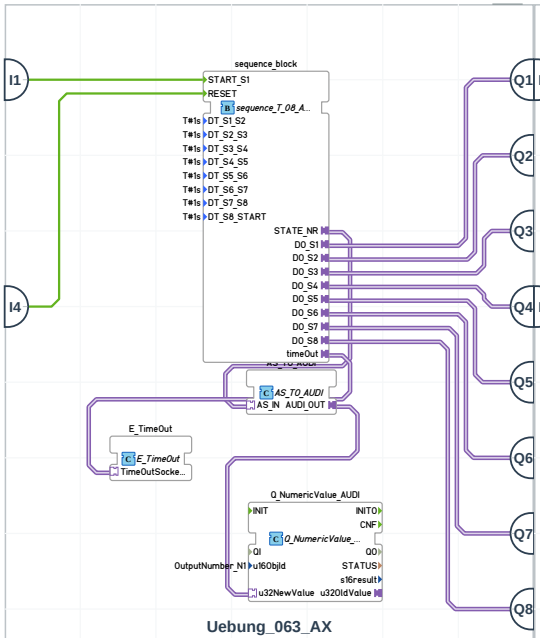
Uebung_061_AX

Reine Zeitsteuerung 4-Kanal (Adapter Version)



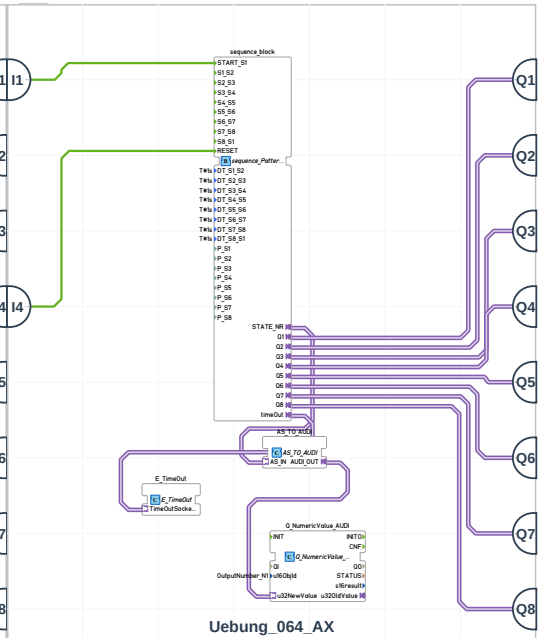
Uebung_062_AX

Reine Zeitsteuerung 5-Kanal (Adapter Version)



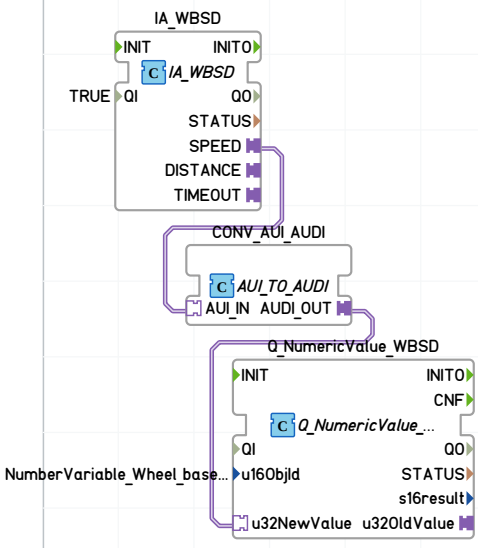
Uebung_063_AX

Reine Zeitsteuerung 8-Kanal (Adapter Version)



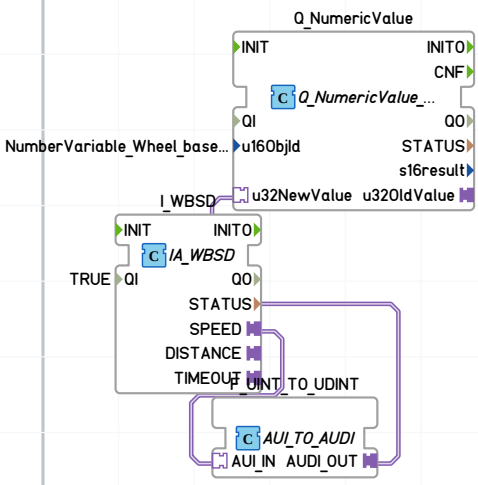
Uebung_064_AX

Musterschrittkette 8-Kanal ENDLOS (Adapter Version)



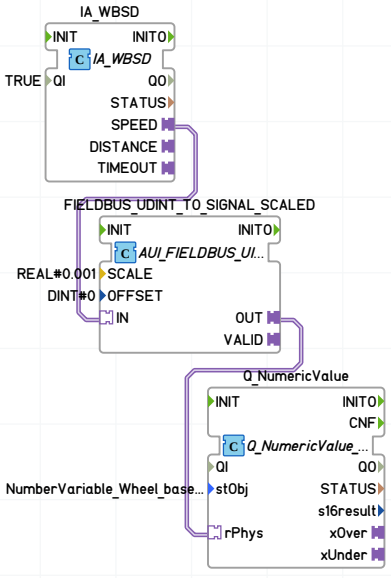
Uebung_070_AUI

WBSD auf UT ausgeben (Adapter Version)



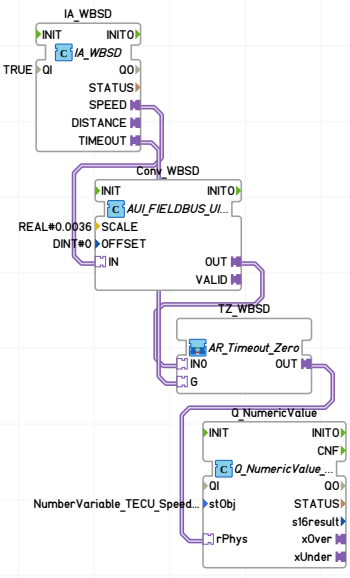
Uebung_070_AX

WBSD auf UT ausgeben



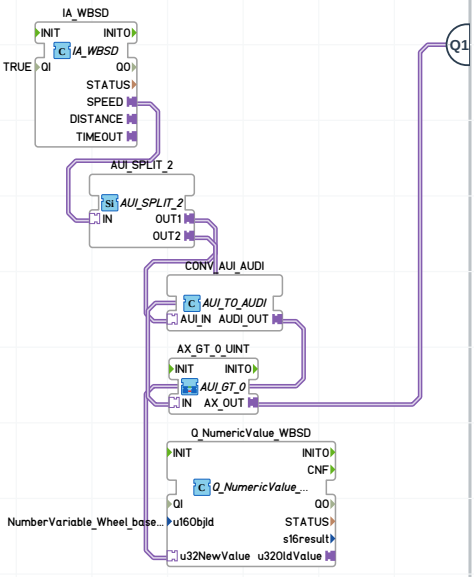
Uebung_070c_AUI

WBSD auf UT ausgeben (Adapter Version)



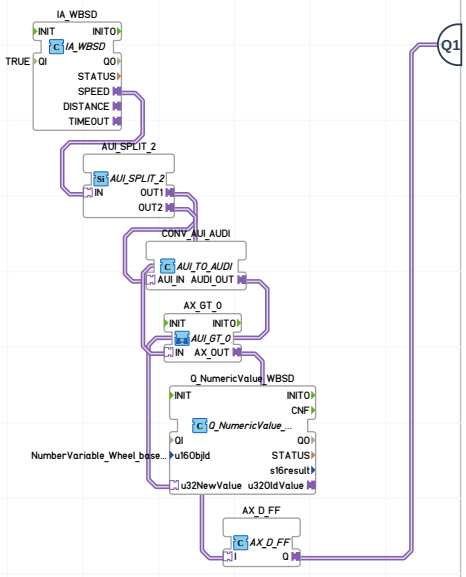
Uebung_070d_AUI

WBSD auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



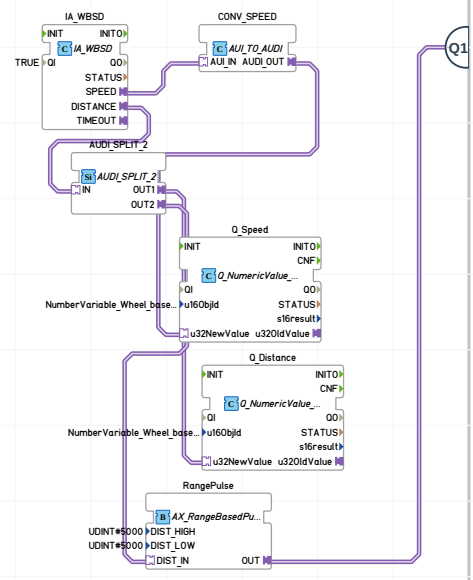
Uebung_071_AUI

WBSD auf UT ausgeben, Q1 Schalten wenn Speed > 0
(Adapter Version)



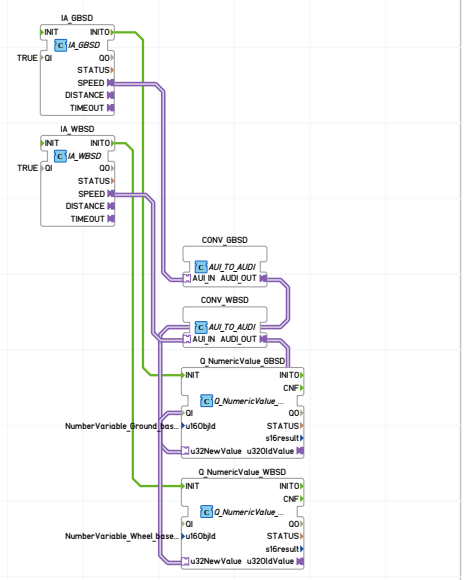
Uebung_071a_AUI

WBSD auf UT ausgeben, Q1 Schalten wenn Speed > 0
(Adapter Version)



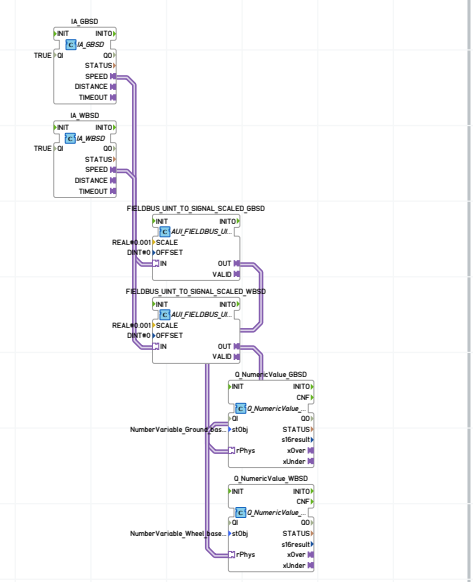
Uebung_071b_AUI

WBSD auf UT ausgeben, Wegabhängige Impulse (Adapter Version)



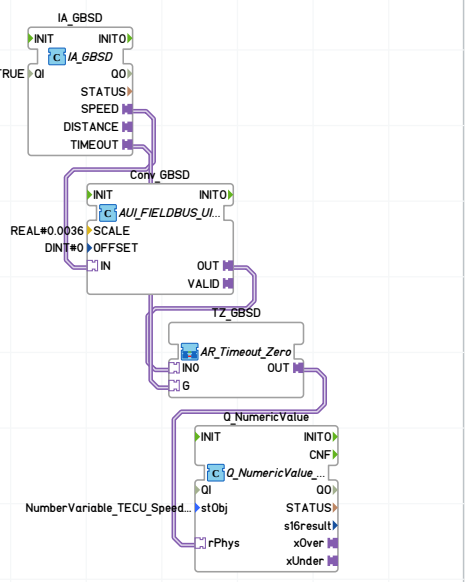
Uebung_072_AUI

GBSD und WBSD auf UT ausgeben (Adapter Version)



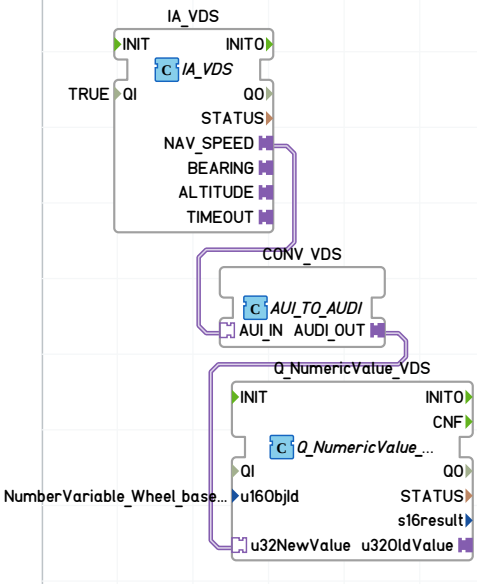
Uebung_072c_AUI

GBSD und WBSD auf UT ausgeben mit PHYS (Adapter Version)



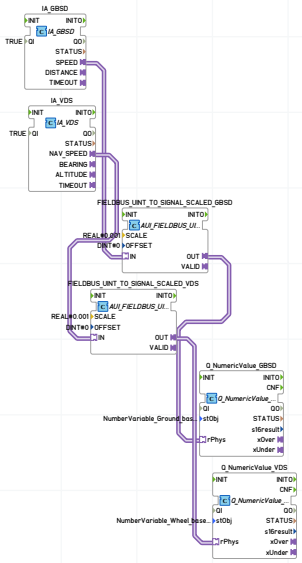
Uebung_072d_AUI

GBSD auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



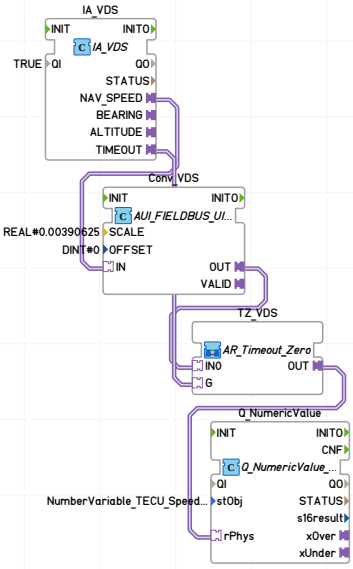
Uebung_073_AUI

VDS auf UT ausgeben (Adapter Version)



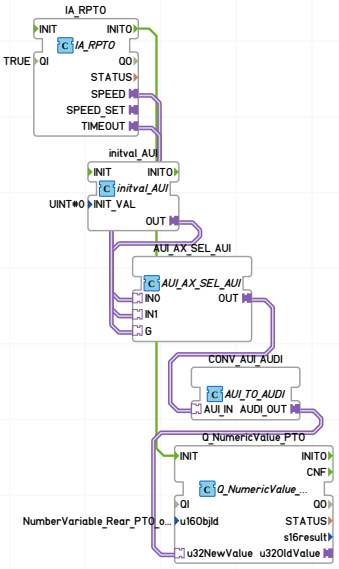
Uebung_073c_AUI

VDS und GBSD auf UT ausgeben mit PHYS (Adapter Version)



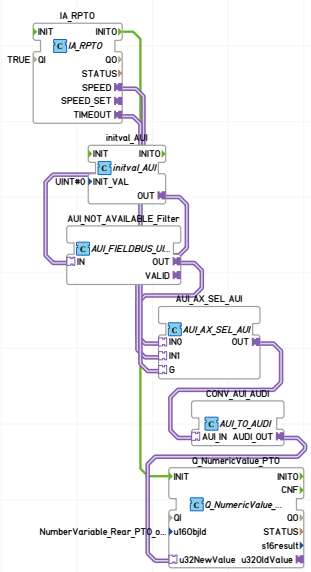
Uebung_073d_AUI

VDS auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



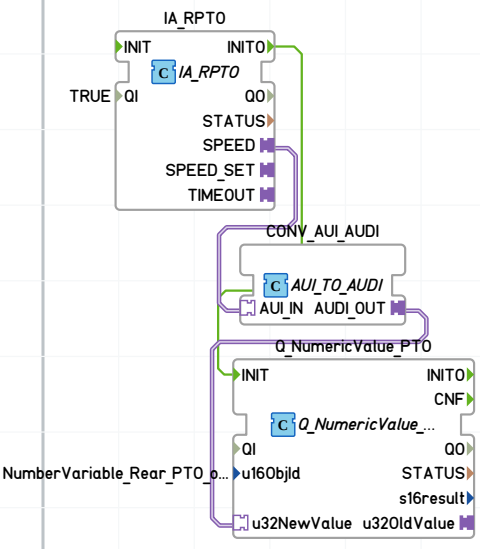
Uebung_074_AUI

RPTO auf UT ausgeben (Adapter Version) mit Fendt-Schaltung



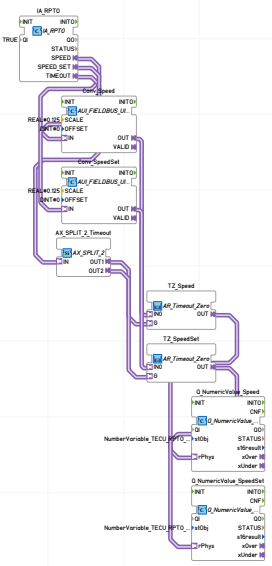
Uebung_074a_AUI

RPTO auf UT ausgeben (Adapter Version) mit Fendt-Schaltung + NOT_AVAILABLE-Filter, FIX



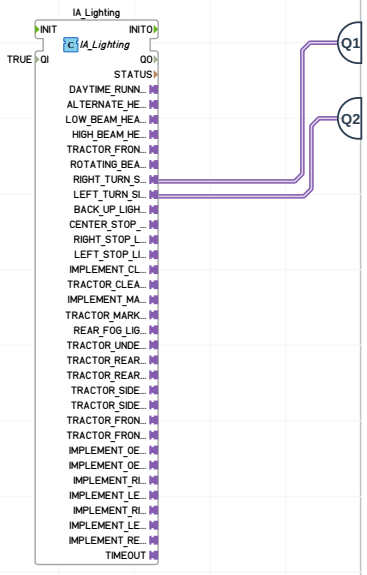
Uebung_074b_AUI

RPTO auf UT ausgeben (Adapter Version) ohne Fendt-Schaltung



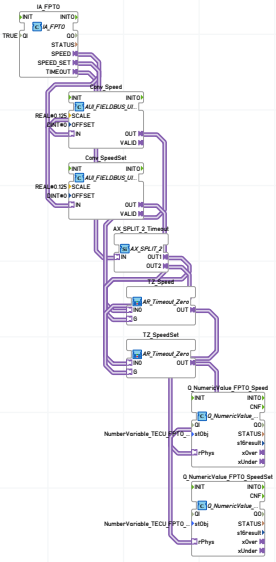
Uebung_074c_AUI

RPTO Speed + SpeedSet auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



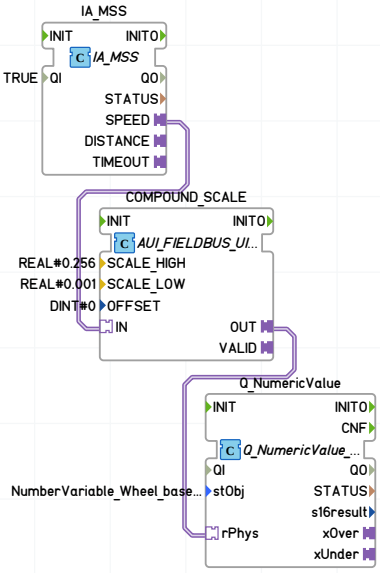
Uebung_075_AX

LD auf Q



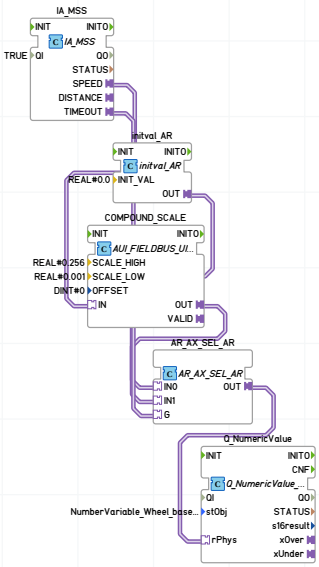
Uebung_075a_AUI

FPTO Speed + SpeedSet auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



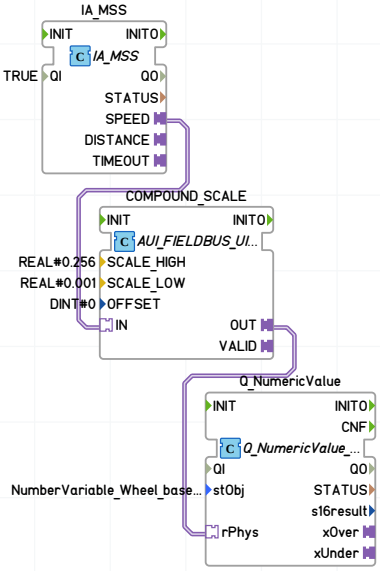
Uebung_076_AUI

MSS mit Compound Scaling auf UT ausgeben (Adapter Version)



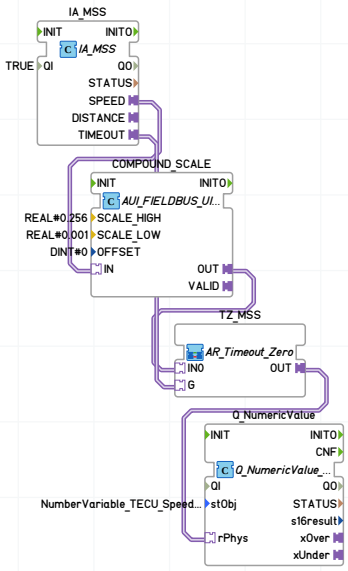
Uebung_076a_AUI

MSS mit Compound Scaling auf UT ausgeben (Adapter Version) mit Fendt-Schaltung + Timeout-Zero, FIX



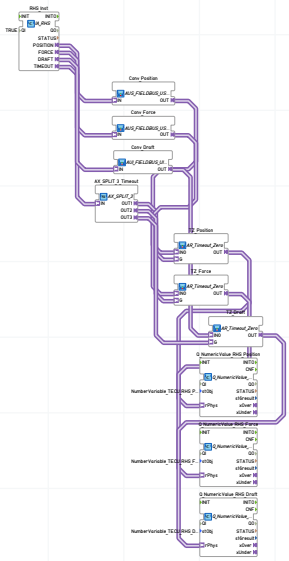
Uebung_076b_AUI

MSS mit Compound Scaling auf UT ausgeben (Adapter Version) ohne Fendt-Schaltung



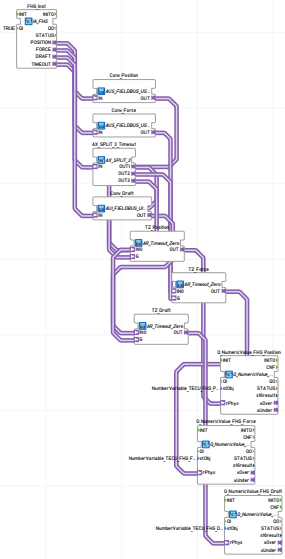
Uebung_076c_AUI

MSS mit Compound Scaling + AR_Timeout_Zero auf UT ausgeben (Adapter Version)



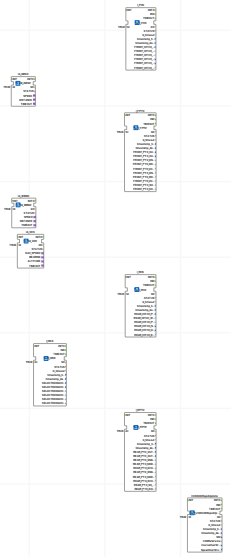
Uebung_077_AUI

RHS Position/Force/Draft auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



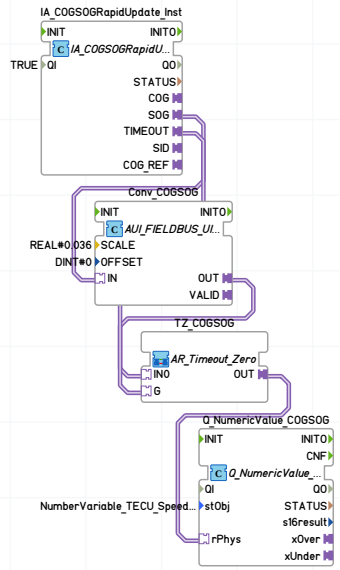
Uebung_078_AUI

FHS Position/Force/Draft auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



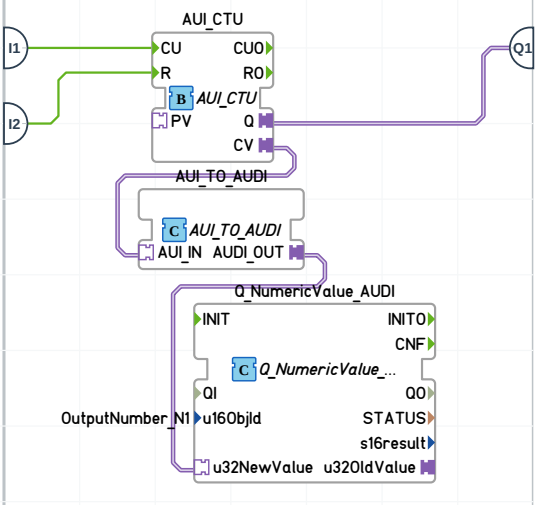
Uebung_079_AX

Alle TECU FBs (Adapter Version)



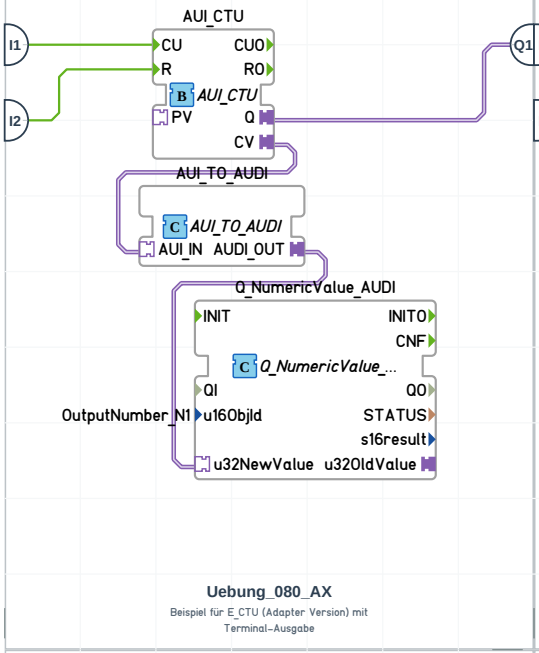
Uebung_079a_AUI

COGSOG Speed Over Ground auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)



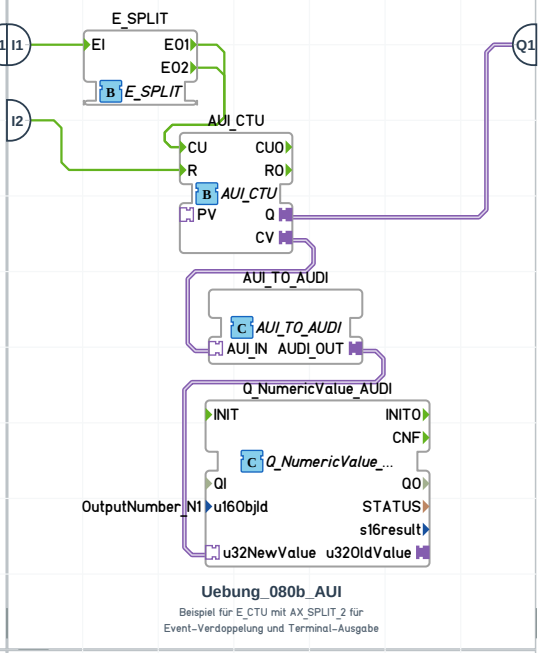
Uebung_080_AUI

Beispiel für E_CTU (Adapter Version) mit Terminal-Ausgabe



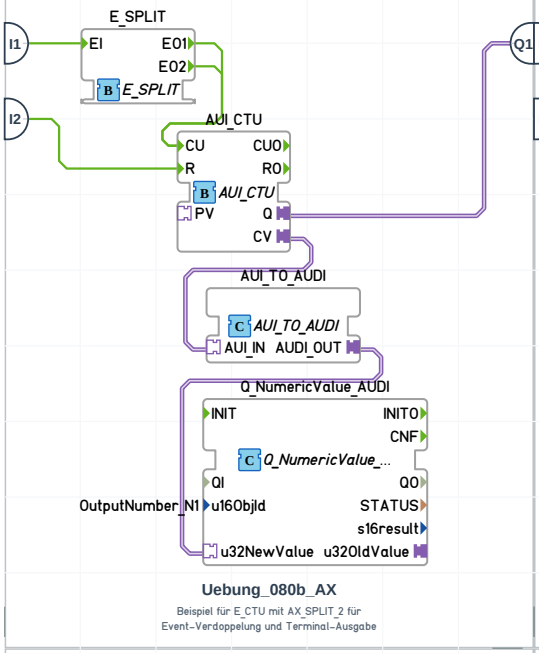
Uebung_080_AX

Beispiel für E_CTU (Adapter Version) mit Terminal-Ausgabe



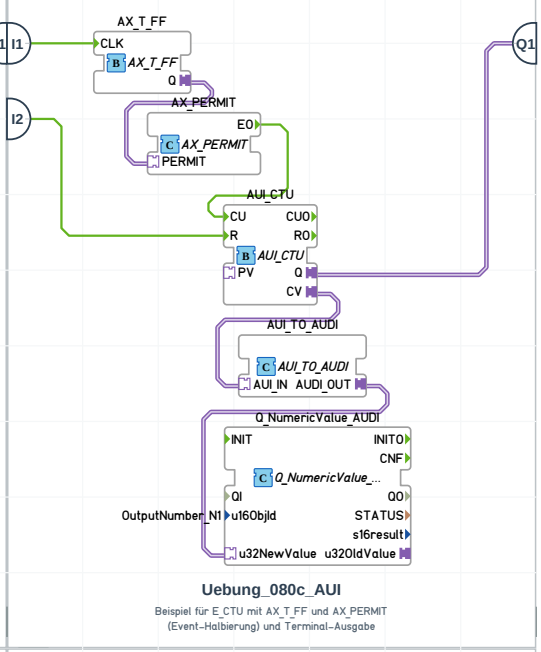
Uebung_080b_AUI

Beispiel für E_CTU mit AX_SPLIT_2 für Event-Verdoppelung und Terminal-Ausgabe



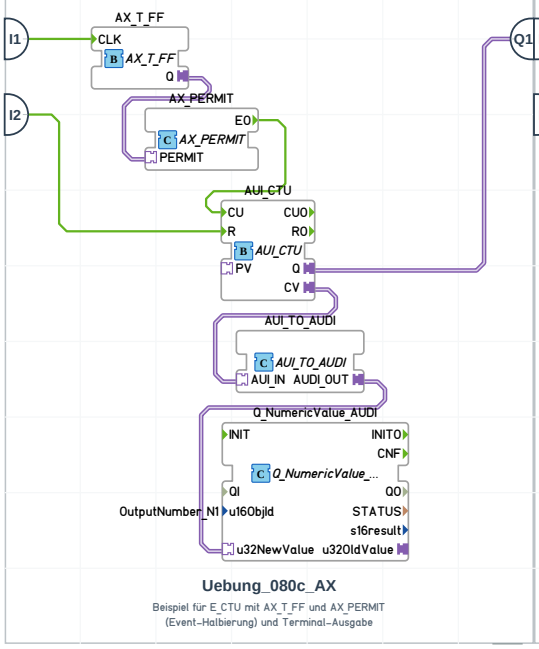
Uebung_080b_AX

Beispiel für E_CTU mit AX_SPLIT_2 für Event-Verdoppelung und Terminal-Ausgabe



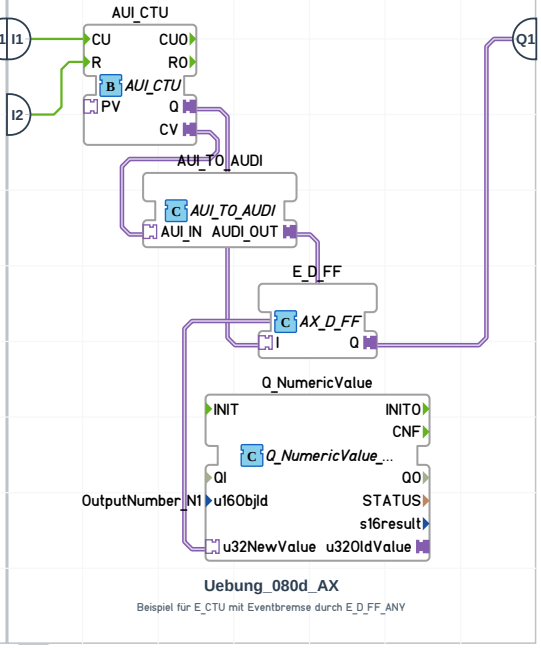
Uebung_080c_AUI

Beispiel für E_CTU mit AX_T_FF und AX_PERMIT (Event-Halbierung) und Terminal-Ausgabe



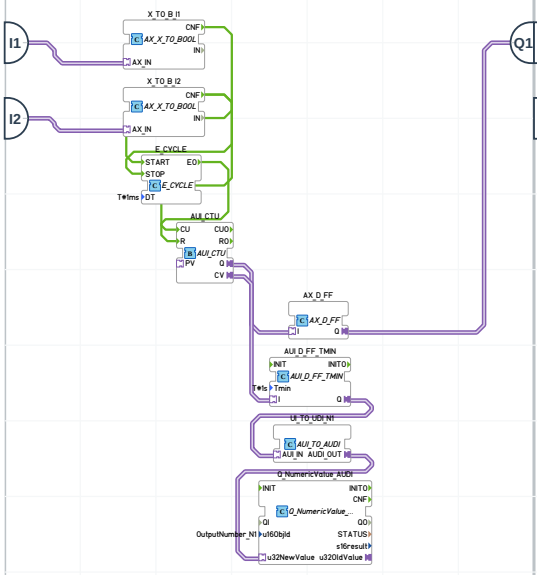
Uebung_080c_AX

Beispiel für E_CTU mit AX_T_FF und AX_PERMIT (Event-Halbierung) und Terminal-Ausgabe



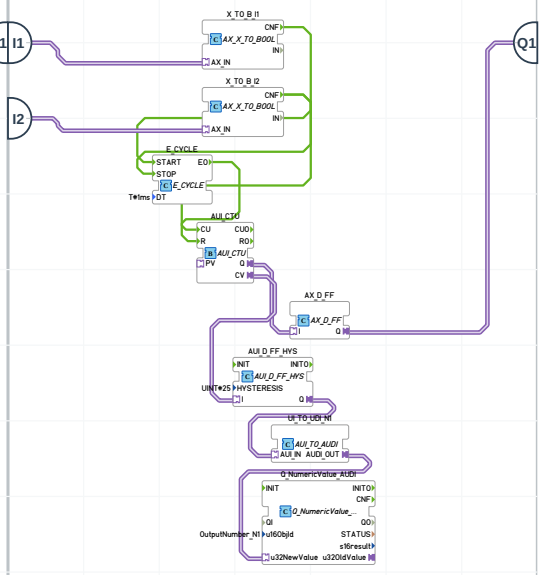
Uebung_080d_AX

Beispiel für E_CTU mit Eventbremse durch E_D_FF_ANY



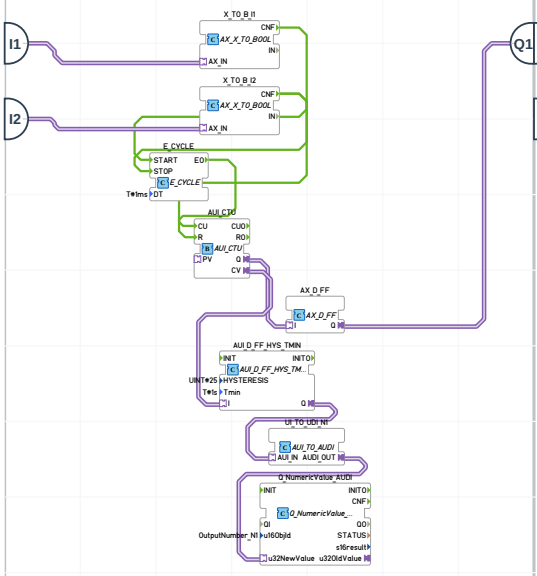
Uebung_080e2_AX

Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_TMIN



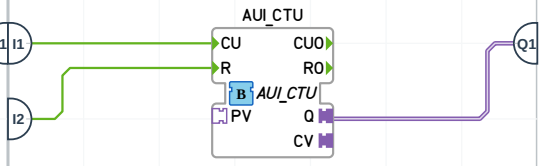
Uebung_080e3_AX

Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_HYS



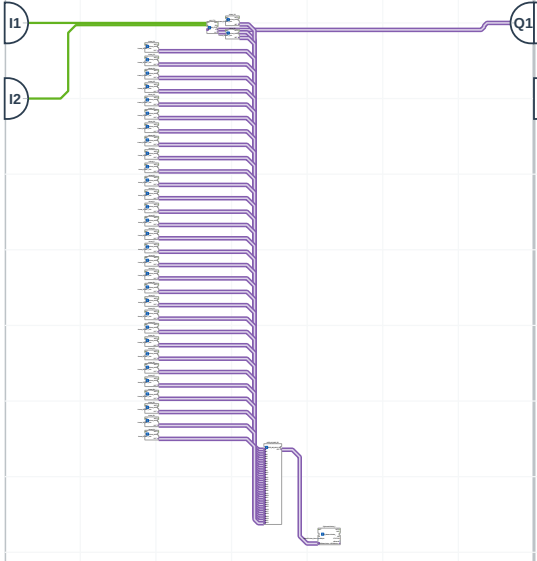
Uebung_080e4_AX

Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_HYS



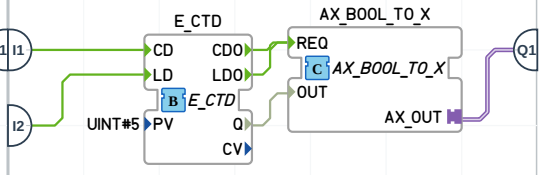
Uebung_080e_AX

Beispiel für E_CTU mit Eventbremse durch E_D_FF_ANY



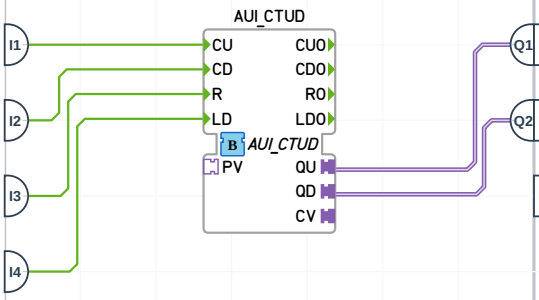
Uebung_080f_AX

Beispiel für E_CTU



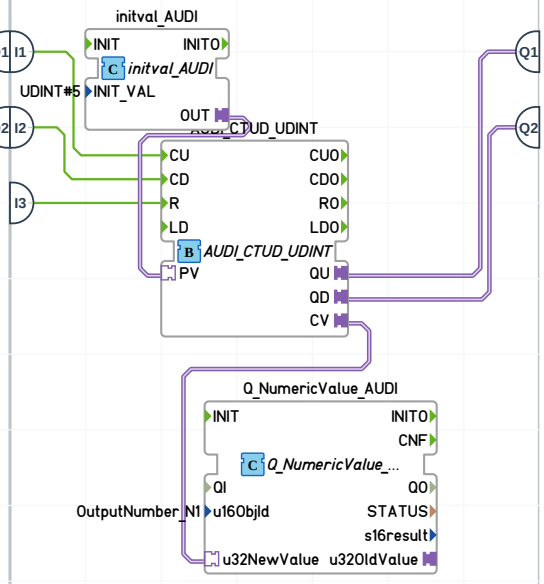
Uebung_081_AX

Beispiel für E_CTD, mit Plug and Socket



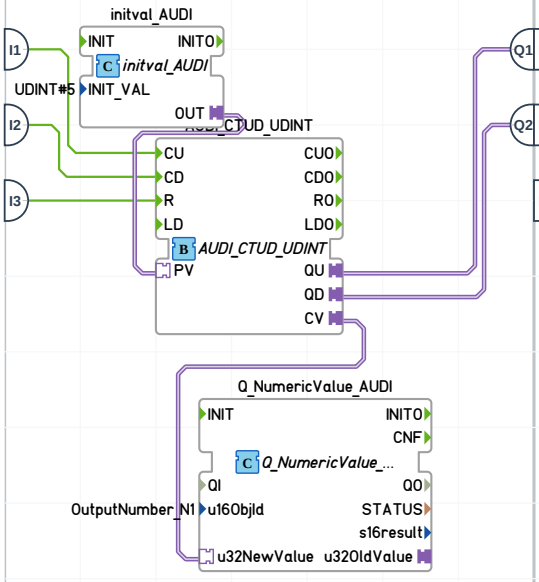
Uebung_082_AX

Beispiel für E_CTUD, mit Plug and Socket



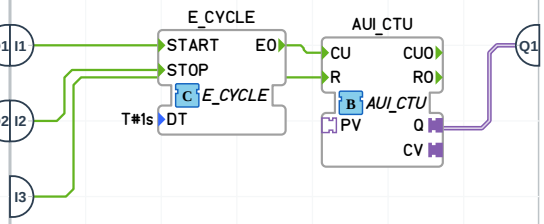
Uebung_083_AUDI

Beispiel für AUDI_CTUD_UDINT (Adapter-basiert)



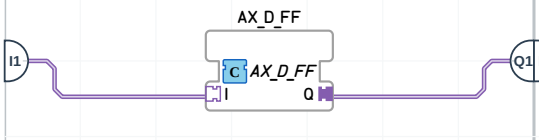
Uebung_083_AX

Beispiel für AUDI_CTUD_UDINT (Adapter-basiert)



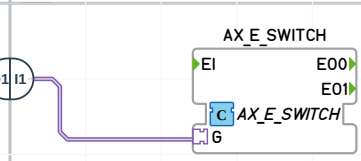
Uebung_084_AX

Beispiel für E_CYCLE



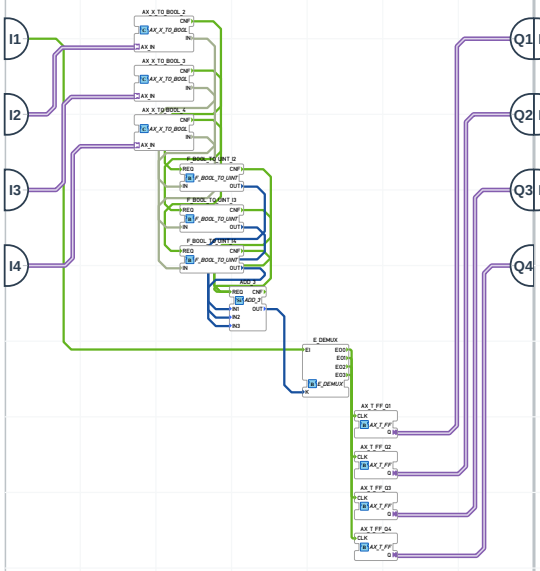
Uebung_085_AX

Beispiel für E_D_FF, mit Plug and Socket



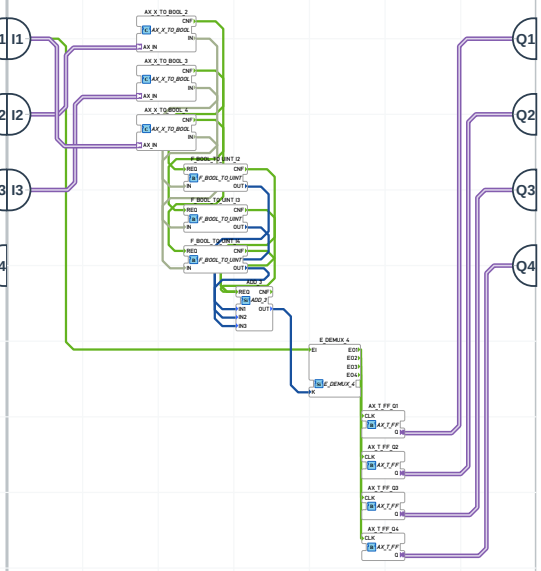
Uebung_086_AX

Beispiel für E_SWITCH, mit Plug and Socket



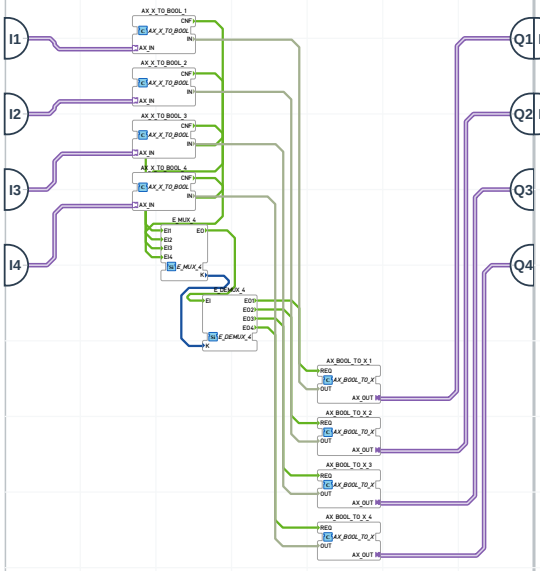
Uebung_087_AX

Beispiel für E_DEMUX, mit Plug and Socket



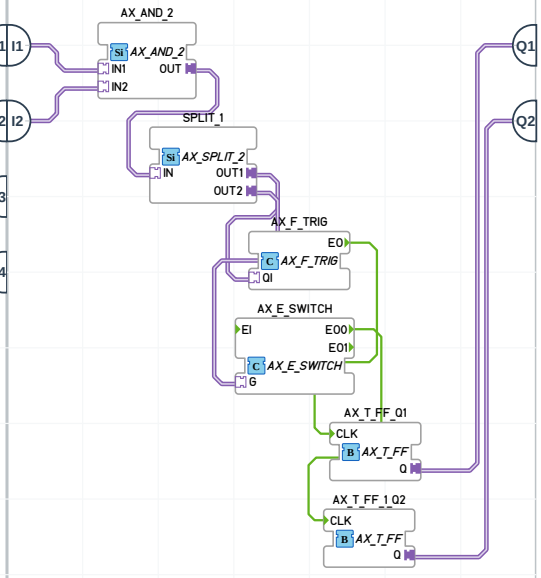
Uebung_087a1_AX

Beispiel für E_DEMUX_4, mit Plug and Socket



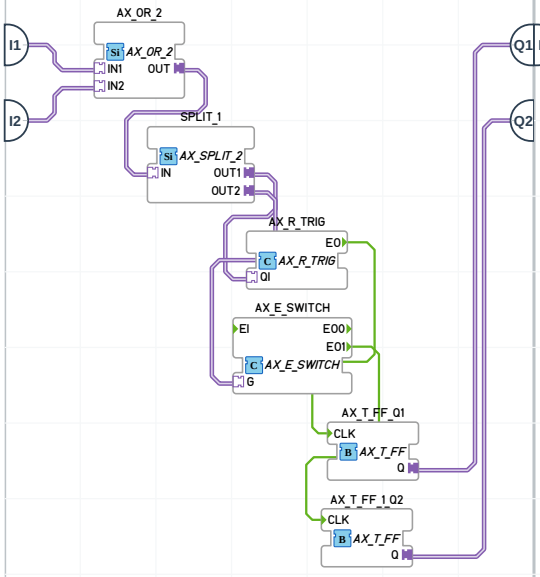
Uebung_087a2_AX

Beispiel für E_DEMUX_4, mit Plug and Socket



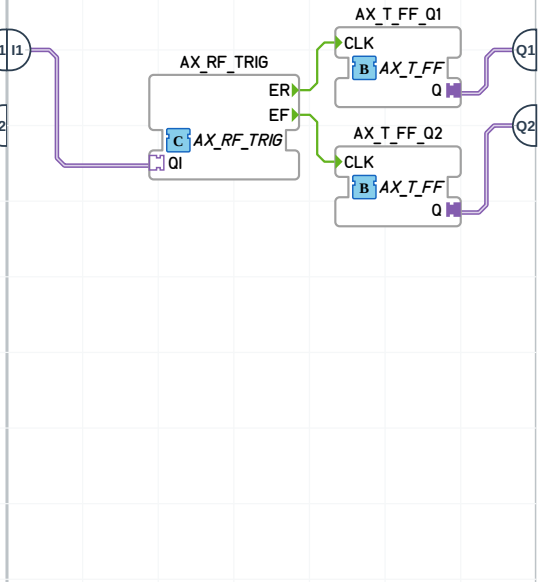
Uebung_088_AX

Beispiel für E_F_TRIG, mit Plug and Socket



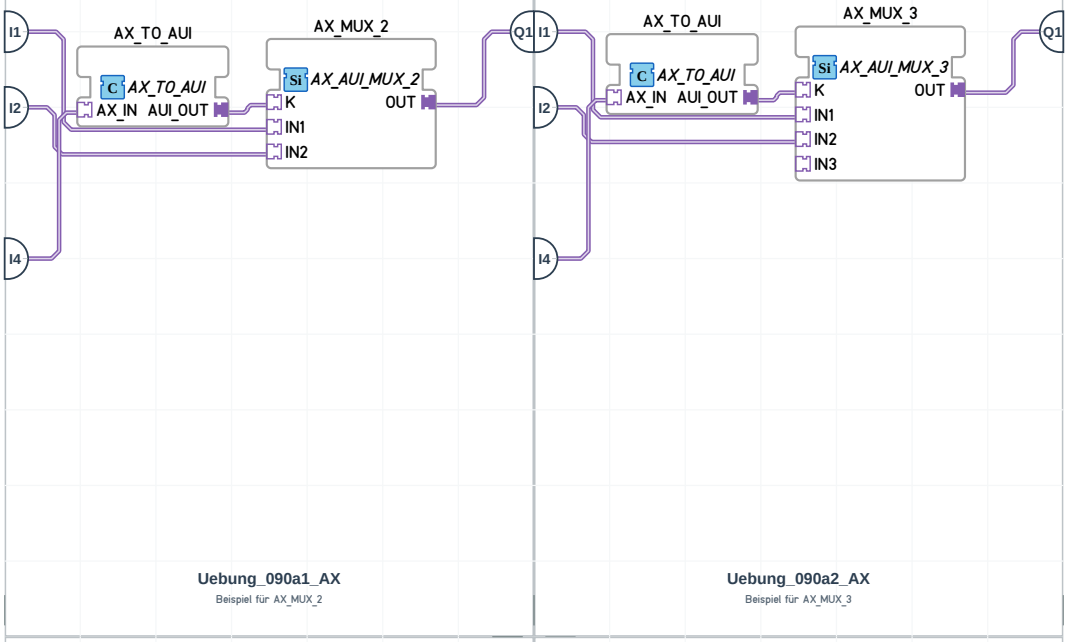
Uebung_089_AX

Beispiel für E_R_TRIG, mit Plug and Socket



Uebung_089a_AX

Beispiel für E_RF_TRIG, mit Plug and Socket

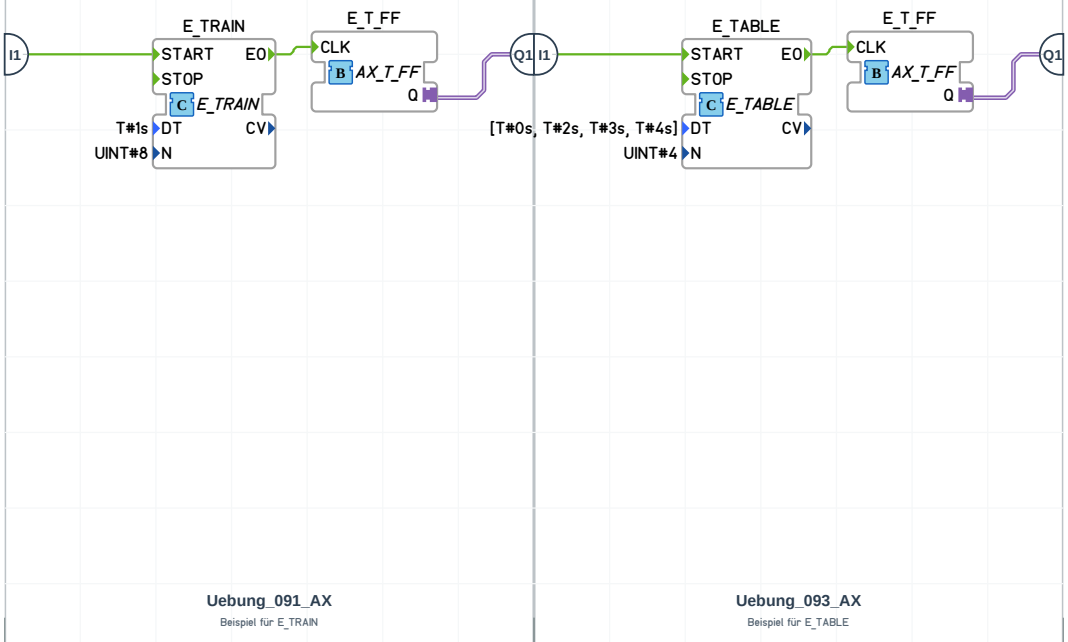


Uebung_090a1_AX

Beispiel für AX_MUX_2

Uebung_090a2_AX

Beispiel für AX_MUX_3

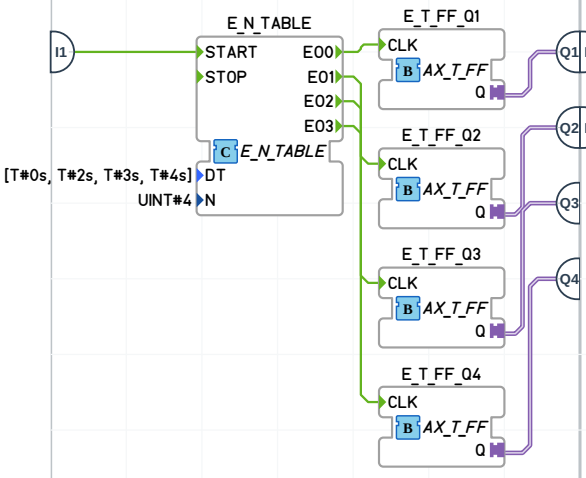


Uebung_091_AX

Beispiel für E_TRAIN

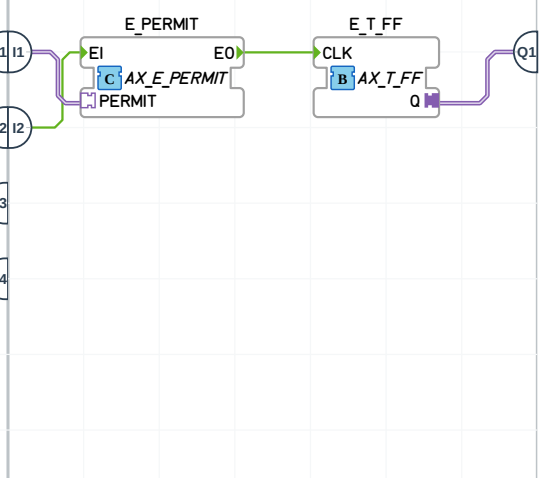
Uebung_093_AX

Beispiel für E_TABLE



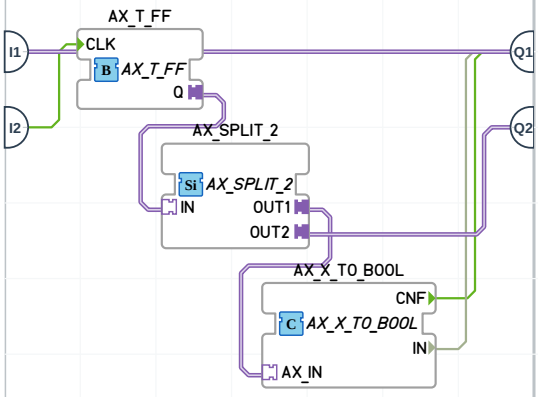
Uebung_093b_AX

Beispiel für E_N_TABLE



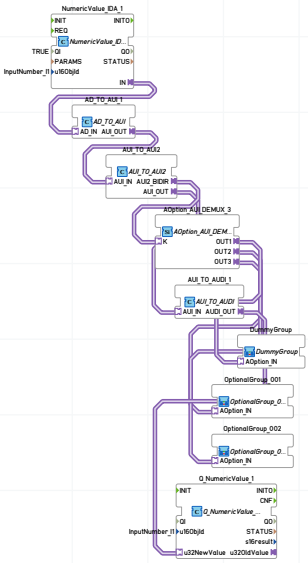
Uebung_094_AX

Beispiel für E_PERMIT



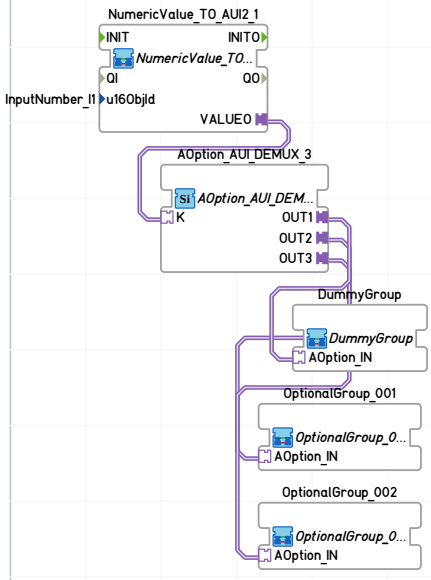
Uebung_094a_AX

Beispiel für QI anstelle Permit



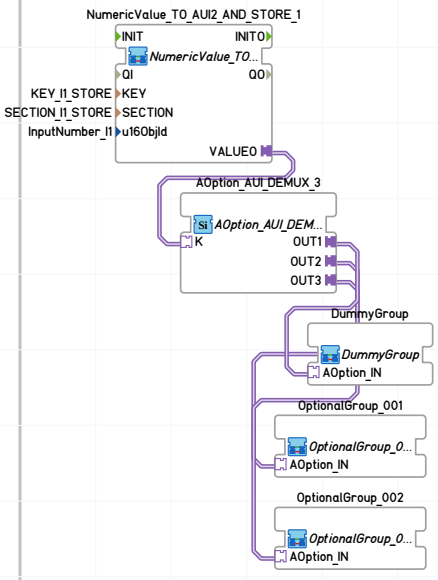
Uebung_094b2_AX

Wie Uebung_094b_AX, aber mit explizitem 'nichts ausgewaehlt'-Zustand: AOption_AUI_DEMUX_3 statt _2 – K=1 geht auf DummyGroup (neutraler, wirkungsloser



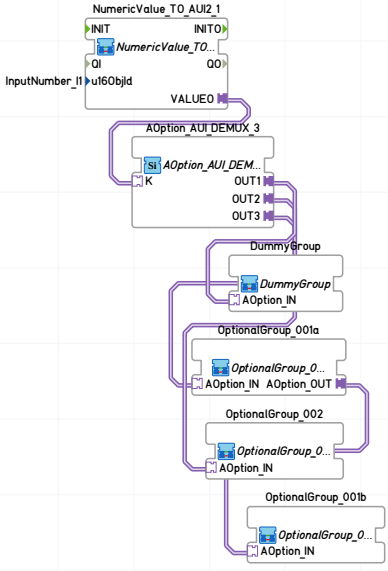
Uebung_094b3_AX

Wie Uebung_094b2_AX, aber mit MyLib::sys::NumericValue_TO_AUI2 als wiederverwendbarem Baustein statt der inline



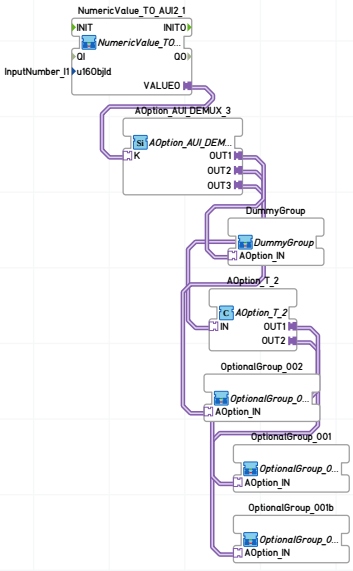
Uebung_094b3b_AX

Wie Uebung_094b3_AX, aber mit MyLib::sys::NumericValue_TO_AUI2_AND_STORE statt NumericValue_TO_AUI2: der Index (K) wird zusaetzlich



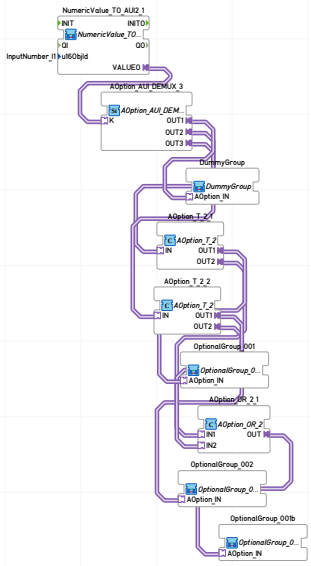
Uebung_094b3c_AX

Wie Uebung_094b3_AX, aber K=2 aktiviert zwei Optionsgruppen NACHEINANDER statt nur einer: OptionalGroup_001a wird vom DEMUX aktiviert und



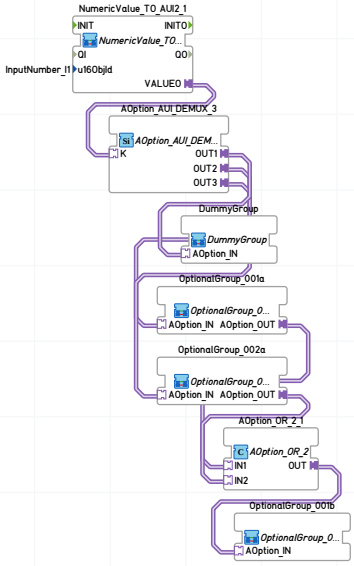
Uebung_094b3d_AX

Wie Uebung_094b3c_AX (K=2 aktiviert zwei Optionsgruppen NACHEINANDER), aber mit dem neuen generischen T-Stueck Option::splitting::AOption_T_2



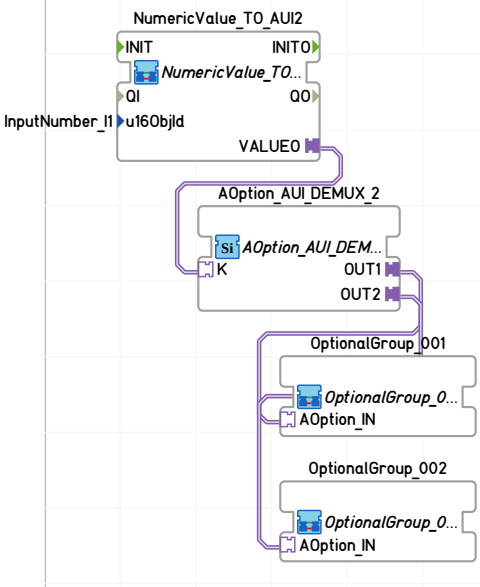
Uebung_094b3e_AX

Wie Uebung_094b3_AX, aber mit dritter Gruppe OptionalGroup_001b, die ueber AOption_T_2 (Abgriff hinter 001 und 002) und AOption_OR_2 aktiv ist,



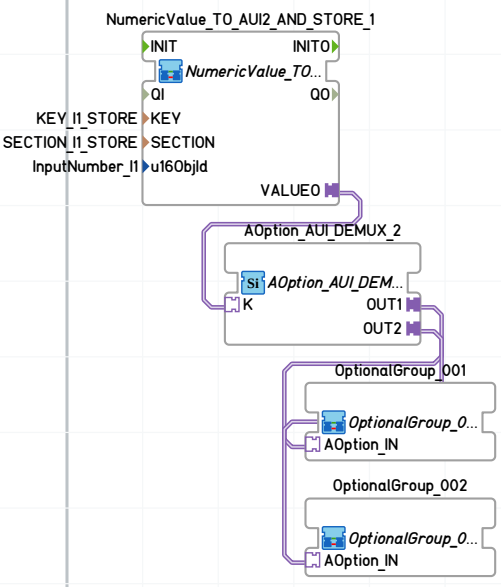
Uebung_094b3f_AX

Wie Uebung_094b3e_AX, aber ohne T-Stuecke: die Gruppen 1 und 2 sind OptionalGroup_001a bzw. OptionalGroup_002a (Muster mit AOption IN und



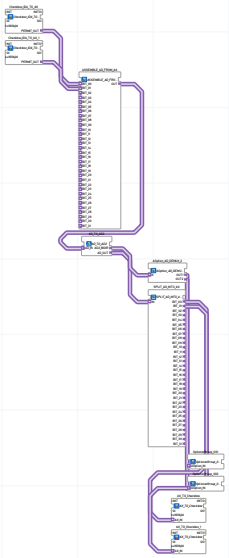
Uebung_094b4_AX

Wie Uebung_094b_AX, aber mit MyLib_AX::sys::NumericValue_TO_AUI2 als wiederverwendbarem Baustein statt der inline



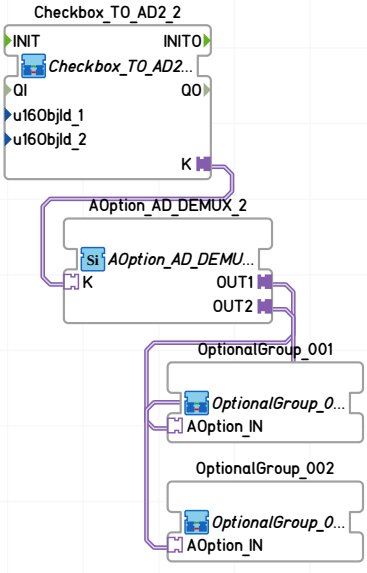
Uebung_094b4b_AX

Wie Uebung_094b_AX, aber mit MyLib::sys::NumericValue_TO_AUI2_AND_STORE als wiederverwendbarem Baustein: wie



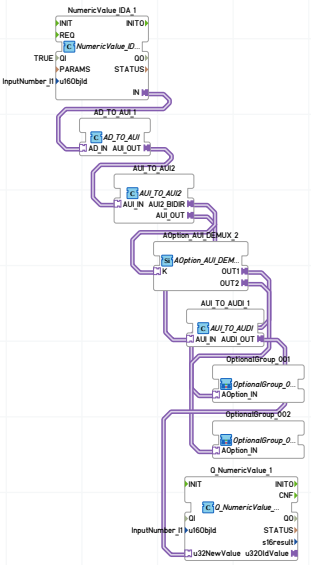
Uebung_094b5_AX

Wie Uebung_094b_AX, aber mit AOption_AD_DEMUX_2 (Bitmaske, Multi-Select) statt AOption_AUI_DEMUX_2 (Index, Single-Select): Bit0 schaltet



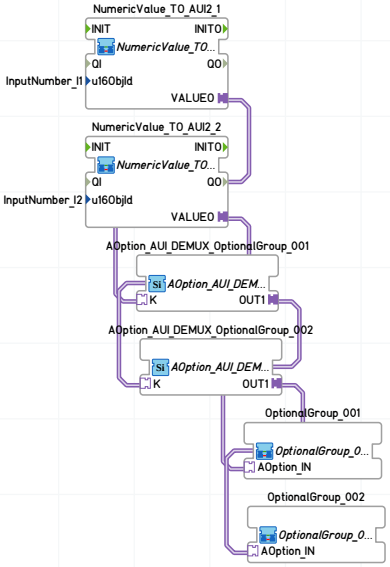
Uebung_094b5b_AX

Wie Uebung_094b5_AX, aber mit MyLib::sys::Checkbox_TO_AD2_2 als wiederverwendbarem Baustein statt der einzeln verdrahteten



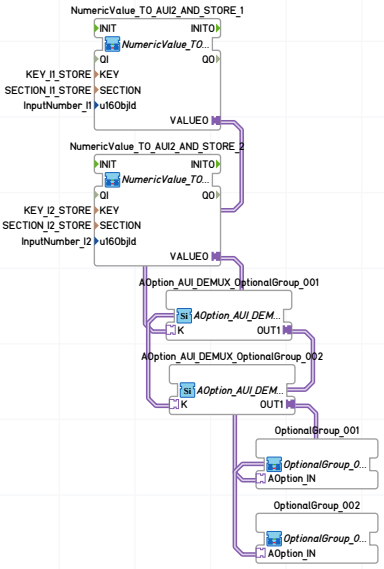
Uebung_094b_AX

Optionsgruppe per VT-Eingabe (Index) auswaehlen:
AD_TO_AUI -> AOption_AUI_DEMUX_2 schaltet genau eine
von zwei OptionalGroup-SubApps per AOption-Adapter



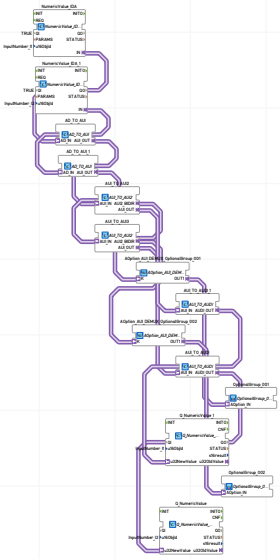
Uebung_094c2_AX

Wie Uebung_094c_AX, aber mit
MyLib::sys::NumericValue_TO_AUI2 als
wiederverwendbarem Baustein statt der 2x inline



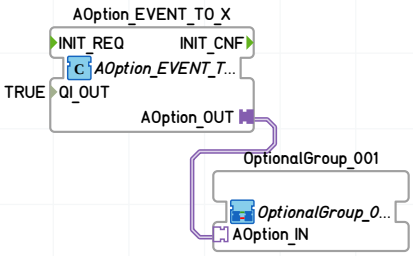
Uebung_094c2b_AX

Wie Uebung_094c2_AX, aber mit
MyLib::sys::NumericValue_TO_AUI2_AND_STORE statt
NumericValue_TO_AUI2: beide Indizes (K) werden



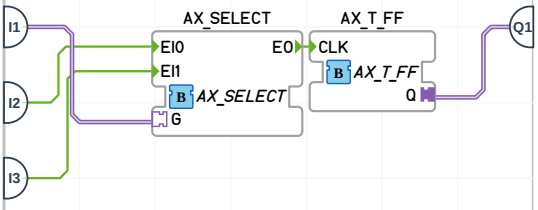
Uebung_094c_AX

Zwei unabhaengige VT-Dropdowns (InputNumber I1/I2)
schalten je eine eigene Optionsgruppe frei:
AD_TO_AUI -> AOption_AUI_DEMUX_1 ->



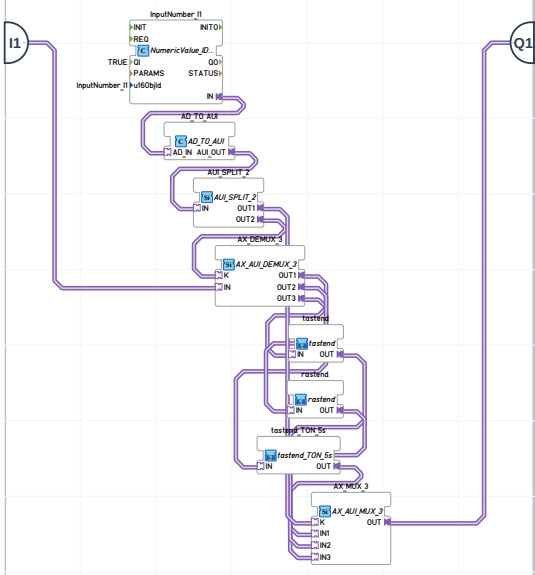
Uebung_094d_AX

OptionalGroup_001 OHNE DEMUX: die Gruppe ist immer
aktiv, kein K/Index noetig - E_RESTART (COLD/WARM)
loest einmalig INIT_REQ mit konstantem QI_OUT=TRUE



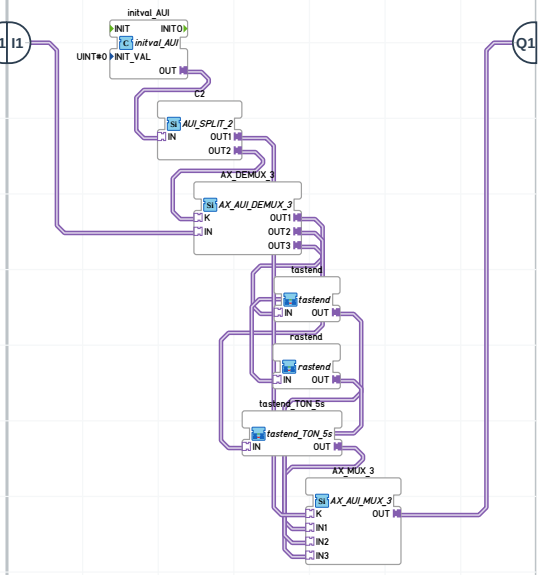
Uebung_095_AX

Beispiel für AX_SELECT



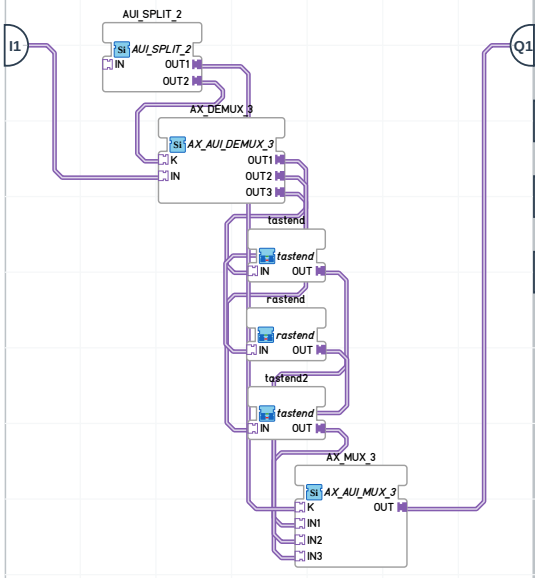
Uebung_103

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket und DEMUX



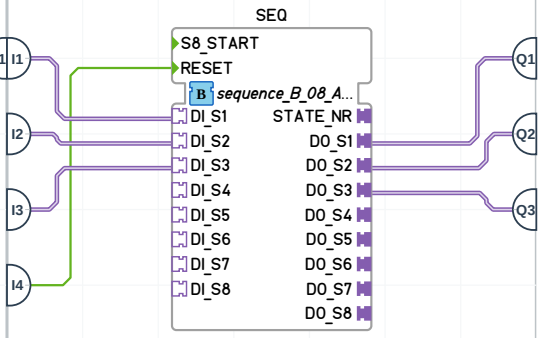
Uebung_103c

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket und DEMUX



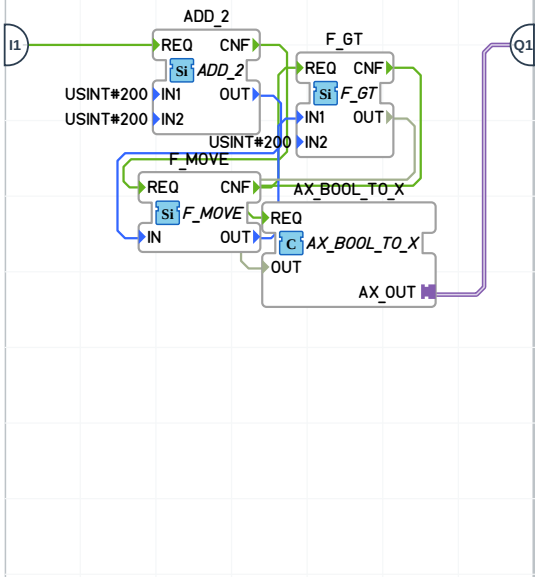
Uebung_103c2

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket und DEMUX



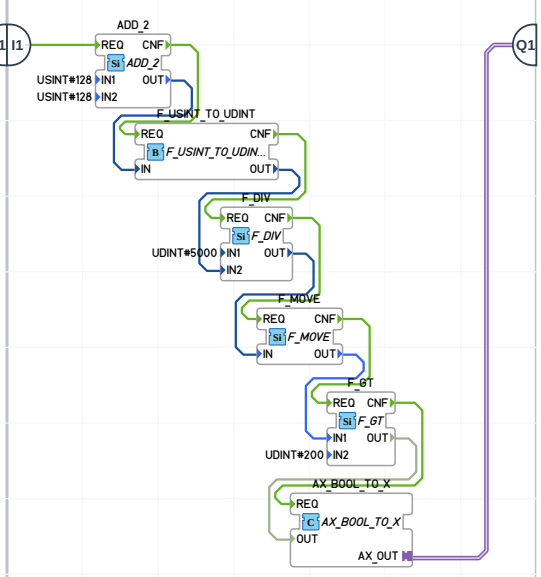
Uebung_104_AX

Zustandsbasierte Schrittkette mit Power-Fail Recovery (AX-Version)



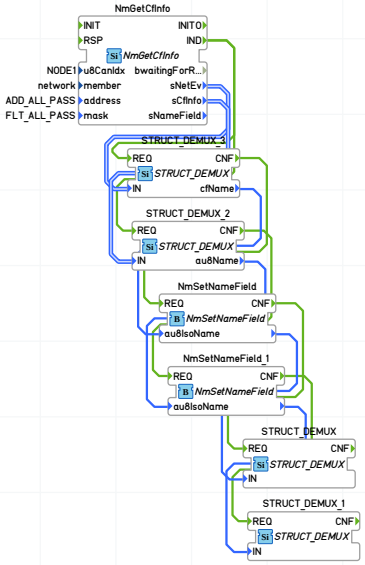
Uebung_110_AX

Übung zum Integer Überlauf



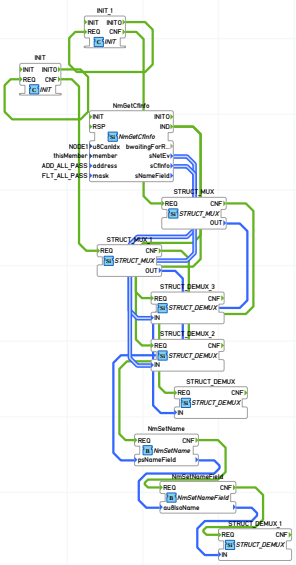
Uebung_111_AX

Übung zum Integer Überlauf



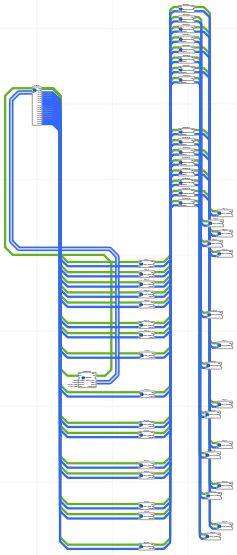
Uebung_120_AX

Übung zu ISOBUS Name



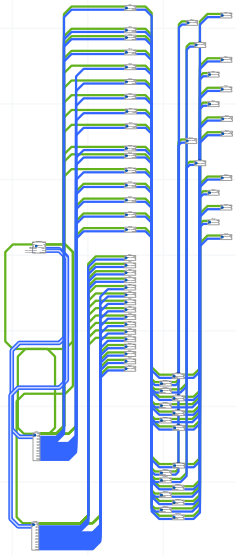
Uebung_121_AX

Übung zu ISOBUS Name



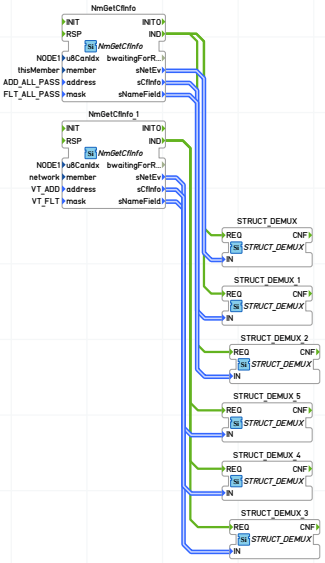
Uebung_122_AX

Übung zu ISOBUS Name



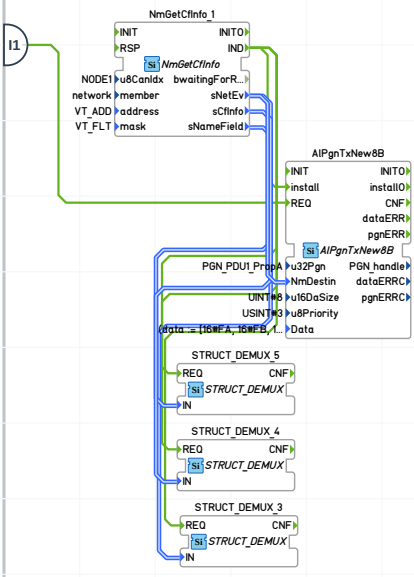
Uebung_122b_AX

Übung zu ISOBUS Name



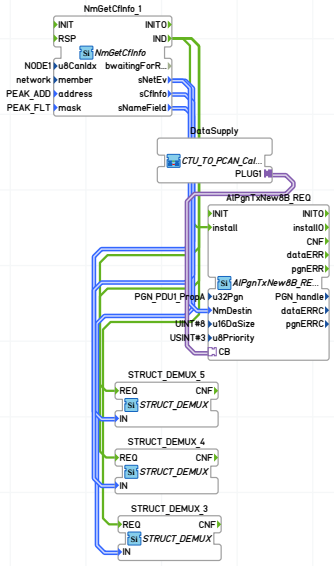
Uebung_123_AX

Übung zu ISOBUS Name



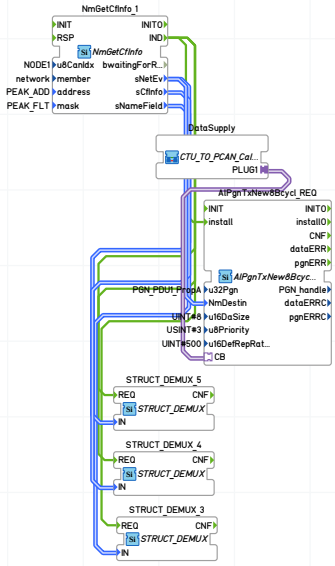
Uebung_124_AX

Übung zu ISOBUS Send Message



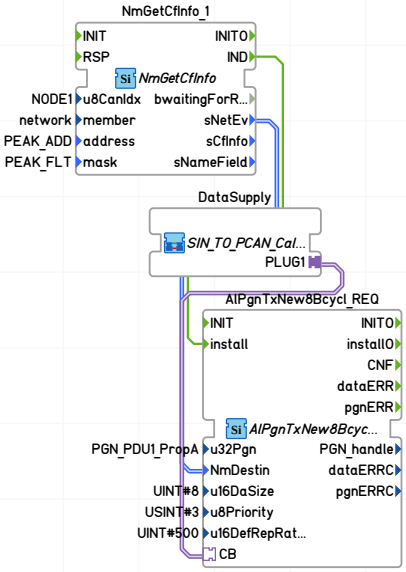
Uebung_125_AX

Übung zu ISOBUS Send Message on Request (mit CB)



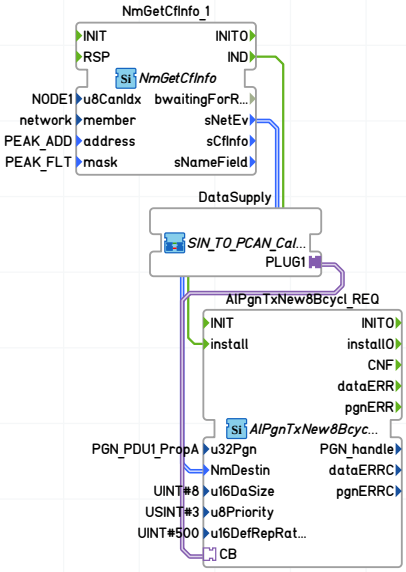
Uebung_126_AX

Übung zu ISOBUS Send Message Cyclic (mit CB)



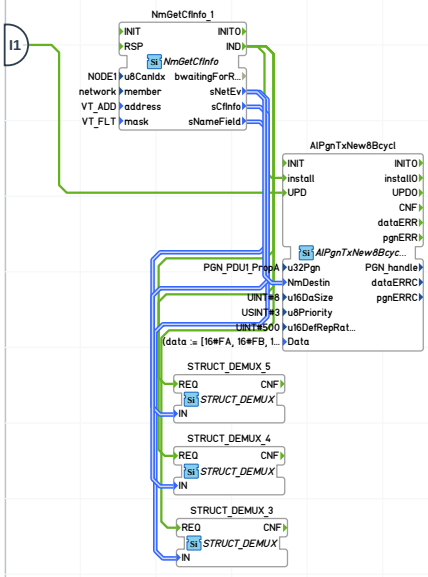
Uebung_126b2_AX

Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten



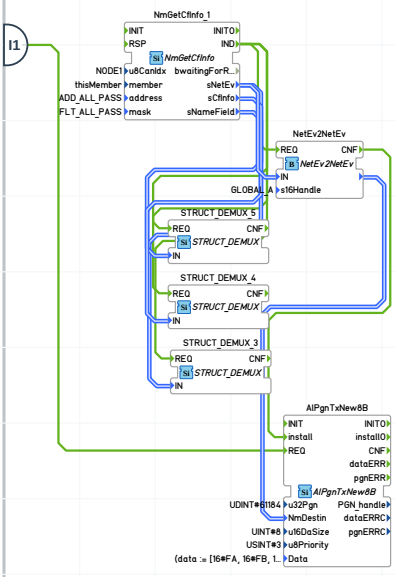
Uebung_126b_AX

Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten



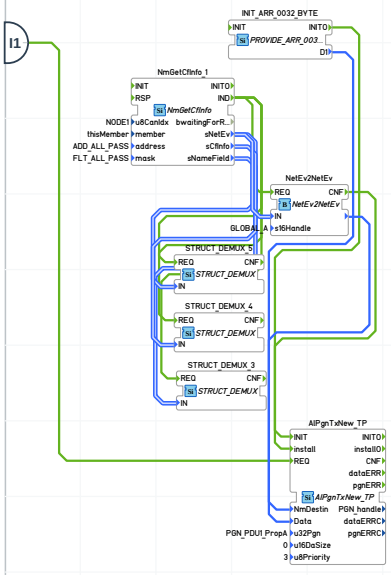
Uebung_127_AX

Übung zu ISOBUS Send Message Cyclic (ohne CB)



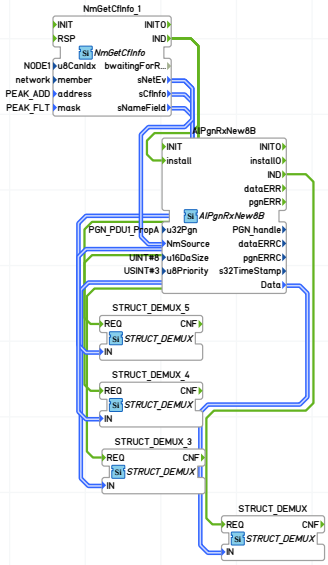
Uebung_128_AX

Übung zu ISOBUS Send Message GLOBAL



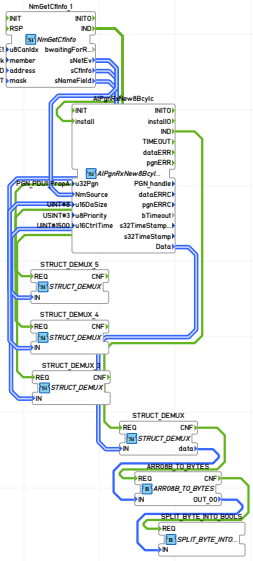
Uebung_128b_AX

Übung zu ISOBUS Send Message GLOBAL TP BAM



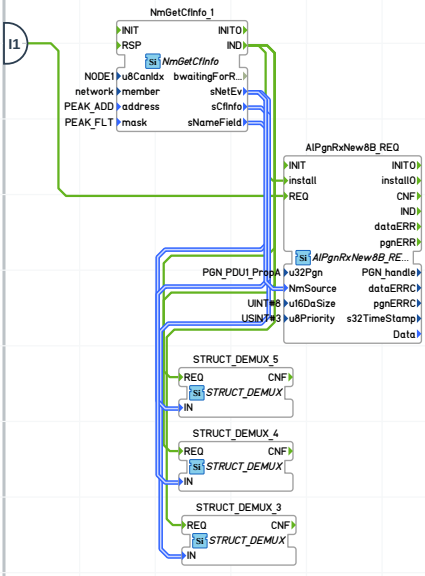
Uebung_130_AX

Übung zu ISOBUS Receive Message



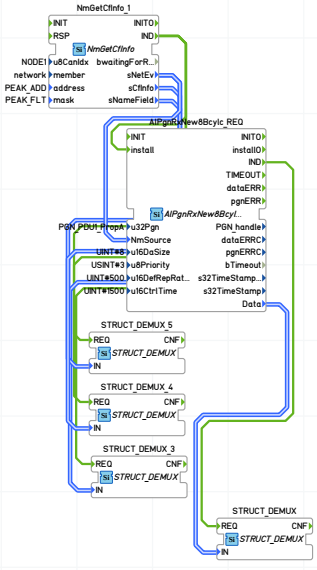
Uebung_131_AX

Übung zu ISOBUS Receive Message Cyclic



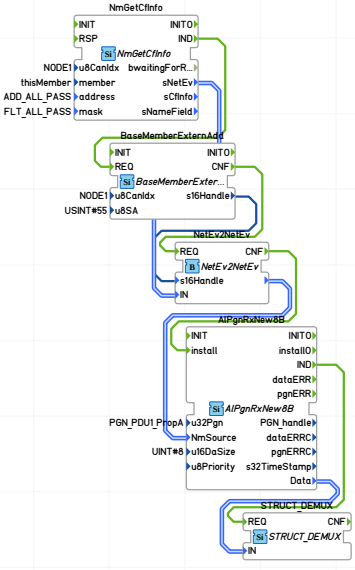
Uebung_132_AX

Übung zu ISOBUS Request Message



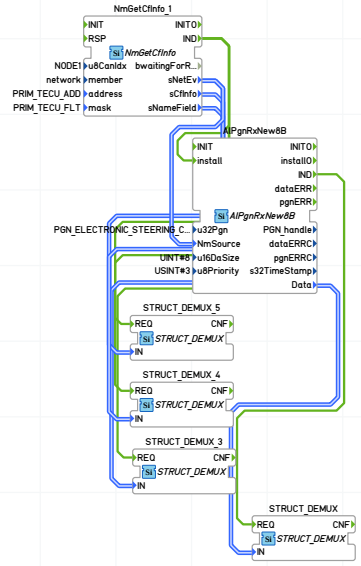
Uebung_133_AX

Übung zu ISOBUS Request Message Cyclic



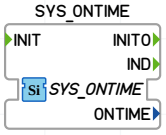
Uebung_134_AX

Übung zu ISOBUS Receive from Unclaimed Partner



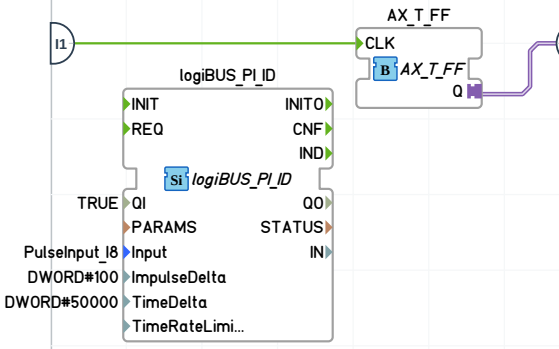
Uebung_135_AX

Übung zu ISOBUS Receive Message



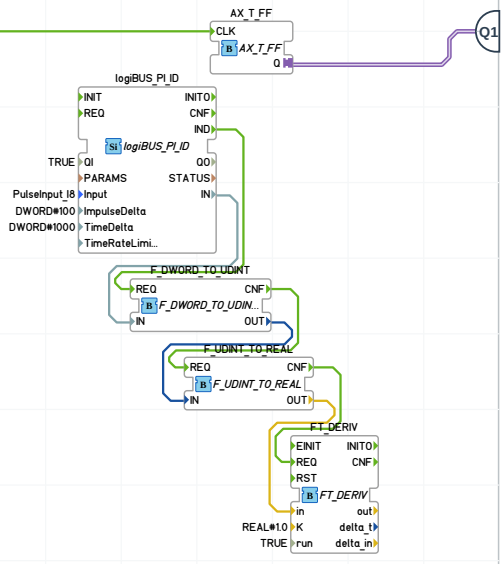
Uebung_140_AX

Übung zu SYS_ONTIME (Betriebsstundenzähler)



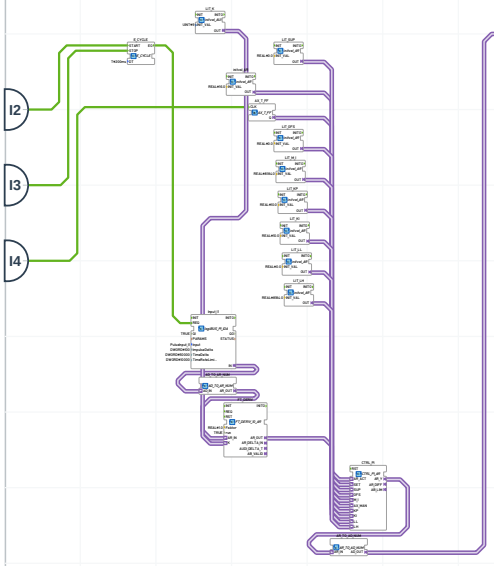
Uebung_150_AX

Impulszähler & DigitalInput_I1 auf DigitalOutput_Q1,
mit Plug and Socket



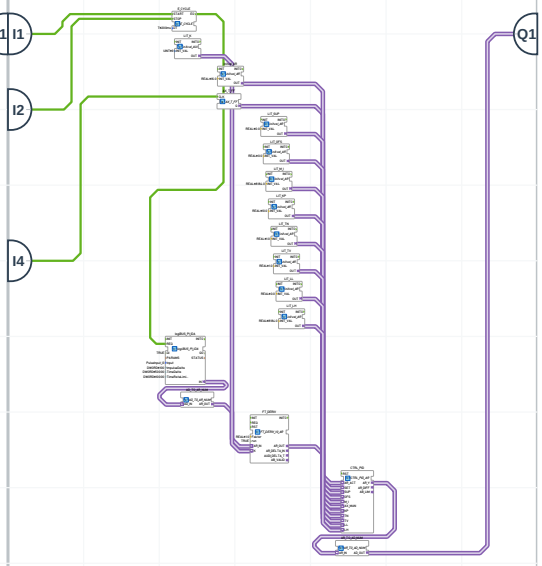
Uebung_151_AX

Impulszähler & DigitalInput_I1 auf DigitalOutput_Q1,
mit Plug and Socket



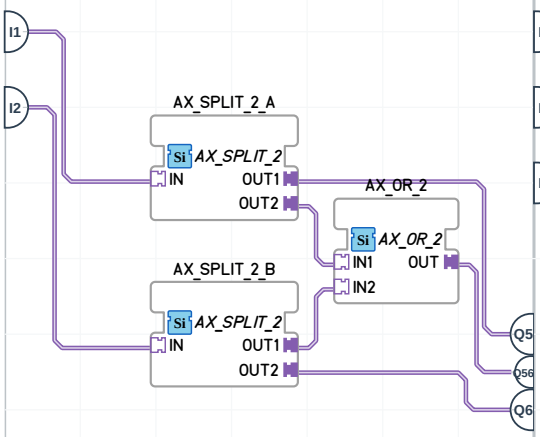
Uebung_152_AX

Impulszähler & PI Regler, mit Plug and Socket
(Adapter-Native)



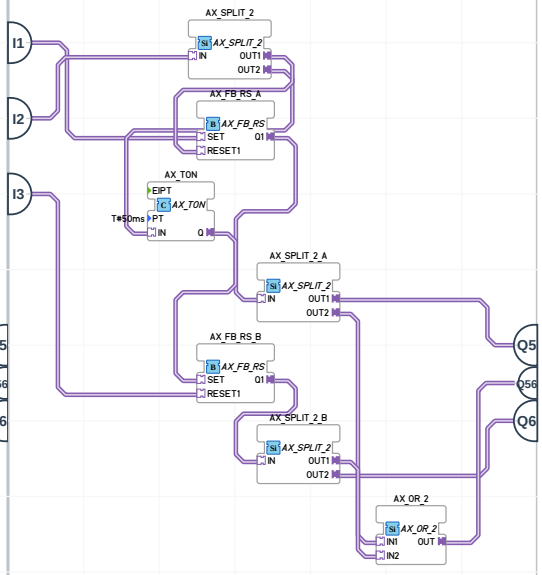
Uebung_153_AX

Impulszähler & PID Regler, mit Plug and Socket
(Adapter-Native)



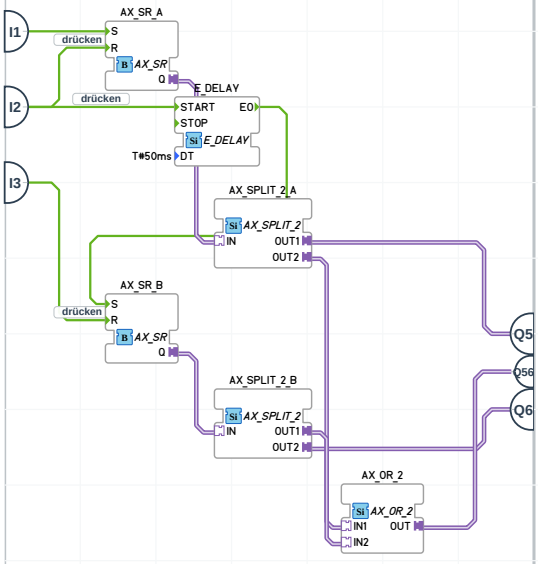
Uebung_160_AX

Motor Links/Rechtslauf



Uebung_160b2_AX

Motor Links/Rechtslauf Automatisierung IXA



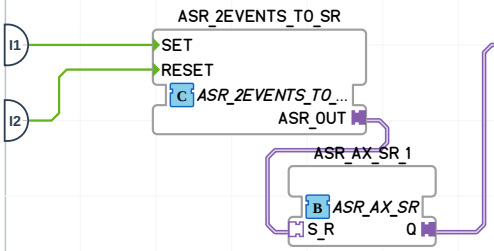
Uebung_160b_AX

Motor Links/Rechtslauf Automatisierung IE



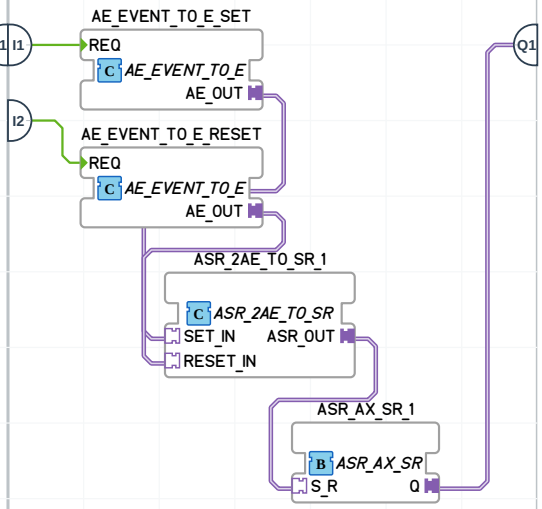
Uebung_170_AX

Exercise for AX_D_FF



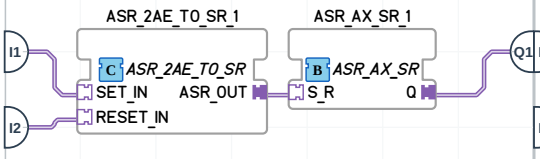
Uebung_171_ASR

Exercise for ASR_AX_SR



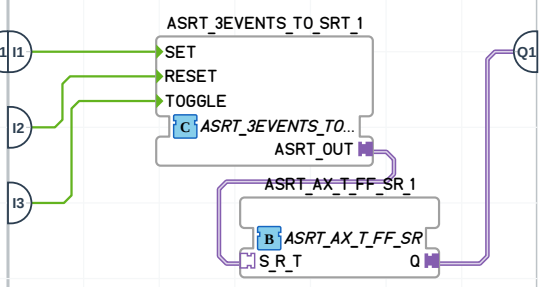
Uebung_171b_ASR

Exercise for ASR_AX_SR, ueber AE-Adapter statt rohe Events (ASR_2AE_TO_SR statt ASR_2EVENTS_TO_SR)



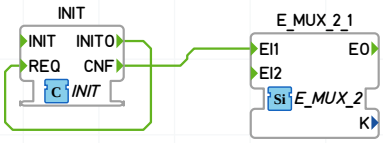
Uebung_171c_ASR

Exercise for ASR_AX_SR, ueber AE-Adapter mit logiBUS_IEA statt logiBUS_IE+AE_EVENT_TO_E



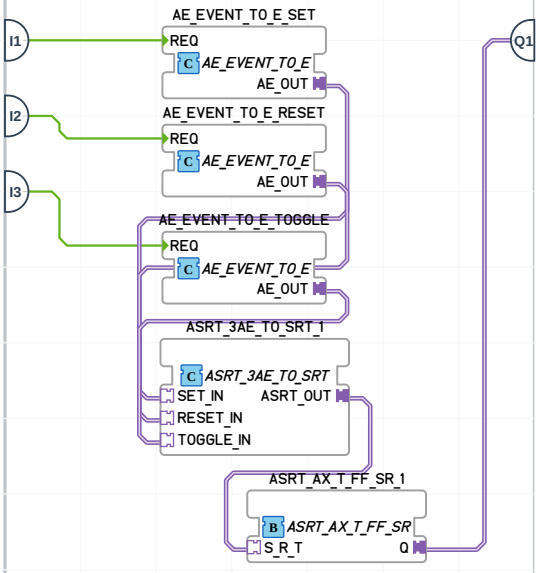
Uebung_172_ASRT

Exercise for ASRT_AX_T_FF_SR



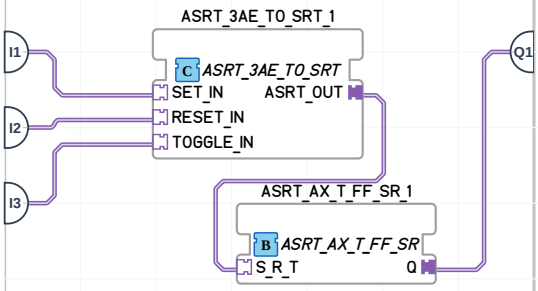
Uebung_172_AX

Exercise for E_MUX_2



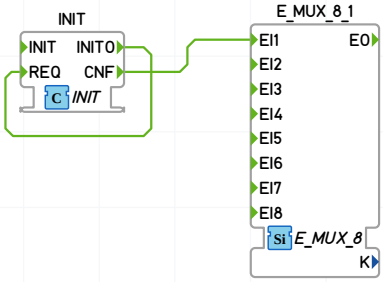
Uebung_172b_ASRT

Exercise for ASRT_AX_T_FF_SR, ueber AE-Adapter statt rohe Events (ASRT_3AE_TO_SRT statt ASRT_3EVENTS_TO_SRT)



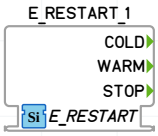
Uebung_172c_ASRT

Exercise for ASRT_AX_T_FF_SR, ueber AE-Adapter mit logiBUS_IEA statt logiBUS_IE+AE_EVENT_TO_E



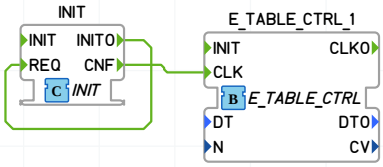
Uebung_173_AX

Exercise for E_MUX_8



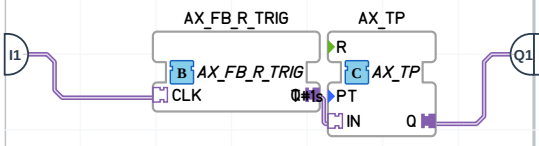
Uebung_174_AX

Exercise for E_RESTART



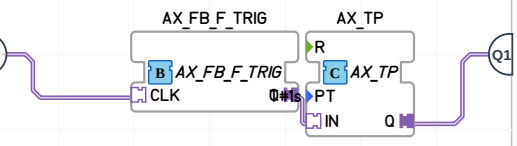
Uebung_175_AX

Exercise for E_TABLE_CTRL



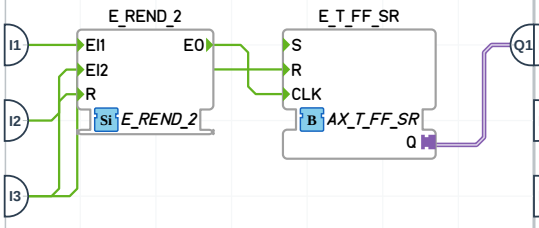
Uebung_177_AX

Beispiel für AX_FB_R_TRIG (Rising Edge Detection)



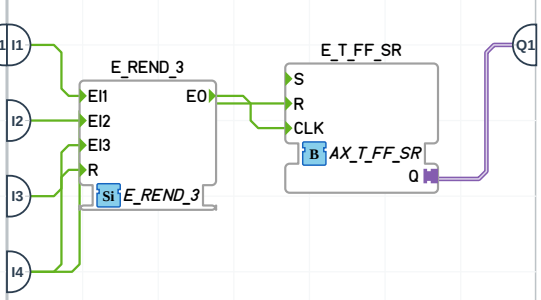
Uebung_178_AX

Beispiel für AX_FB_F_TRIG (Falling Edge Detection)



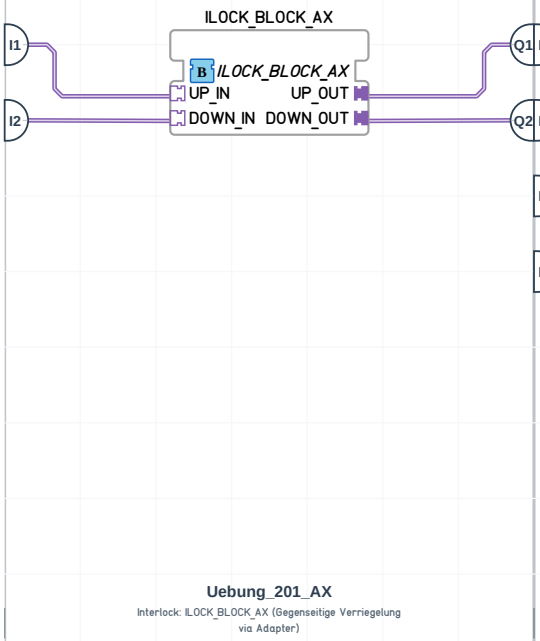
Uebung_179_AX

Beispiel für E_REND_2 (Rendezvous von zwei Events)



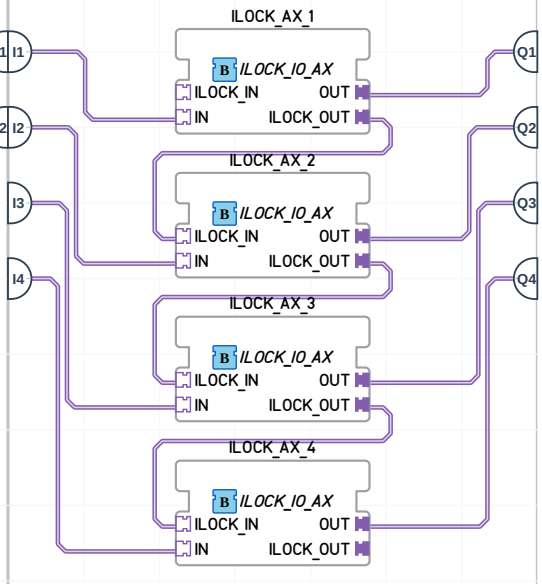
Uebung_180_AX

Beispiel für E_REND_3 (Rendezvous von drei Events)



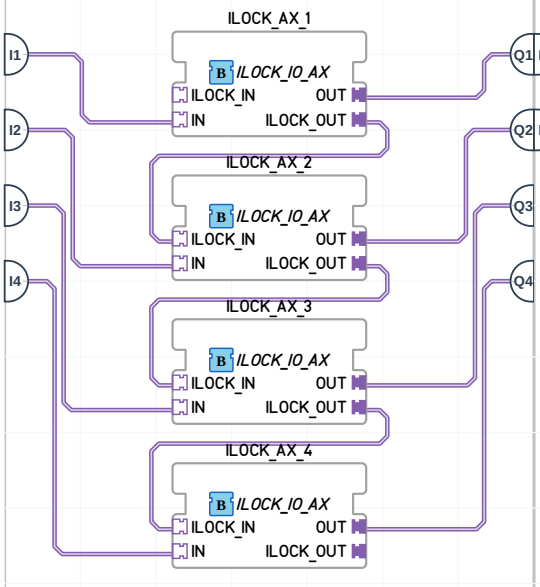
Uebung_201_AX

Interlock: ILOCK_BLOCK_AX (Gegenseitige Verriegelung via Adapter)



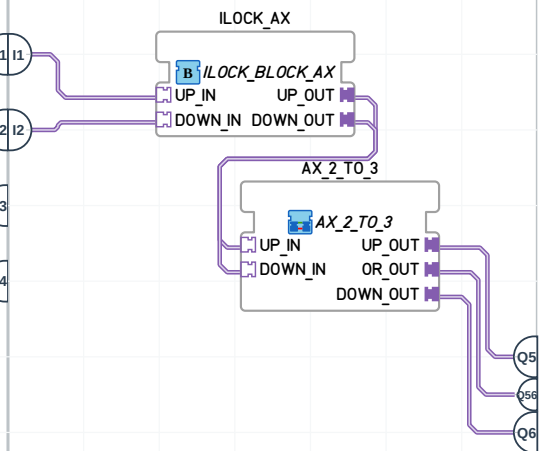
Uebung_201_Interlock_AX

Interlock advanced exercise (AX)



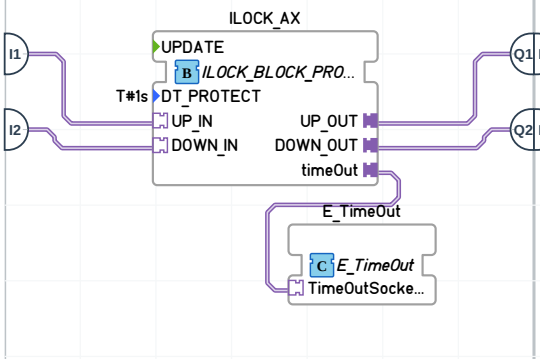
Uebung_201_Interlock_BOOL_AX

Interlock basic exercise (BOOL)



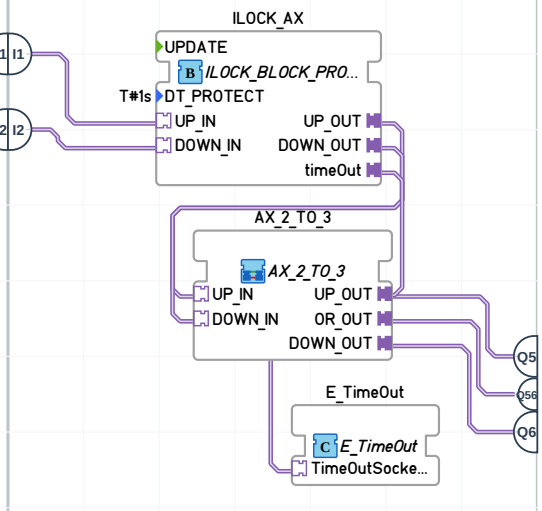
Uebung_201b_AX

Interlock: ILOCK_BLOCK_AX (Motor Rechts/Linkslauf via Adapter)



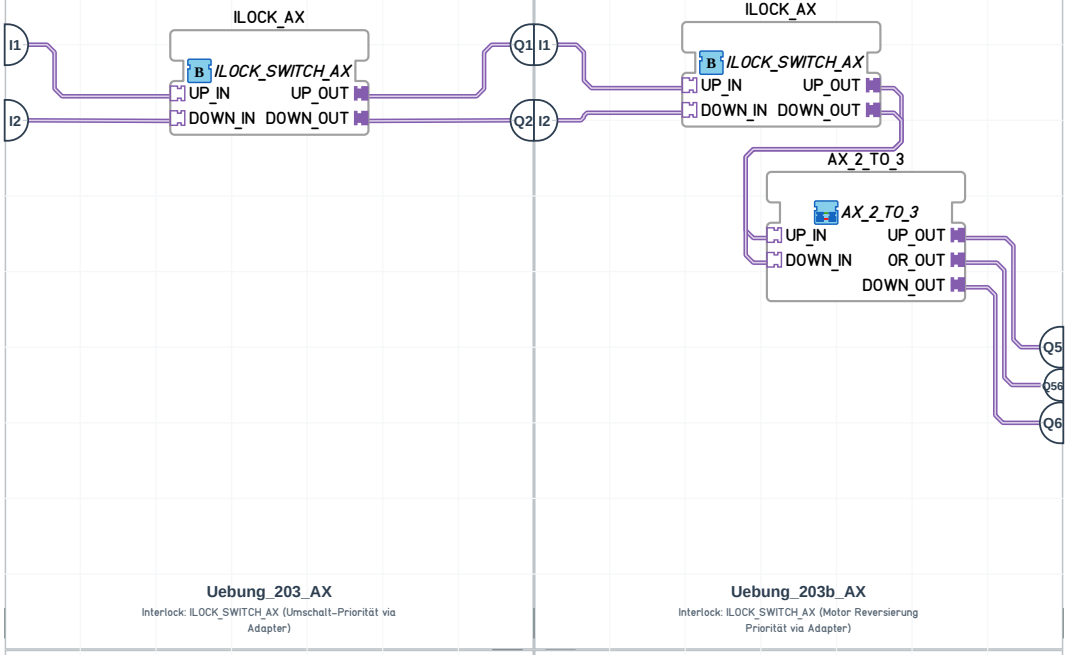
Uebung_202_AX

Interlock: ILOCK_BLOCK_PROTECT_AX (Verriegelung mit Schutzzeit via Adapter)



Uebung_202b_AX

Interlock: ILOCK_BLOCK_PROTECT_AX (Motor Reversierung mit Schutzzeit via Adapter)

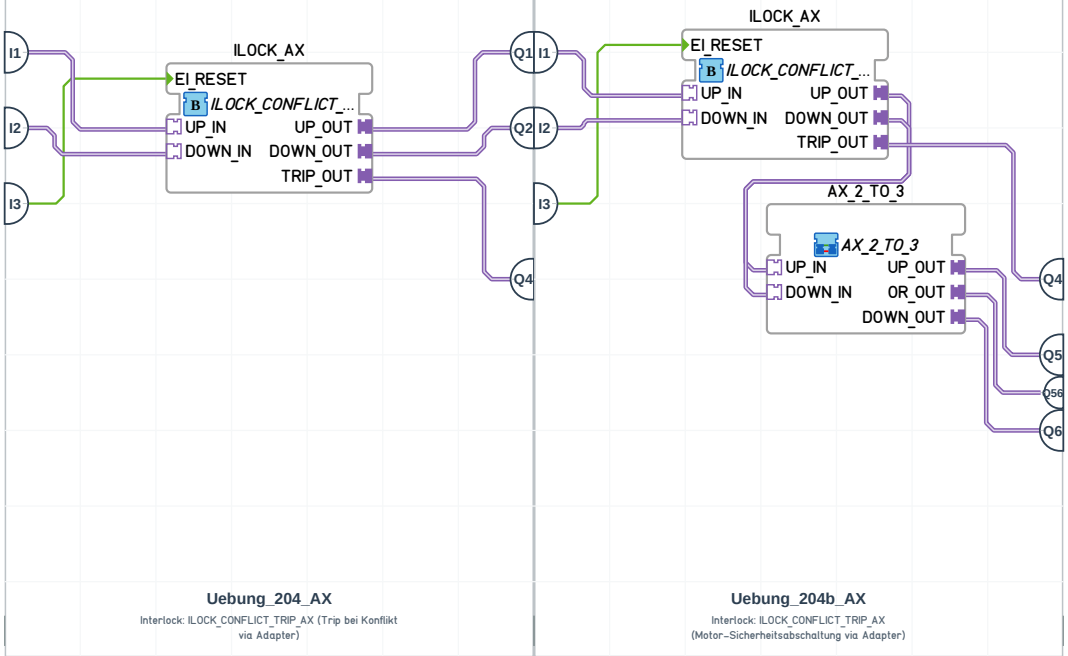


Uebung_203_AX

Interlock: ILOCK_SWITCH_AX (Umschalt-Priorität via Adapter)

Uebung_203b_AX

Interlock: ILOCK_SWITCH_AX (Motor Reversierung Priorität via Adapter)

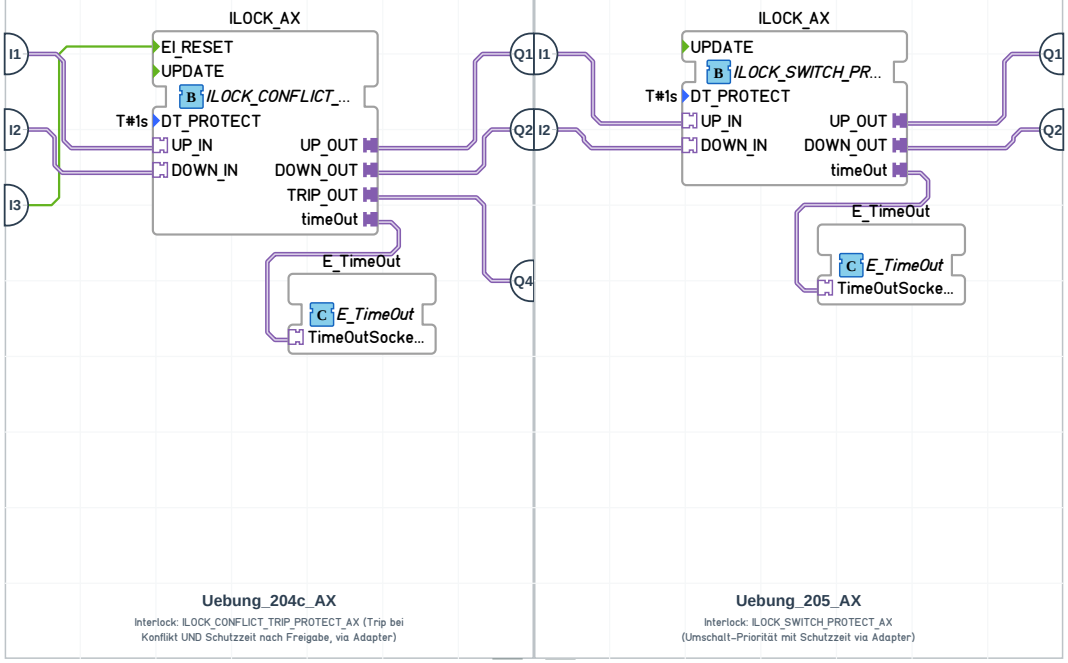


Uebung_204_AX

Interlock: ILOCK_CONFLICT_TRIP_AX (Trip bei Konflikt via Adapter)

Uebung_204b_AX

Interlock: ILOCK_CONFLICT_TRIP_AX (Motor-Sicherheitsabschaltung via Adapter)

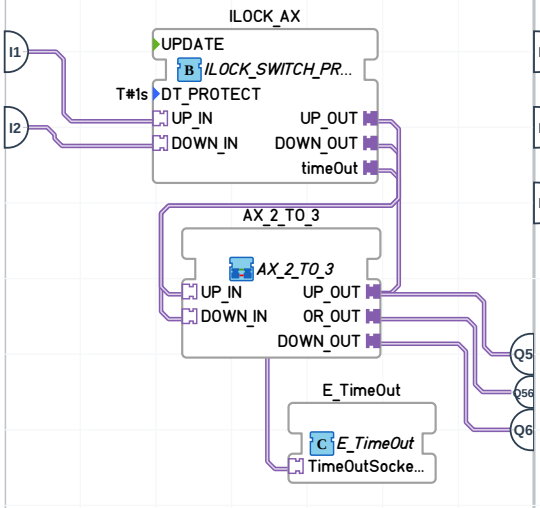


Uebung_204c_AX

Interlock: ILOCK_CONFLICT_TRIP_PROTECT_AX (Trip bei Konflikt UND Schutzzeit nach Freigabe, via Adapter)

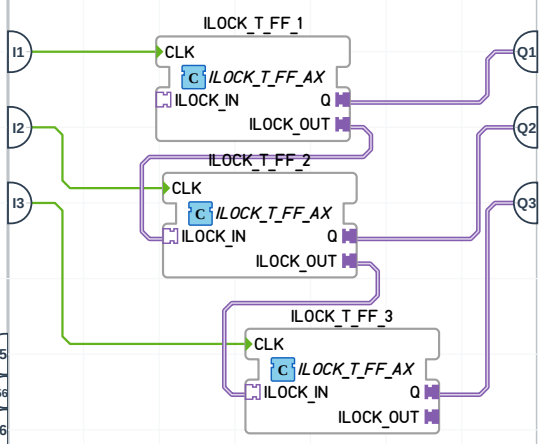
Uebung_205_AX

Interlock: ILOCK_SWITCH_PROTECT_AX (Umschalt-Priorität mit Schutzzeit via Adapter)



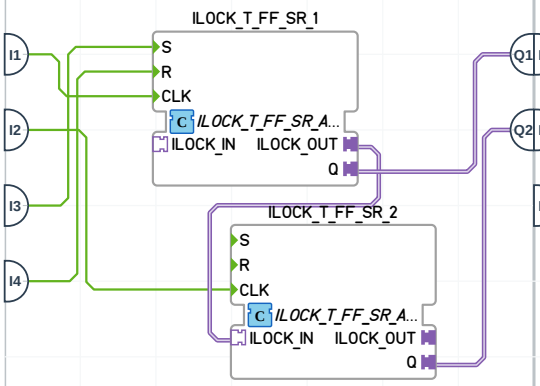
Uebung_205b_AX

Interlock: ILOCK_SWITCH_PROTECT_AX (Motor Reversierung mit Schutzzeit via Adapter)



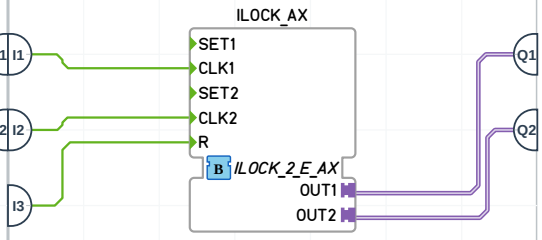
Uebung_206_AX

Interlock: ILOCK_T_FF_AX (Drei gegenseitig verriegelte Toggle-Flip-Flops via Adapter-Kette)



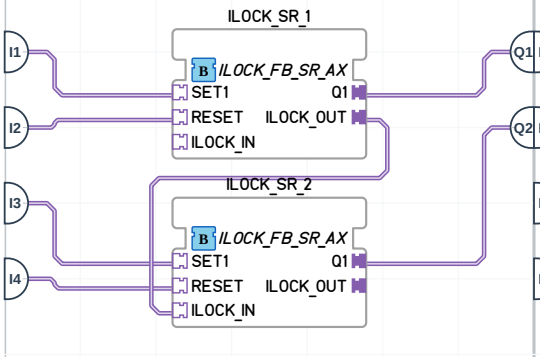
Uebung_206b_AX

Interlock: ILOCK_T_FF_SR_AX (2 gegenseitig verriegelte Toggle-Flip-Flops via Adapter-Kette, zusaetzlich direktes Set/Reset auf FF1)



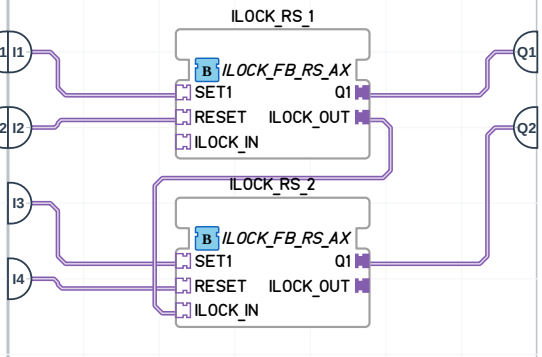
Uebung_207_AX

Interlock: ILOCK_2_E_AX (Event-gesteuertes bistabiles Relais mit gegenseitiger Verriegelung via Adapter)



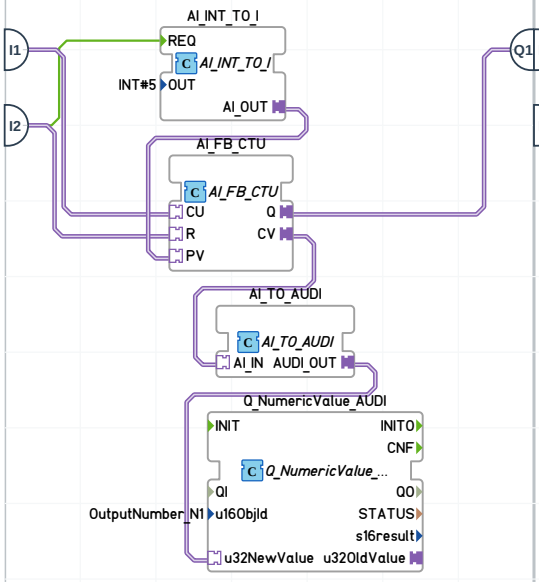
Uebung_208_AX

Interlock: ILOCK_FB_SR_AX (Zwei gegenseitig verriegelte Set-Dominante Latches via AX/AX2-Adapter)



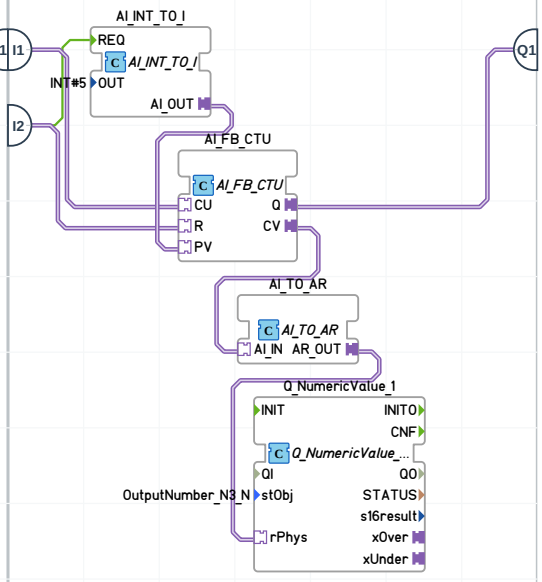
Uebung_209_AX

Interlock: ILOCK_FB_RS_AX (Zwei gegenseitig verriegelte Reset-Dominante Latches via AX/AX2-Adapter)



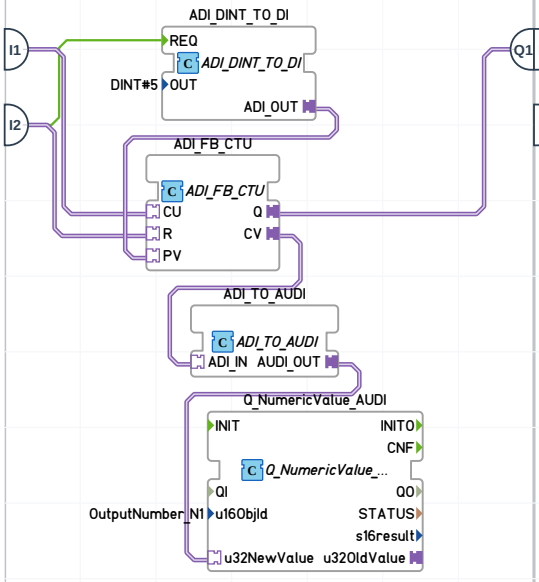
Uebung_210_AI

Standard IEC 61131-3 AI_FB_CTU (Adapter Version, Vorwärtszähler, INT) mit Terminal-Ausgabe



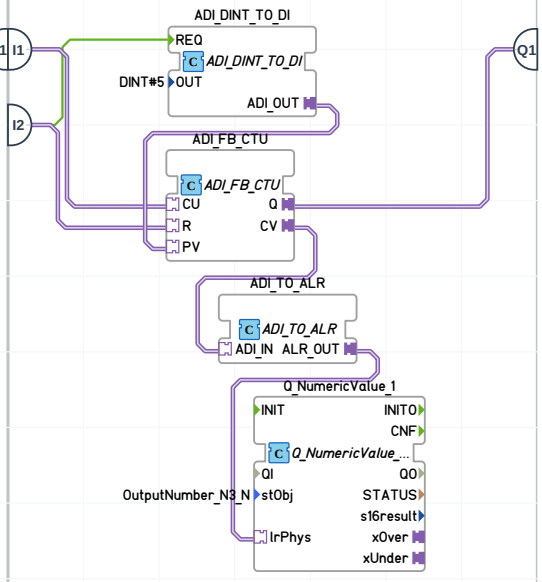
Uebung_210b_AR

Standard IEC 61131-3 AI_FB_CTU (Adapter Version, Vorwärtszähler, INT) mit Terminal-Ausgabe (PHYS)



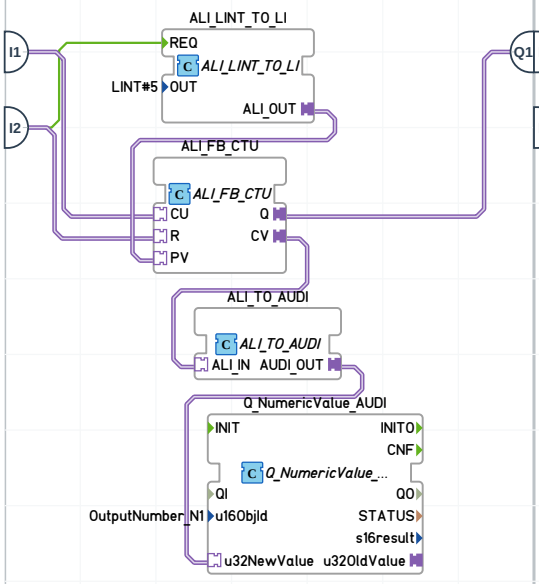
Uebung_211_ADI

Standard IEC 61131-3 ADI_FB_CTU (Adapter Version, Vorwärtszähler, DINT) mit Terminal-Ausgabe



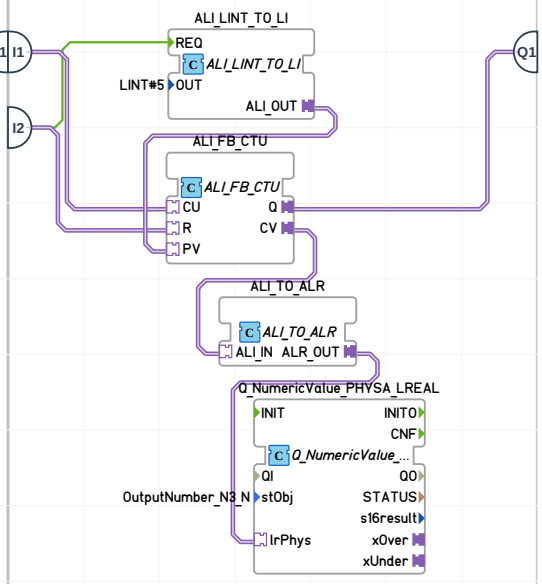
Uebung_211b_ALR

Standard IEC 61131-3 ADI_FB_CTU (Adapter Version, Vorwärtszähler, DINT) mit Terminal-Ausgabe (PHYS)



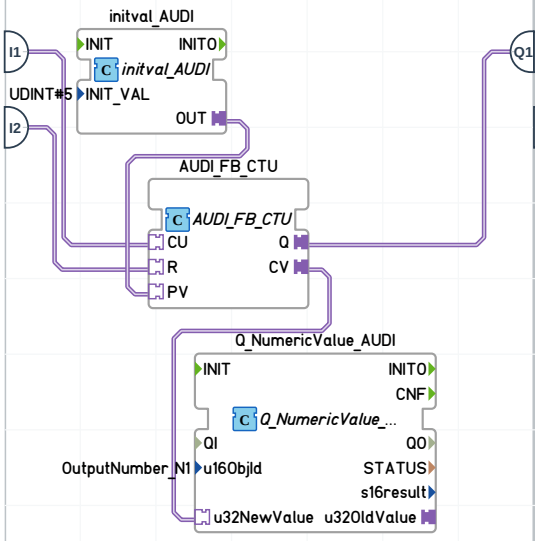
Uebung_212_ALI

Standard IEC 61131-3 ALI_FB_CTU (Adapter Version, Vorwärtszähler, LINT) mit Terminal-Ausgabe



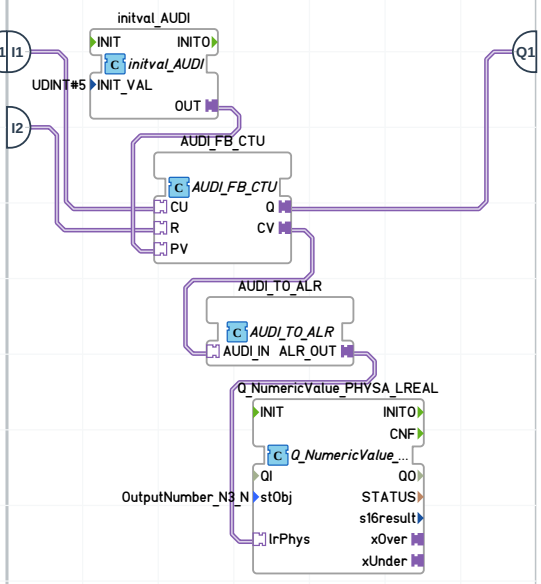
Uebung_212b_ALR

Standard IEC 61131-3 ALI_FB_CTU (Adapter Version, Vorwärtszähler, LINT) mit Terminal-Ausgabe (PHYS)



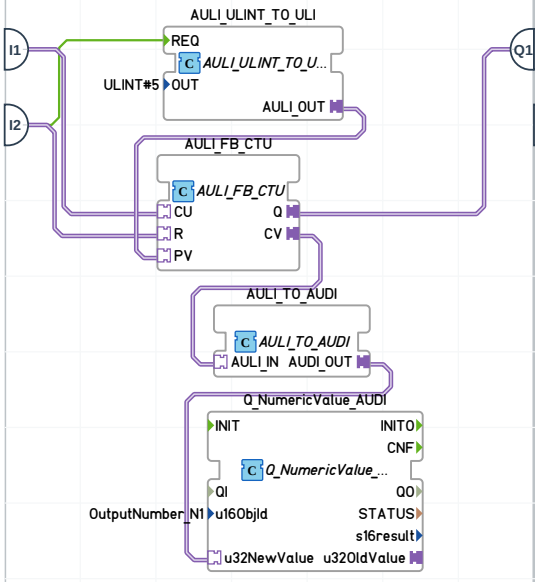
Uebung_213_AUDI

Standard IEC 61131-3 AUDI_FB_CTU (Adapter Version, Vorwärtszähler, UDINT) mit Terminal-Ausgabe



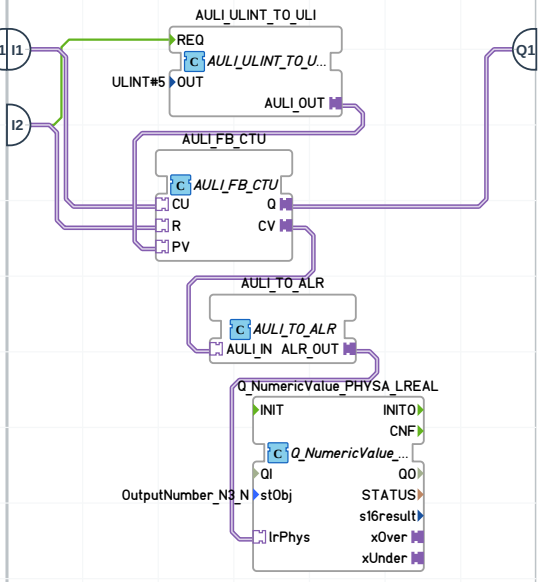
Uebung_213b_ALR

Standard IEC 61131-3 AUDI_FB_CTU (Adapter Version, Vorwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA_LREAL)



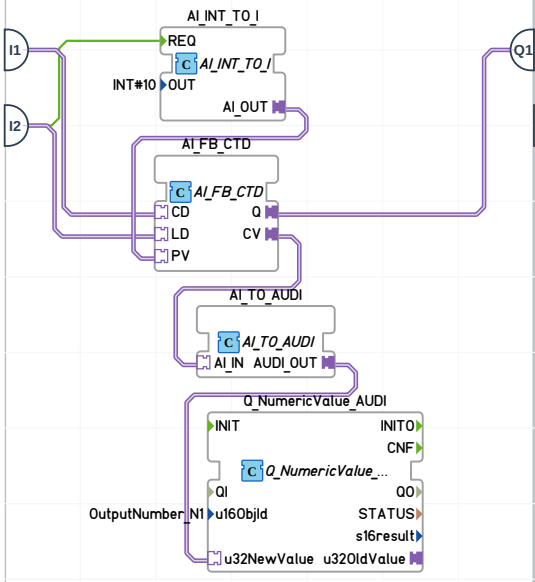
Uebung_214_AULI

Standard IEC 61131-3 AULI_FB_CTU (Adapter Version, Vorwärtszähler, ULINT) mit Terminal-Ausgabe



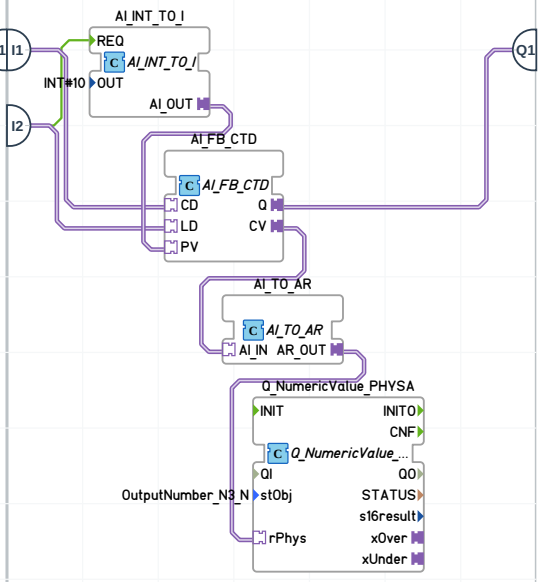
Uebung_214b_ALR

Standard IEC 61131-3 AULI_FB_CTU (Adapter Version, Vorwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA_LREAL)



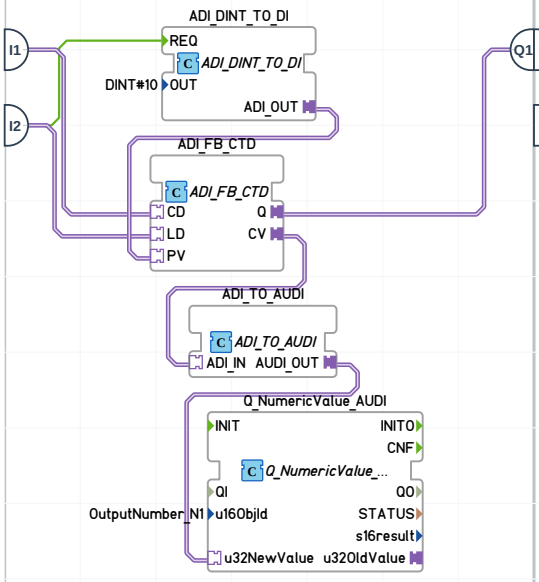
Uebung_215_AI

Standard IEC 61131-3 AI_FB_CTD (Adapter Version, Rückwärtszähler, INT) mit Terminal-Ausgabe



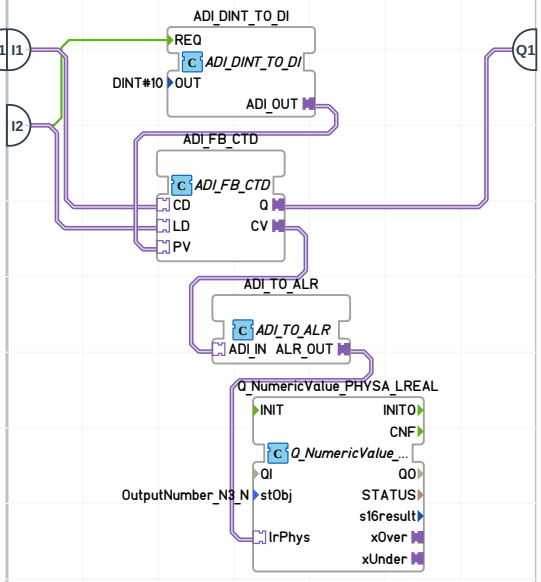
Uebung_215b_AR

Standard IEC 61131-3 AI_FB_CTD (Adapter Version, INT) mit Terminal-Ausgabe (PHYS)



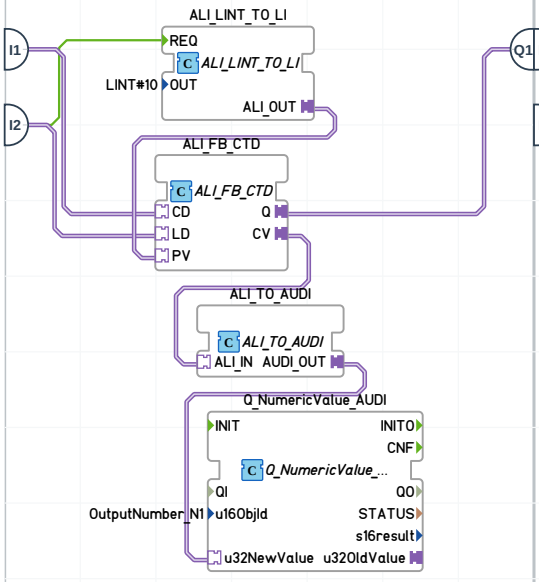
Uebung_216_ADI

Standard IEC 61131-3 ADI_FB_CTD (Adapter Version, Rückwärtszähler, DINT) mit Terminal-Ausgabe



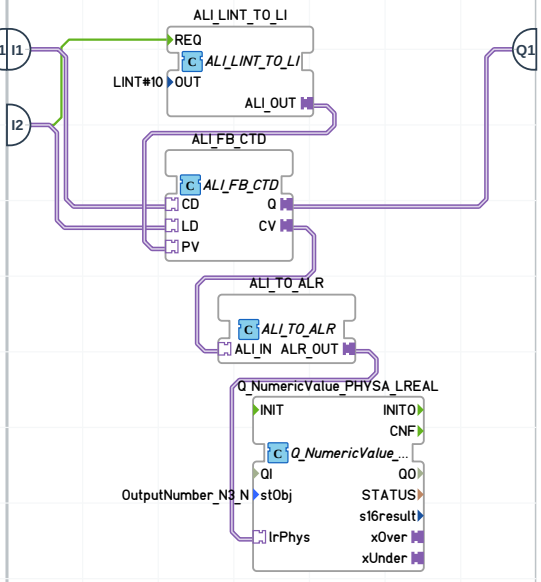
Uebung_216b_ALR

Standard IEC 61131-3 ADI_FB_CTD (Adapter Version, DINT) mit Terminal-Ausgabe (PHYS)



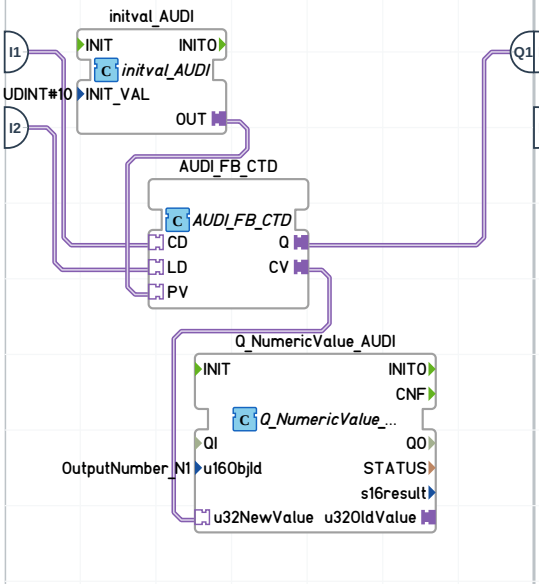
Uebung_217_ALI

Standard IEC 61131-3 ALI_FB_CTD (Adapter Version, Rückwärtszähler, LINT) mit Terminal-Ausgabe



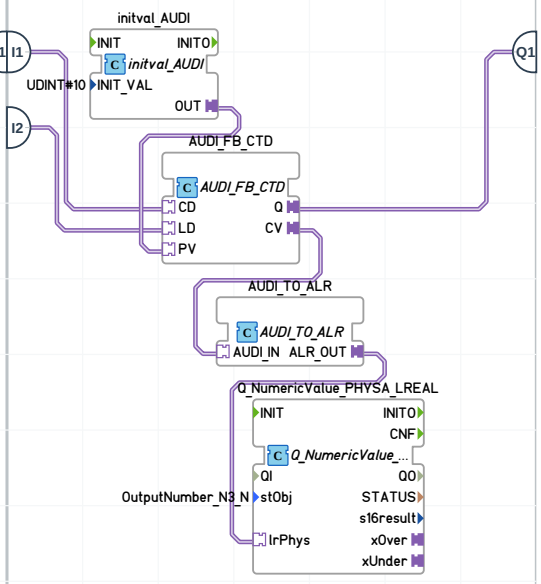
Uebung_217b_ALR

Standard IEC 61131-3 ALI_FB_CTD (Adapter Version, LINT) mit Terminal-Ausgabe (PHYSA_LREAL)



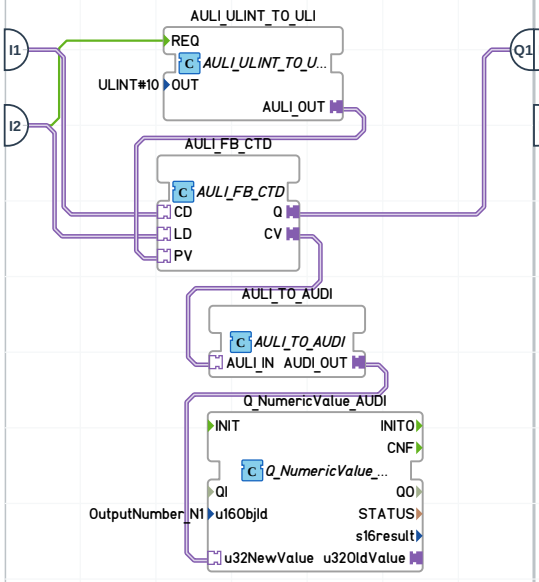
Uebung_218_AUDI

Standard IEC 61131-3 AUDI_FB_CTD (Adapter Version, Rückwärtszähler, UDINT) mit Terminal-Ausgabe



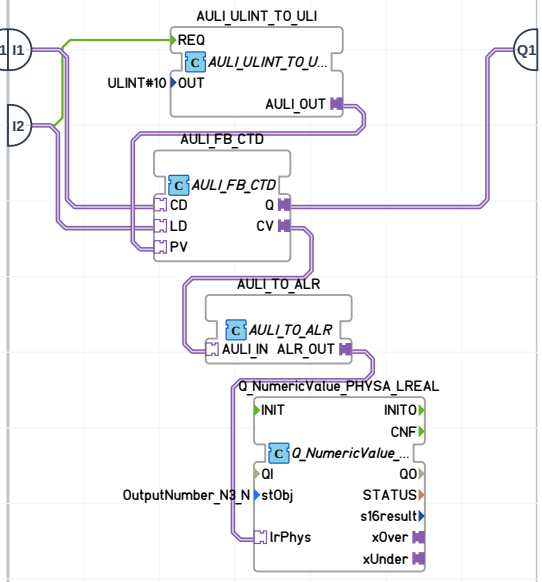
Uebung_218b_ALR

Standard IEC 61131-3 AUDI_FB_CTD (Adapter Version, Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA_LREAL)



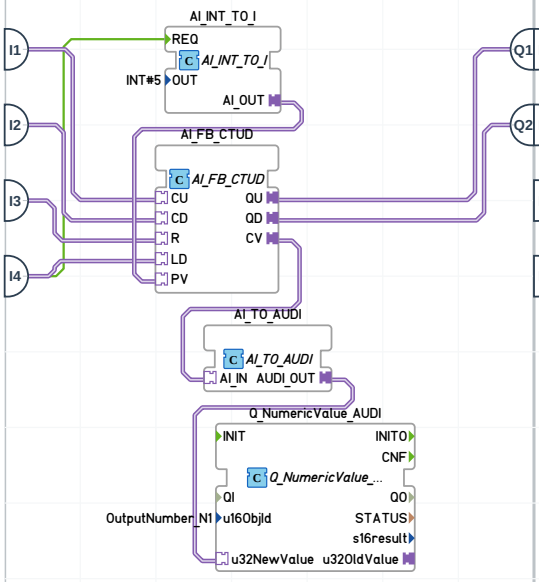
Uebung_219_AULI

Standard IEC 61131-3 AULI_FB_CTD (Adapter Version, Rückwärtszähler, ULINT) mit Terminal-Ausgabe



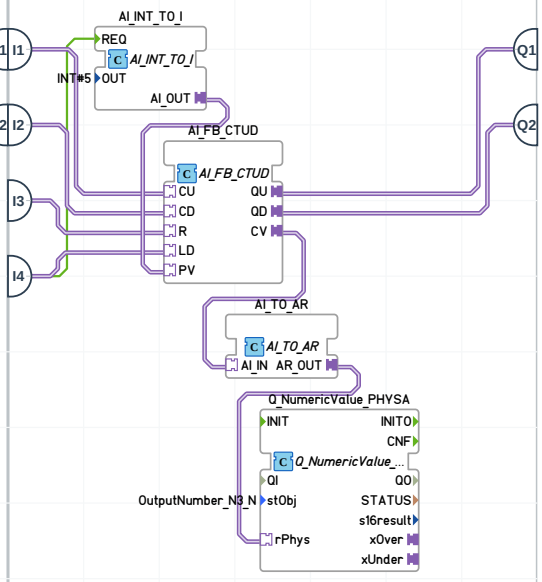
Uebung_219b_ALR

Standard IEC 61131-3 AULI_FB_CTD (Adapter Version, Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA_LREAL)



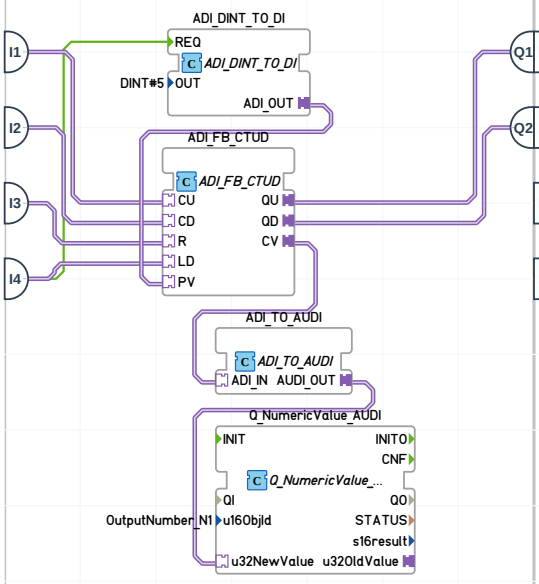
Uebung_220_AI

Standard IEC 61131-3 AI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, INT) mit Terminal-Ausgabe



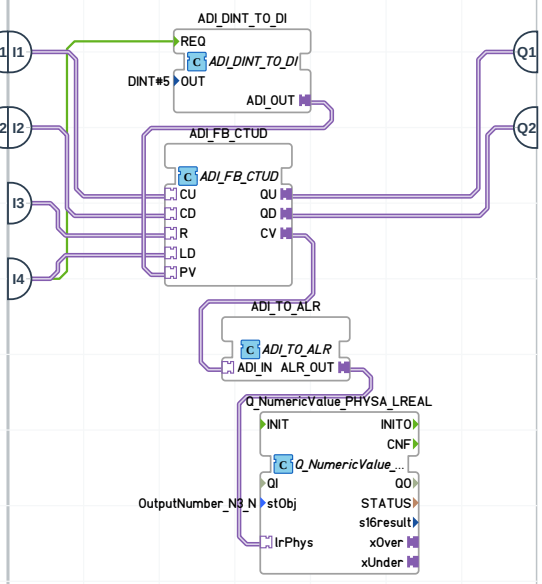
Uebung_220b_AR

Standard IEC 61131-3 AI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, INT) mit Terminal-Ausgabe (PHYS)



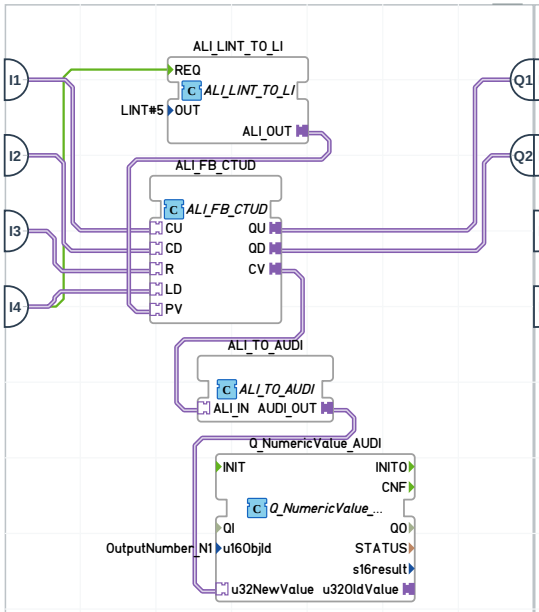
Uebung_221_ADI

Standard IEC 61131-3 ADI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, DINT) mit Terminal-Ausgabe



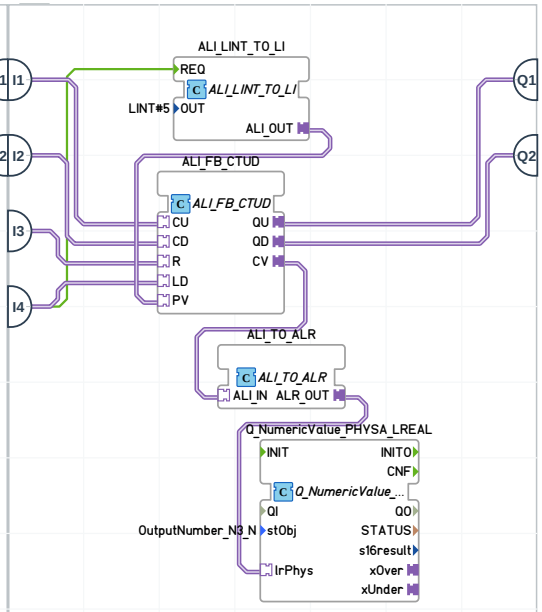
Uebung_221b_ALR

Standard IEC 61131-3 ADI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, DINT) mit Terminal-Ausgabe (PHYS)



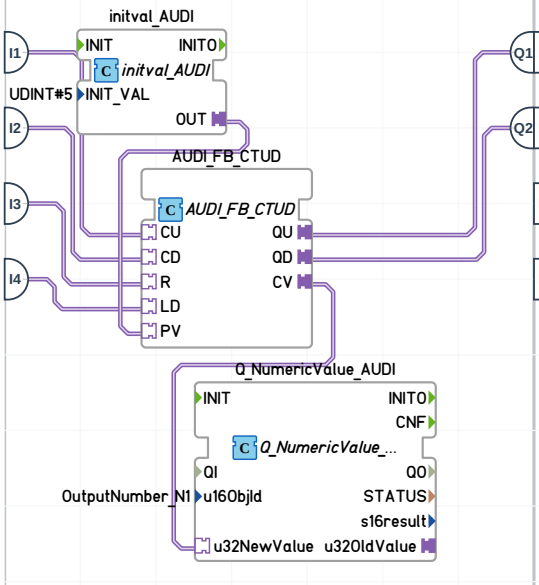
Uebung_222_ALI

Standard IEC 61131-3 ALI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, LINT) mit Terminal-Ausgabe



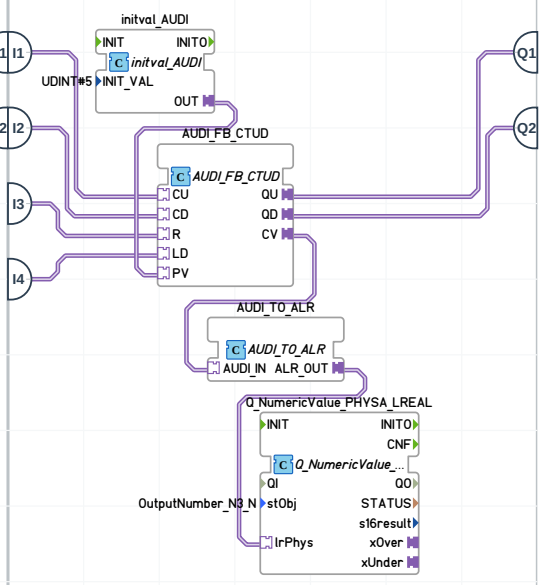
Uebung_222b_ALR

Standard IEC 61131-3 ALI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, LINT) mit Terminal-Ausgabe (PHYSA_LREAL)



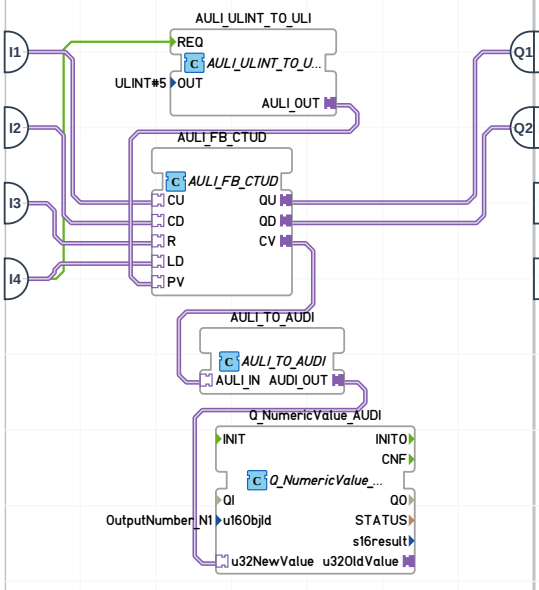
Uebung_223_AUDI

Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe



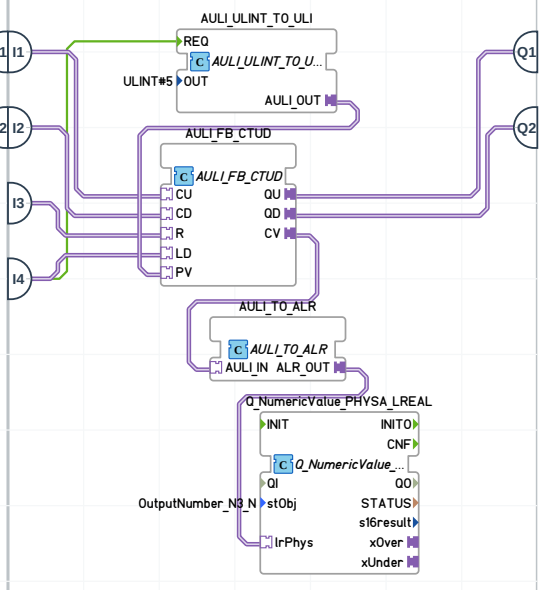
Uebung_223b_ALR

Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA_LREAL)



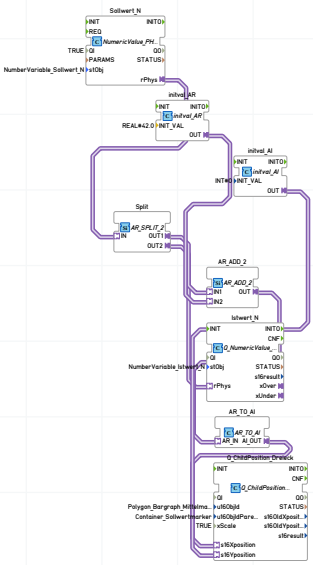
Uebung_224_AULI

Standard IEC 61131-3 AULI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe



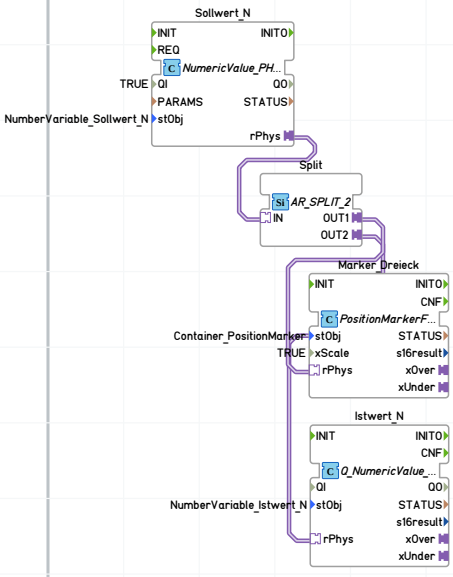
Uebung_224b_ALR

Standard IEC 61131-3 AULI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA_LREAL)



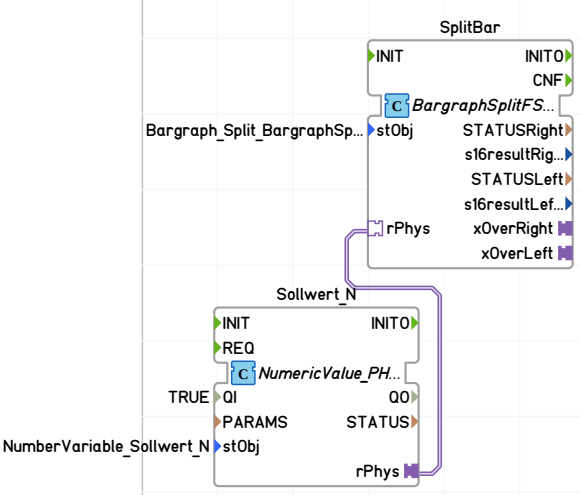
Uebung_225_AX

Dreieck-Sollwertmarker (wie Uebung_225 in test_B, aber ein reiner Adapter-Spiegel: NumericValue_PHYSA, AR_ADD_2, AR_TO_AI, Q_ChildPosition_AI,



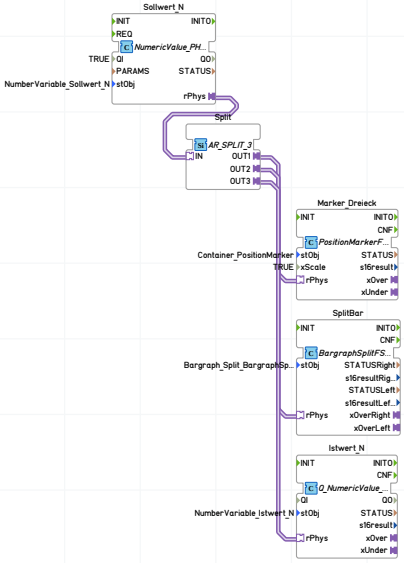
Uebung_225b_AX

Dreieck-Sollwertmarker (wie Uebung_225b in test_B, aber mit AR-Adapter-Bausteinen: NumericValue_PHYSA, PositionMarkerFSA, Q_NumericValue_PHYSA)



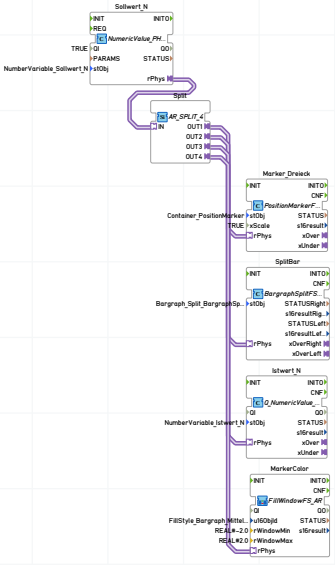
Uebung_226_AX

Split-Bargraph (wie Uebung_226 in test_B, aber mit AR-Adapter-Baustein BargraphSplitFS_AR statt REQ/rValue)



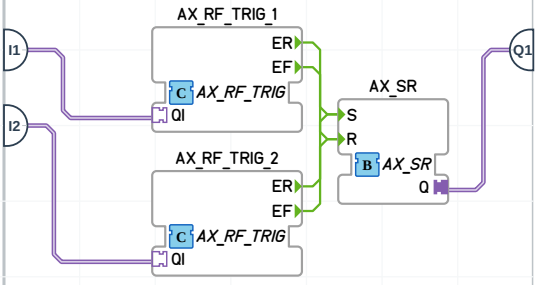
Uebung_227_AX

Kombiniert 225b_AX (Dreieck-Sollwertmarker) und 226_AX (Split-Bargraph): ein Sollwert treibt beide Anzeigen, komplett über AR-Adapter-Bausteine



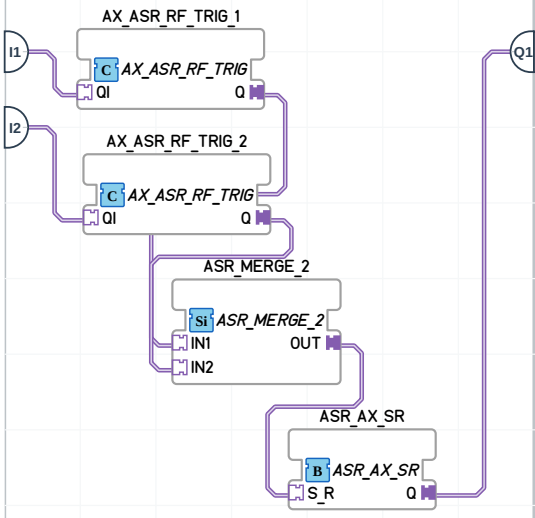
Uebung_228_AX

Ableitung von 227_AX: das Dreieck wird ROT außerhalb des Fensters -2..+2, GRÜN innerhalb



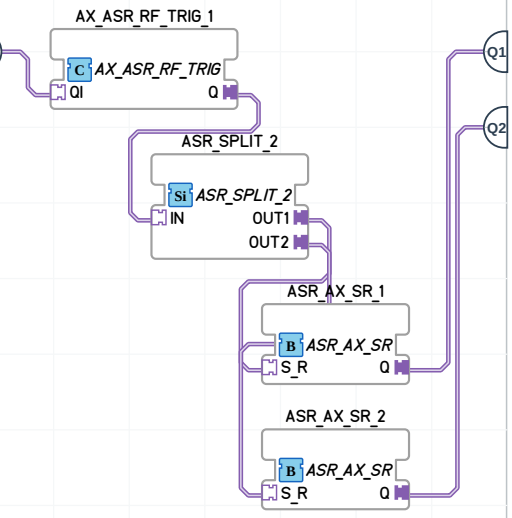
Uebung_229_AX

2 Taster (I1/I2) auf 1 SR-Latch (Last-Wins) via 2x AX_RF_TRIG + AX_SR, Ausgang Q1



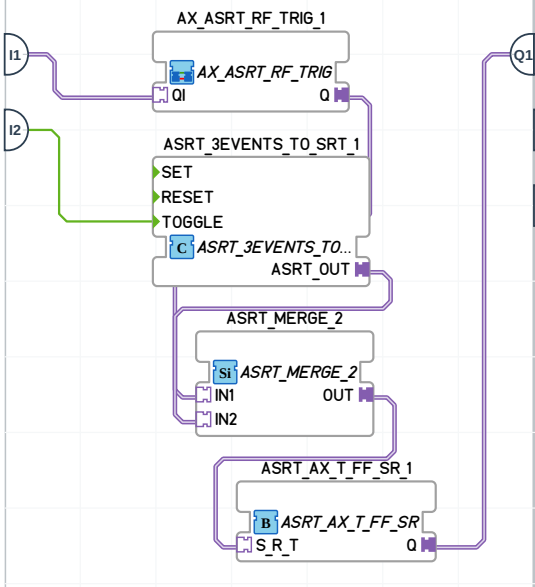
Uebung_230_AX

Wie Uebung_229_AX, aber mit ASR-Adapter-Umbau: 2x AX_ASR_RF_TRIG + ASR_MERGE_2 + ASR_AX_SR statt loser Event-Verkabelung



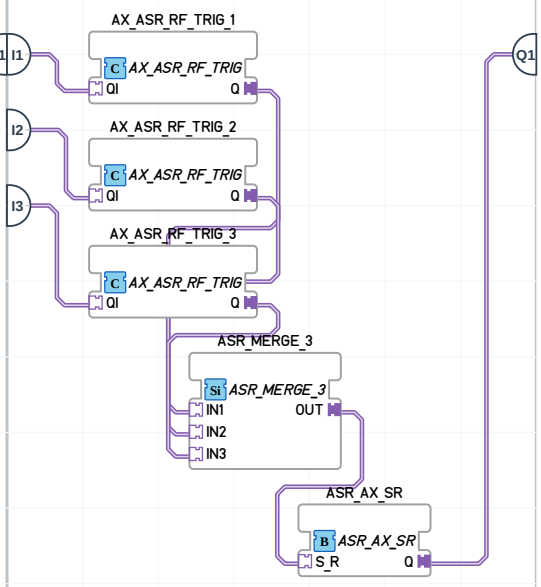
Uebung_231_AX

1 Taster via AX_ASR_RF_TRIG + ASR_SPLIT_2 auf 2 unabhaengige ASR_AX_SR-Latches (Q1+Q2) – Spiegelbild zu Uebung_230_AX (dort MERGE N->1, hier SPLIT 1->N)



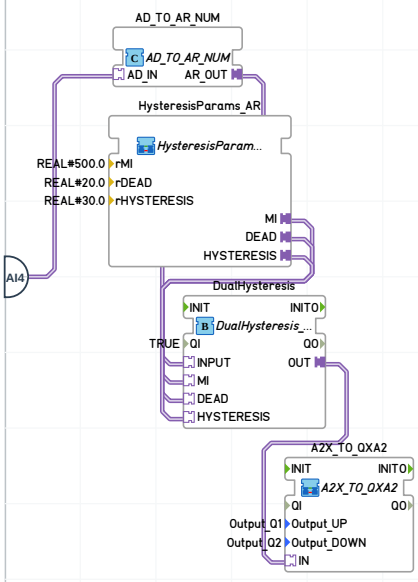
Uebung_232_AX

2 ASRT-Quellen (Taster Last-Wins + echter Toggle-Taster) auf 1 ASRT-Latch via ASRT_MERGE_2, Ausgang Q1



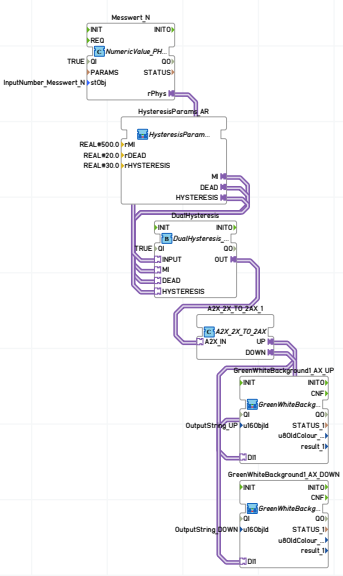
Uebung_233_AX

3 Taster (I1/I2/I3) auf 1 SR-Latch (Last-Wins) via 3x AX_ASR_RF_TRIG + ASR_MERGE_3 + ASR_AX_SR, Ausgang Q1



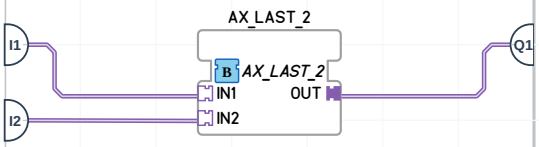
Uebung_234_AX

Zweipunktregler (Fuellstand/Thermostat-Muster) via DualHysteresis_AR_A2X + A2X_TO_QXA2 – reiner Hardware-Aufbau ohne VT



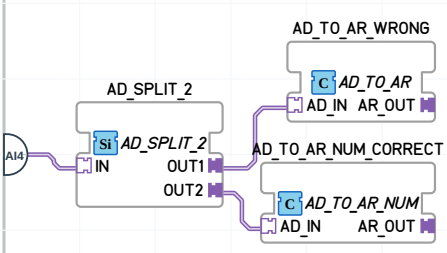
Uebung_235_AX

Zweipunktregler (Fuellstand/Thermostat-Muster) via DualHysteresis_AR_A2X, Messwert simuliert per VT-Eingabefeld, UP/DOWN als Hintergrundfarbe auf 2



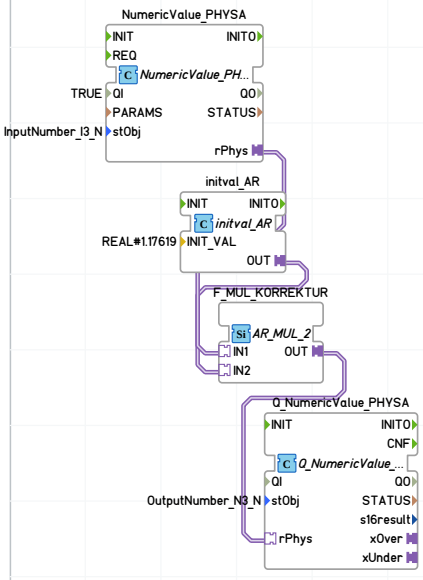
Uebung_236_AX

AX_LAST_2: derselbe
Last-Wins-auf-jeder-Flanke-Effekt wie
Uebung 229/230/233 AX, aber ohne Edge-Detektor-Latch



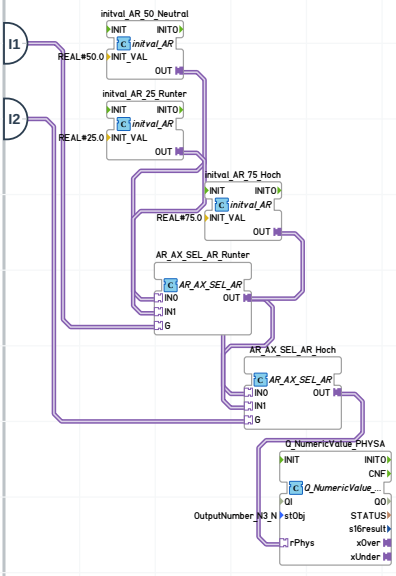
Uebung_237_AX

AD_TO_AR-Bitfalle: derselbe Analogwert einmal FALSCH
ueber AD_TO_AR (Bit-Reinterpretation) und einmal
RICHTIG ueber AD_TO_AR_NUM (echter numerischer Cast)



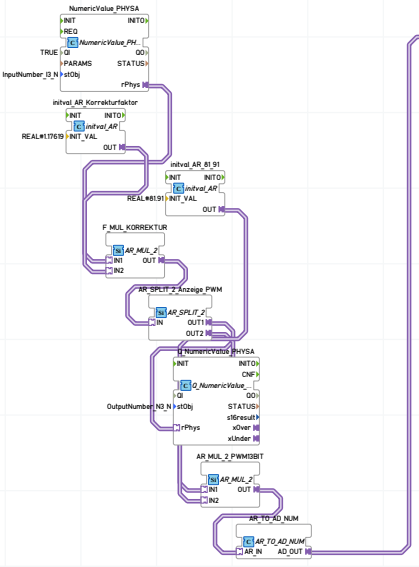
Uebung_245_AX

Sollwert mit festem Korrekturfaktor skalieren
(AR_MUL_2 = initval_AR) – Lastteiler-Kompensation,
I3 nach N3



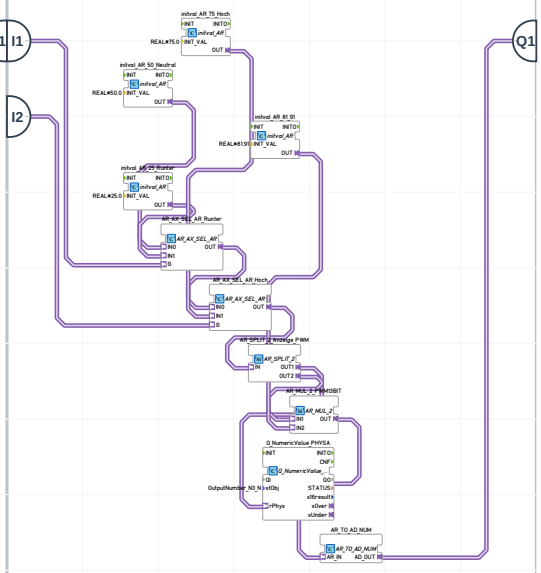
Uebung_246_AX

logiBUS_IXA I1(Hoch)/I2(Runner) auf
3-Stufen-Sollwert 75%/25%/50%(Neutral), zwei
AR_AX_SEL_AR verkettet



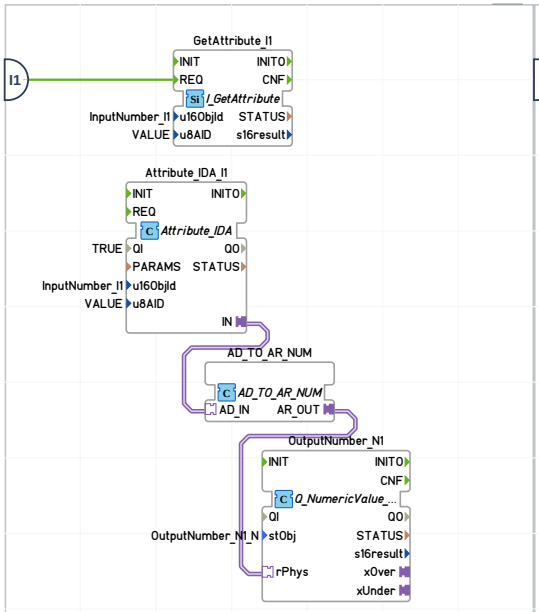
Uebung_247_AX

Wie Uebung_245_AX (Korrekturfaktor via
AR_MUL_2/initval_AR), zusaetzlich echter
logiBUS_QDA_PWM-Ausgang Q1 zum Nachmessen



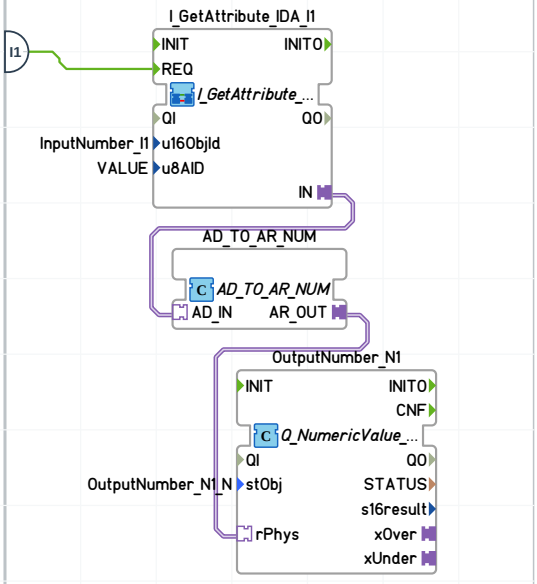
Uebung_248_AX

Wie Uebung_246_AX (Hoch/Runner auf 75/50/25%),
zusaetzlich echter logiBUS_QDA_PWM-Ausgang Q1 zum
Nachmessen



Uebung_249_AX

Asynchrone Objektattribut-Abfrage mittels I_GetAttribute (Kommando) und Attribute_IDA (asynchroner Empfang) bei Tasterdruck (Input_I1) und



Uebung_249b_AX

Asynchrone Objektattribut-Abfrage mittels gebündelter SubApp I_GetAttribute_IDA (Kommando + asynchroner Empfang in einem Baustein) bei